Real S^{12} and S^{24} Routes from Alpöge's Candidate $(3, 4, \infty)$ Two-Torus Fibration

Exact sphere carriers, four-dimensional cusp limits, and their sharp scope

A self-contained mathematical note

based on the candidate S^6 construction of Levent Alpöge

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Abstract

Starting from the geometric input of Alpöge's candidate $(3, 4, \infty)$ complex two-torus fibration—reconstructed and separately verified in the accompanying workbench, but imported here as stated input—we derive the requested S^{12}- and S^{24}-based four-dimensional decompactification routes directly from its native $T^2 \times \mathbb{R}^2$ cusp geometry. The cusp-local rank-two quotient in two copies and the unique full-monodromy-invariant rank-one quotient in four copies are optimal among native connected torus quotients with unscaled Euclidean quotient metric. The two-copy route discards the bounded T^2 in each original fibre and combines the two surviving rank-two quotients; it does not retain all four directions of one fibre. After a based choice $X \cong_{\mathrm{diff}} S^6$, they give continuous surjective quotient maps

$$S^{12} \twoheadrightarrow S^6, \qquad S^{24} \twoheadrightarrow S^8.$$

The four-copy map $S^{24} \to S^8$ is nevertheless based null-homotopic: more generally, if $f = \widehat{q}_\gamma : S^6 \to S^2$, then $[f^{\wedge r}] = 0 \in \pi_{6r}(S^{2r})$ for every $r \geq 2$. The distinct two-copy map $S^{12} \to S^6$ is also based null-homotopic: more strongly, Toda's external-smash formula and the exact low-stem groups show that $x \wedge y : S^{12} \to S^6$ is based null-homotopic for every pair of based maps $x, y : S^6 \to S^3$. Neither conclusion computes either one-copy sphere-map class, and neither implies nullity of a map into one of the earlier conically stratified targets before the final sphere-valued postcomposition. For the specified $S^6 \to S^3$ map we now compute that datum: its complete regular inverse image and actual derivative framing bound an explicit collared $D^2 \times T^2$ inside one original cusp chart times an interval. The full holomorphic period correction is retained by an exact invertible chart map. Thus the one-copy map itself is based null-homotopic, as are all its Cartesian powers. The general framing-torsor calculation remains valid, but its actual coefficient for this map is zero. The separate one-copy map to S^2 remains open. Retaining the angle before the final sphere quotient instead gives $H = p_3 q_2 : X \to S^4$ with $[H] = \eta_4 \eta_5 \neq 0$. We compute its entire normally four-framed inverse torus and its Arf invariant one, proving that the original conical map q_2 is nonnull. The ordered Hopf smash $\eta_2 \wedge H : S^9 \to S^6$ represents $12\nu_6 \neq 0$ in $\mathbb{Z}/24$, and its threefold suspension represents $12\nu_9$ in $\pi_{12}(S^9)$. Its exact peripheral character has values $(12, 0, 12)$ and kernel $\chi^{-1}(2\mathbb{Z}/12)$. These statements use the specific retained-angle map, not the null angle-forgetting postcomposition. Their targets contain moving embedded metric four-spheres whose radius-R balls are exactly Euclidean for all sufficiently deep cusp parameters. The same radial quotients also form explicit Hausdorff second-countable five-manifolds with boundary $\mathbb{R}^4$. Each admits a continuous surjection to a compact cone on T^4, homeomorphic away from that boundary and constant on it. These comparison maps are nonproper and are not quotient maps; in particular they are not topological blow-downs or resolutions, and the compact cone does not carry the topology obtained by collapsing the boundary $\mathbb{R}^4$.

We prove the point-set quotient statements, optimality, target homotopy types, local homology, and metric assertions in full. For one fixed cusp-adapted metric on each compact target, the four-dimensional slices appear as vertical $\mathbb{R}^4$ subspaces of moving-base-point bubbles; the full bubbles are $\mathbb{R}^8$ for the S^{12} route and $\mathbb{R}^{12}$ for the S^{24} routes. Any unrescaled fixed compatible target metric instead

collapses the slices to the cone vertex, and no metric on a source sphere is produced. A separate finite-cover construction also decompactifies the original four-tori to $\mathbb{R}^4$, but it is ancillary: it neither uses nor replaces the S^{12}/S^{24} sphere-carrier construction studied here. We also give an explicit $SU(3)$ clutching calculation: nonzero second-Chern bundles on the moving T^4 slices cannot extend over the contractible cone carrier or over a null ambient four-sphere, while pullback along either null sphere map is trivial. These facts are then compared with the currently separately constructed higher-rung Gram quotient $M_3(\mathbb{C})/U(3) \cong \mathrm{Herm}_3^+$, whose one-point compactification is a nine-ball with rank-deficient boundary an S^8. The smash-map, product, join, bundle, almost-complex, and fixed-metric obstructions do not remove that topological carrier; they do rule out the exact stronger identifications to which they apply. The chosen based homeomorphism of the two abstract S^8's and the resulting null composite are the strongest bridge proved here. Constructing a map that also respects rank strata, metrics, or bundle data remains an active bridge problem, not a claim of ontological separation. Nothing below constructs a complex S^{12} or S^{24}, an ambient four-dimensional compact target, or a physical spacetime. For S^8, the almost-complex obstruction follows from the p_2-formula and the Hirzebruch signature theorem (or, for the standard sphere, stable tangent triviality). For S^{12}, the Borel–Serre mod-5 reduced-power calculation, Bott integrality, and the stable-tangent p_3-identity give three independent obstructions.

Status and provenance. The compact complex threefold X, its torus fibration, its period functions, its three local fillings, and the recognition of its underlying smooth manifold as S^6 belong to the unpublished candidate construction of Levent Alpöge [Alpöge, 2026]. They are the geometric input of this note, have been reconstructed and separately verified in the accompanying workbench, and are restated below in the exact form used rather than reproved here. The quotient, smash, join, framing, clutching, homology, metric, optimality, and obstruction statements are proved here from that displayed input. The symbols S^{12} and S^{24} refer to based homeomorphism types of smash powers of the underlying real S^6, not to Cartesian powers and not to new complex manifolds. The central positive result is an exact four-dimensional limit carried by moving target slices and by a noncompact decompactification boundary. It is not an assertion that either compact source sphere, or either fixed compact quotient target, becomes $\mathbb{R}^4$.

1 The exact input data

1.1 The candidate fibration and its periods

Let $B \cong \mathbb{P}^1$ have three distinguished values p_0, p_1, p_2, with p_0 the cusp and with finite local orders $m_1 = 3$ and $m_2 = 4$. The candidate construction supplies a connected compact complex threefold and a proper surjective holomorphic map

$$f : X \longrightarrow B \tag{1.1}$$

with connected fibres. Its cusp fibre $W = f^{-1}(p_0)$ is reduced, irreducible, nonnormal, and toroidal; the fibres at p_1, p_2 have scheme-theoretic multiplicities $3, 4$. The same construction identifies the underlying based smooth manifold with the standard sphere:

$$(X, x_0) \cong_{\mathrm{diff}} (S^6, *), \qquad x_0 \in W. \tag{1.2}$$

Only this real based diffeomorphism is used below.

Put $B^\circ = B \setminus \{p_0, p_1, p_2\}$ and $J = f^{-1}(B^\circ)$. On the marked universal cover of B° take the ordered lattices

$$V = \mathbb{Z}\langle \gamma, u, w, \delta \rangle, \qquad \Lambda = V^* = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle. \tag{1.3}$$

There are holomorphic functions τ, μ, β for which

$$\Pi = [Z \mid I_2], \qquad Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix} \tag{1.4}$$

maps Λ isomorphically to a lattice in $\mathbb{C}^2$. Thus a marked regular fibre is $\mathbb{C}^2/\Pi(z)\Lambda$. The nondegeneracy inequalities are

$$\Im\tau > 0, \qquad \Im\beta - \frac{6(\Im\mu)^2}{\Im\tau} < 0. \tag{1.5}$$

The equivariant descent fixes the zero section. Consequently J is a holomorphic family of complex Lie groups, rather than merely an affine torus bundle.

Choose a cusp coordinate t with $t(p_0) = 0$. The canonical cusp calculation gives functions $\mu(t), b(t), h(t)$ holomorphic on a full disc, where

$$b = \beta + \tau, \qquad s = \tau - h, \qquad e^{2\pi i s} = t, \tag{1.6}$$

and the exact identity is

$$Z = s\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \begin{pmatrix} 6\mu & h \\ b-h & \mu \end{pmatrix}. \tag{1.7}$$

On the positive radial ray set $t = e^{-2\pi T}$ and choose $s = iT$. Write

$$\begin{aligned} &\mu_T = \alpha_T + i m_T, && h_T = \eta_T + i H_T, && b_T = \xi_T + i B_T, \\ &c_T = \xi_T - \eta_T, && L_T = T + H_T, && q_T = L_T - B_T, \\ &D_T = L_T q_T + 6m_T^2. \end{aligned} \tag{1.8}$$

If $m = \Im\mu(0)$, $H = \Im h(0)$, and $B = \Im b(0)$, holomorphy gives

$$m_T = m + O(e^{-2\pi T}), \quad H_T = H + O(e^{-2\pi T}), \quad B_T = B + O(e^{-2\pi T}). \tag{1.9}$$

Identify $\mathbb{C}^2$ with $\mathbb{R}^4$ in the order (x_1, y_1, x_2, y_2). The exact real period matrix is

$$P_T = \begin{pmatrix} 6\alpha_T & \eta_T & 1 & 0 \\ 6m_T & L_T & 0 & 0 \\ c_T & \alpha_T & 0 & 1 \\ -q_T & m_T & 0 & 0 \end{pmatrix}, \qquad \det P_T = -D_T, \qquad D_T > 0. \tag{1.10}$$

Equip

$$F_T = \mathbb{R}^4/P_T\mathbb{Z}^4 \tag{1.11}$$

with the quotient of the coordinate Euclidean metric. This is a metric chosen on the marked radial fibres; it is not asserted to extend as one canonical global metric on X.

1.2 Integral monodromy and its invariant flag

In the ordered basis of Λ, the three monodromies are

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}, \quad M_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}. \tag{1.12}$$

They satisfy $A_1^3 = A_2^4 = I$ and $A_1 A_2 M_0 = I$. Direct multiplication gives

$$(M_0 - I)(a, b, c, d)^{\mathsf{T}} = (0, 0, b, -a)^{\mathsf{T}}. \tag{1.13}$$

Hence the saturated cusp lattice

$$K_\Lambda := \ker(M_0 - I) = \mathrm{im}(M_0 - I) = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle \tag{1.14}$$

is both the invariant and vanishing image lattice for the unipotent cusp operator.

Define the primitive covector and its kernel by

$$\varepsilon_\gamma(a\widehat{\gamma} + b\widehat{u} + c\widehat{w} + d\widehat{\delta}) = a, \qquad L_\Lambda := \ker \varepsilon_\gamma = \mathbb{Z}\langle \widehat{u}, \widehat{w}, \widehat{\delta} \rangle. \tag{1.15}$$

The first row of each matrix in equation (1.12) shows $\varepsilon_\gamma A_j = \varepsilon_\gamma$ and $\varepsilon_\gamma M_0 = \varepsilon_\gamma$. Moreover

$$\begin{aligned} &A_1\widehat{u} = -\widehat{w} + \widehat{\delta}, \quad A_1\widehat{w} = \widehat{u} - \widehat{w}, \quad A_1\widehat{\delta} = \widehat{\delta}, \\ &A_2\widehat{u} = \widehat{w}, \qquad\qquad A_2\widehat{w} = -\widehat{u} + \widehat{\delta}, \quad A_2\widehat{\delta} = \widehat{\delta}. \end{aligned} \tag{1.16}$$

Thus L_Λ is invariant and is the smallest invariant primitive lattice containing K_Λ. Solving invariance for a row covector (a, b, c, d) gives $b = 2c$, $d = 3c$ from A_1 and $b = c$, $d = 2c$ from A_2; hence $b = c = d = 0$. Therefore

$$\mathrm{Hom}(\Lambda, \mathbb{Z})^{\pi_1(B^\circ)} = \mathbb{Z}\varepsilon_\gamma. \tag{1.17}$$

1.3 The native cusp quotient and the invariant circle quotient

Let $D_0 = \{|t| \le \epsilon\}$ be small enough to contain no singular value except p_0, and put $N_0 = f^{-1}(D_0)$. Set

$$\overline{\Lambda} = \Lambda / K_\Lambda, \qquad \overline{T}^2 = (\overline{\Lambda} \otimes \mathbb{R})/\overline{\Lambda}, \tag{1.18}$$

with ordered basis induced by $(\widehat{\gamma}, \widehat{u})$.

Lemma 1.1 (Native quotient on the punctured cusp)**.** *There is a real-analytic fibrewise group epimorphism*

$$\overline{\pi} : N_0 \setminus W \longrightarrow D_0^* \times \overline{T}^2 \tag{1.19}$$

with connected two-torus kernel. Its target local system is trivial. If

$$Y_T = \begin{pmatrix} 6m_T & L_T \\ -q_T & m_T \end{pmatrix}, \qquad Q_T = \mathbb{R}^2 / Y_T \mathbb{Z}^2, \tag{1.20}$$

then

$$\iota_T : \overline{T}^2 \longrightarrow Q_T, \qquad [(a, b)] \longmapsto [Y_T(a, b)^{\mathsf{T}}] \tag{1.21}$$

is an isomorphism and

$$\pi_T := \iota_T \circ \overline{\pi}|_{F_T} : F_T \longrightarrow Q_T \tag{1.22}$$

is the quotient induced by $\rho(x_1, x_2, x_3, x_4) = (x_2, x_4)$. With the Euclidean quotient metric,

$$\mathrm{sys}(Q_T) \ge T - C \tag{1.23}$$

for a constant C independent of sufficiently large T.

Proof. On the universal cover of D_0^* use the real-analytic flat marking $(z, x + \Lambda) \mapsto [z, \Pi(z)x]$. A deck generator acts by $(z, x + \Lambda) \mapsto (g_0 z, M_0 x + \Lambda)$. By equation (1.13), M_0 preserves K_Λ and induces the identity on $\overline{\Lambda}$. Hence

$$[z, \Pi(z)x] \longmapsto ([z], [x \bmod K_\Lambda])$$

is deck invariant and descends to equation (1.19). The sublattice K_Λ is generated by two members of a basis and is primitive, so the fibre kernel is connected.

The four columns of P_T are $v_{\gamma,T}, v_{u,T}, e_1, e_3$. Projection by ρ kills e_1, e_3 and maps the first two columns to the columns of Y_T. Since

$$\det Y_T = L_T q_T + 6m_T^2 = D_T > 0, \tag{1.24}$$

Y_T is invertible, equation (1.21) is a torus isomorphism and π_T has kernel $(\mathbb{R}e_1 \oplus \mathbb{R}e_3)/(\mathbb{Z}e_1 \oplus \mathbb{Z}e_3)$. Finally

$$Y_T = T\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \begin{pmatrix} 6m_T & H_T \\ B_T - H_T & m_T \end{pmatrix}. \tag{1.25}$$

The second matrix is uniformly bounded by equation (1.9). If C bounds its operator norm, the reverse triangle inequality gives

$$\|Y_T(a,b)^{\mathsf{T}}\| \geq (T - C)\sqrt{a^2 + b^2}$$

for every $(a,b) \in \mathbb{Z}^2$. This proves equation (1.23). □

The full monodromy has a different, one-dimensional quotient. In the flat marking define

$$\chi_\gamma([z, \Pi(z)x]) = ([z], \varepsilon_\gamma(x) \bmod \mathbb{Z}). \tag{1.26}$$

Equation (1.17) makes this a global real-analytic fibrewise group epimorphism

$$\chi_\gamma : J \longrightarrow B^\circ \times (\mathbb{R}/\mathbb{Z}) \tag{1.27}$$

and put

$$\theta_\gamma := \mathrm{pr}_{\mathbb{R}/\mathbb{Z}} \circ \chi_\gamma : J \longrightarrow \mathbb{R}/\mathbb{Z}. \tag{1.28}$$

Thus $\chi_\gamma = (f, \theta_\gamma)$; in particular, scalar circle coordinates below use θ_γ, not the product-valued map χ_γ. with connected three-torus kernel determined by L_Λ. It is a real quotient, not a holomorphic quotient of complex tori.

On F_T, its differential is the dual vector

$$P_T^{-\mathsf{T}} e_\gamma = \left(0, \frac{m_T}{D_T}, 0, -\frac{L_T}{D_T}\right)^{\mathsf{T}}, \qquad e_\gamma = (1,0,0,0)^{\mathsf{T}}. \tag{1.29}$$

Thus the quotient circle, written in arclength coordinates as

$$C_T = \mathbb{R}/\ell_T\mathbb{Z}, \tag{1.30}$$

has circumference

$$\ell_T = \frac{1}{\|P_T^{-\mathsf{T}} e_\gamma\|} = \frac{D_T}{\sqrt{L_T^2 + m_T^2}} = T + H - B + O(T^{-1}). \tag{1.31}$$

Indeed, equation (1.9) gives $D_T = T^2 + (2H - B)T + O(1)$ and $\sqrt{L_T^2 + m_T^2} = T + H + O(T^{-1})$.

1.4 The two finite Seifert fillings

We record the filling data because the full-base cone construction uses the actual target overlap maps. Let Δ_j be a linearizing disc above D_j with coordinate s, where $s \mapsto e^{-2\pi i/m_j} s$. Put

$$\begin{aligned} (m_1, v_1, \kappa_1) &= (3, \widehat{\gamma} + 2\widehat{u} - 4\widehat{w}, 1), \\ (m_2, v_2, \kappa_2) &= (4, -\widehat{\gamma} - 3\widehat{u} + 3\widehat{w}, -1), \end{aligned} \qquad \kappa_j = \varepsilon_\gamma(v_j). \tag{1.32}$$

Matrix multiplication using equation (1.12) gives $A_j v_j = v_j$. On the un-conjugated flat family the generator is

$$\Phi_j(s,x) = \left(e^{-2\pi i/m_j} s, A_j x + \frac{v_j}{m_j} \right). \tag{1.33}$$

It has order m_j. Every nonidentity power is free: away from $s = 0$ the base point moves, while over $s = 0$ a fixed point would imply that the primitive invariant character changes integrally, contrary to

$$\varepsilon_\gamma \left(A_j^k x + \frac{k v_j}{m_j} \right) = \varepsilon_\gamma(x) + \frac{k\kappa_j}{m_j} \pmod{\mathbb{Z}}, \quad 0 < k < m_j. \tag{1.34}$$

The quotient is the smooth finite filling $N_j \to D_j$.

The character descends equivariantly to

$$C_j = \frac{\Delta_j \times (\mathbb{R}/\mathbb{Z})}{(s,\theta) \sim (e^{-2\pi i/m_j} s, \theta + \kappa_j/m_j)}. \tag{1.35}$$

The action is free because $\gcd(\kappa_j, m_j) = 1$. To see the topology without an external classification theorem, put $u = e^{2\pi i\theta}$ and choose $r_j\kappa_j \equiv 1 \pmod{m_j}$; one may take $r_1 = 1, r_2 = -1$. The invariant coordinates

$$w = s u^{r_j}, \qquad U = u^{m_j} \tag{1.36}$$

identify C_j real-analytically with a solid torus, and

$$t_j = s^{m_j} = w^{m_j} U^{-r_j}. \tag{1.37}$$

Thus $C_j \to D_j$ has a multiple central circle of order m_j. The map $N_j \to C_j$ has fibres modeled on the connected torus $(L_\Lambda \otimes \mathbb{R})/L_\Lambda$ and is a smooth proper $\mathbb{T}^3$-bundle; no group zero is asserted on the affine central filling.

It remains important that this local target agrees with the product circle on the punctured overlap. Let $\Pi_{\mathbb{R}}(z) : \Lambda \otimes \mathbb{R} \to \mathbb{C}^2$ be the underlying real isomorphism. For a logarithm branch ℓ define $b_j^\ell(z)$ uniquely by

$$\Pi_{\mathbb{R}}(z) b_j^\ell(z) = \frac{\ell(z)}{2\pi i} \Pi(z) v_j, \qquad \beta_j(z) = \varepsilon_\gamma(b_j^\ell(z)) \bmod \mathbb{Z}. \tag{1.38}$$

Changing ℓ by $2\pi i n$ adds $n v_j$ and changes the last expression by $n\kappa_j \in \mathbb{Z}$, so β_j is branch independent. Directly from the changed logarithm branch,

$$\beta_j(g_j z) = \beta_j(z) - \frac{\kappa_j}{m_j} \pmod{\mathbb{Z}}. \tag{1.39}$$

Therefore

$$h_j^{S^1}(z,\theta) = (z, \theta + \beta_j(z)) \tag{1.40}$$

conjugates the affine target action in equation (1.35) to the linear product-circle action. If $h_j : N_j|_{D_j^*} \to J|_{D_j^*}$ is the logarithmic source overlap and $\overline{\chi}_j : N_j \to C_j$ the descended map, equivariance gives the exact identity

$$\chi_\gamma \circ h_j = h_j^{S^1} \circ \overline{\chi}_j. \tag{1.41}$$

1.5 Why the literal circle map cannot cross the cusp

The cone used later changes the codomain at p_0; this is forced by a pointwise obstruction. On the radial ray write

$$Z_T = iTB_0 + C(e^{-2\pi T}), \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{1.42}$$

For $a = (a_1, a_2)^{\mathsf{T}} \in \mathbb{R}^2$, let $x_T(a)$ be represented by $Z_T a$ with zero short-period coordinates. In the dense cusp torus, $x_T(a) = \exp(2\pi i Z_T a)$, and its normalized logarithmic position is

$$y_T(a) = B_0 a + \frac{\Im C(e^{-2\pi T})a}{T}, \qquad \chi_{\gamma,T}(x_T(a)) = a_1 \bmod \mathbb{Z}. \tag{1.43}$$

In the affine toric chart of $\mathrm{conv}(0, e_1, e_2)$ the coordinates are

$$z_0 = \frac{t}{x_1 x_2}, \qquad z_1 = x_1, \qquad z_2 = x_2. \tag{1.44}$$

If $B_0 a = (y_1, y_2)$ is in the interior of this triangle, the three absolute values are $|t|^{1-y_{T,1}-y_{T,2}}, |t|^{y_{T,1}}, |t|^{y_{T,2}}$ and all tend to zero. Hence all such paths approach the same point of W.

Take

$$a = (-\tfrac{1}{4}, \tfrac{1}{4}), \qquad a' = (-\tfrac{1}{2}, \tfrac{1}{4}). \tag{1.45}$$

Their limiting positions are $(1/4, 1/4)$ and $(1/4, 1/2)$, both in the triangle interior, whereas their character values are $3/4$ and $1/2$. Thus χ_γ has no continuous extension through W. Repeating one of these two alternatives in one factor and fixing the paths in the other factors proves the same for every product retaining this literal character. The conical target below succeeds precisely because it collapses the entire limiting boundary torus.

Lemma 1.2 (Exact balls in Euclidean and collapsed-set quotients)**.** *Let $L \subset \mathbb{R}^n$ be a full lattice, let*

$$d_L([x],[y]) = \min_{\lambda \in L} \|x - y + \lambda\|_2, \qquad \mathrm{sys}(L) = \min_{0 \neq \lambda \in L} \|\lambda\|_2,$$

and let $q_L : \mathbb{R}^n \to \mathbb{R}^n/L$ be the quotient projection. If $\mathrm{sys}(L) > 4R$, then q_L restricts to an isometry from the Euclidean closed ball $\overline{B}_R(0)$ onto the closed radius-R ball about $[0]$ in $(\mathbb{R}^n/L, d_L)$.

If (E, d) is a compact length space, $A \subset E$ is closed, and $q_A : E \to E/A$ is equipped with the quotient length metric, then

$$d_{E/A}(q_A(x), q_A(y)) = \min\{d(x,y), d(x,A) + d(y,A)\}. \tag{1.46}$$

If $d(p, A) > 2R$, the restriction of q_A is an isometry from $\overline{B}_R(p)$ onto $\overline{B}_R(q_A(p))$. Thus any Euclidean metric ball on the left remains exactly the same Euclidean ball after the collapse.

Proof. For $x, y \in \overline{B}_R(0)$ and $0 \neq \lambda \in L$,

$$\|x - y + \lambda\|_2 \geq \|\lambda\|_2 - \|x - y\|_2 > 4R - 2R \geq \|x - y\|_2.$$

Hence the zero translate is the unique shortest translate on the ball. Conversely, the minimum defining $d_L([0],[y])$ is attained because L is discrete. If that distance is at most R, a minimizing representative lies in $\overline{B}_R(0)$. This proves both isometry and surjectivity onto the quotient ball.

For the second assertion, every quotient path either avoids the collapsed point and has length at least $d(x, y)$, or meets it and has length at least $d(x, A) + d(y, A)$; both bounds are attained as infima by paths in E. This proves equation (1.46). If $x, y \in \overline{B}_R(p)$, then

$$d(x, A) + d(y, A) \geq 2d(p, A) - d(p, x) - d(p, y) > 2R \geq d(x, y),$$

so no path through the collapsed point shortens their distance. If $d_{E/A}(q_A(p), q_A(y)) \leq R$, the second entry in the minimum is greater than $2R$, and therefore $d(p, y) \leq R$. This proves the onto-ball assertion. □

2 The native two-torus route and its S^{12} source

Define the compact Hausdorff space

$$\mathcal{A} = \frac{[0,\epsilon] \times S^1 \times \overline{T}^2}{\{0\} \times S^1 \times \overline{T}^2 \sim v_0, \quad \{\epsilon\} \times S^1 \times \overline{T}^2 \sim v_1}. \tag{2.1}$$

Rescaling the interval gives

$$\mathcal{A} \cong \mathrm{Susp}(S^1 \times \overline{T}^2) \cong \mathrm{Susp}(T^3). \tag{2.2}$$

The first circle remembers the base angle. Forgetting it would give a smaller target with the same map on a fixed radial fibre, but would not record the quotient over the whole punctured cusp.

Theorem 2.1 (Exact localized S^{12} route)**.** *There is a continuous surjective based quotient map*

$$q_2 : (X, x_0) \longrightarrow (\mathcal{A}, v_0) \tag{2.3}$$

defined by

$$\begin{aligned} q_2(x) &= [r, e^{i\phi}, \overline{\pi}_f(x)] && \text{if } x \in N_0 \setminus W, \quad t(f(x)) = re^{i\phi},\ 0 < r < \epsilon, \\ q_2(W) &= v_0, && q_2(X \setminus \operatorname{int} N_0) = v_1, \end{aligned} \tag{2.4}$$

where $\overline{\pi}_f$ is the fibre component of equation (1.19); the boundary $f^{-1}(|t| = \epsilon)$ maps to v_1.

Its smash square is a global based map

$$q_2^{\wedge 2} : X \wedge X \longrightarrow \mathcal{A} \wedge \mathcal{A}, \qquad X \wedge X \cong S^{12}. \tag{2.5}$$

Let $\sigma_Z^{(m)} : Z^m \to Z^{\wedge m}$ denote the smash projection. On a synchronized radial product, the restrictions

$$\begin{aligned} e^{(2)}_{X,T} &:= \sigma^{(2)}_X|_{F^2_T} : F^2_T \hookrightarrow X^{\wedge 2}, \\ e^{(2)}_{\mathcal{A},T}(y_1, y_2) &:= \sigma^{(2)}_{\mathcal{A}}([r_T, 1, \iota_T^{-1}(y_1)], [r_T, 1, \iota_T^{-1}(y_2)]) : Q^2_T \hookrightarrow \mathcal{A}^{\wedge 2} \end{aligned}$$

are embeddings, where $r_T = e^{-2\pi T}$. The restriction of the smash map factors exactly as

$$q_2^{\wedge 2} \circ e^{(2)}_{X,T} = e^{(2)}_{\mathcal{A},T} \circ (\pi_T \times \pi_T). \tag{2.6}$$

The product quotient $\pi_T \times \pi_T$ is not an embedding: its connected kernel is the four-torus $\ker(\pi_T)^2 \cong T^4$*. Equip $\mathbb{R}^4 = \mathbb{R}^2 \oplus \mathbb{R}^2$ with its Euclidean ℓ^2 norm and equip*

$$Q^2_T = \mathbb{R}^4 / \operatorname{diag}(Y_T, Y_T)\mathbb{Z}^4$$

with the explicit quotient distance

$$d_{Q^2_T}([x], [y]) = \min_{n \in \mathbb{Z}^4} \|x - y + \operatorname{diag}(Y_T, Y_T)n\|_2. \tag{2.7}$$

This is the intrinsic Euclidean ℓ^2-product metric; it is assigned to the slice independently of any ambient metric on $\mathcal{A}^{\wedge 2}$. With this metric,

$$\mathrm{sys}(Q^2_T) \geq T - C, \qquad (Q^2_T, (0,0)) \longrightarrow (\mathbb{R}^4, 0) \tag{2.8}$$

in pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff topology. Once $T - C > 4R$, the universal quotient maps the Euclidean radius-R ball isometrically onto *the corresponding metric ball in Q^2_T.*

The reduced integral homology is

$$\widetilde{H}_k(\mathcal{A};\mathbb{Z}) \cong \begin{cases} \mathbb{Z}^3, & k = 2,3, \\ \mathbb{Z}, & k = 4, \\ 0, & \textit{otherwise}, \end{cases} \tag{2.9}$$

and

$$\widetilde{H}_k(\mathcal{A} \wedge \mathcal{A};\mathbb{Z}) \cong \begin{cases} \mathbb{Z}^9, & k = 4, \\ \mathbb{Z}^{18}, & k = 5, \\ \mathbb{Z}^{15}, & k = 6, \\ \mathbb{Z}^6, & k = 7, \\ \mathbb{Z}, & k = 8, \\ 0, & \textit{otherwise}. \end{cases} \tag{2.10}$$

In particular, the target is not S^4, S^{12}, or a compact four-manifold.

Proof. The equivalence relation in equation (2.1) is the union of the diagonal and the squares of the two closed end sections, hence is closed. A compact Hausdorff space modulo a closed equivalence relation is compact Hausdorff, and the quotient is exactly the unreduced suspension.

The interior formula is single valued by theorem 1.1; a turn in ϕ changes the marking by M_0, which is the identity modulo K_Λ. Let U be a neighborhood of v_0. Its inverse image in the cylinder contains the compact end section $\{0\} \times S^1 \times \overline{T}^2$, and hence contains $[0,\delta) \times S^1 \times \overline{T}^2$ for some $\delta > 0$. Therefore $q_2^{-1}(U)$ contains $f^{-1}(|t| < \delta)$, a neighborhood of W. The same compactness argument at v_1 gives uniform continuity at the outer boundary; together with the interior continuity and closed pasting, this proves continuity on X.

Every interior target point has a preimage by fibrewise surjectivity of $\overline{\pi}$, while the nonempty cusp fibre and exterior map to the two vertices. Thus q_2 is surjective. Since X is compact and $\mathcal{A}$ Hausdorff, it is a quotient map.

Because $q_2(x_0) = v_0$, functoriality of the smash quotient produces equation (2.5). The standard quotient cell structure gives $S^m \wedge S^n \cong S^{m+n}$; applying the based diffeomorphism equation (1.2) yields $X \wedge X \cong S^{12}$ [Hatcher, 2002, Chapter 0, Smash Products].

The radial fibre does not contain $x_0 \in W$, and its image lies in an interior cross-section of $\mathcal{A}$, away from v_0. Hence the source and target product slices miss the respective fat wedges. The restrictions of the smash projections to those compact slices are continuous and one-to-one. Their targets are Hausdorff because each fat wedge is closed in the corresponding compact Hausdorff product, so the restrictions are the displayed embeddings. Functoriality of the smash quotient and the definitions of q_2 and ι_T give precisely equation (2.6).

The period lattice of the product target is $\operatorname{diag}(Y_T, Y_T)\mathbb{Z}^4$, whose systole equals that of $Y_T\mathbb{Z}^2$. If $T - C > 4R$, theorem 1.2 applied to equation (2.7) says that the universal quotient projection is an exact isometry *onto* its radius-R quotient ball. Since this holds for every fixed R, so it proves all three convergences.

Reduced suspension shifts reduced homology by one. The homology of T^3 has ranks $3, 3, 1$ in positive degrees $1, 2, 3$, proving equation (2.9). All groups are free, so the smash Künneth theorem has no Tor terms. Convolving the ranks $(3, 3, 1)$ with themselves gives

$$9, \quad 18, \quad 2(3)(1) + 9 = 15, \quad 6, \quad 1$$

in degrees $4, 5, 6, 7, 8$, which proves equation (2.10). □

3 Two circle routes and their S^{24} sources

3.1 The localized one-circle construction

Define

$$\mathcal{B} = \frac{[0,\epsilon] \times S^1 \times S^1}{\{0\} \times S^1 \times S^1 \sim b_0, \quad \{\epsilon\} \times S^1 \times S^1 \sim b_1} \cong \mathrm{Susp}(T^2). \tag{3.1}$$

The two interior circle coordinates are the base angle and θ_γ.

Proposition 3.1 (Localized S^{24} route)**.** *There is a continuous surjective based quotient map*

$$q_{1,\mathrm{cusp}} : (X, x_0) \longrightarrow (\mathcal{B}, b_0) \tag{3.2}$$

given by

$$\begin{aligned} q_{1,\mathrm{cusp}}(x) &= [r, e^{i\phi}, \theta_\gamma(x)] && (x \in N_0 \setminus W,\ t(f(x)) = re^{i\phi}), \\ q_{1,\mathrm{cusp}}(W) &= b_0, && q_{1,\mathrm{cusp}}(X \setminus \operatorname{int} N_0) = b_1. \end{aligned} \tag{3.3}$$

Its fourth smash power has source $X^{\wedge 4} \cong S^{24}$. Identify the normalized character circle with the arclength circle by $\vartheta_T : \mathbb{R}/\mathbb{Z} \to C_T$, $\vartheta_T([\theta]) = [\ell_T \theta]$. In the following radial factorizations, $\chi_{\gamma,T}$ denotes the C_T-valued map $\vartheta_T \circ \theta_\gamma|_{F_T}$. Define

$$e^{(4)}_{X,T} := \sigma^{(4)}_X|_{F_T^4} : F_T^4 \hookrightarrow X^{\wedge 4},$$

$$e^{(4)}_{\mathcal{B},T}(c_1, \dots, c_4) := \bigwedge_{i=1}^{4} [r_T, 1, \vartheta_T^{-1}(c_i)] : C_T^4 \hookrightarrow \mathcal{B}^{\wedge 4}.$$

Then the synchronized radial restriction factors exactly as

$$q_{1,\mathrm{cusp}}^{\wedge 4} \circ e^{(4)}_{X,T} = e^{(4)}_{\mathcal{B},T} \circ \chi_{\gamma,T}^{\times 4}. \tag{3.4}$$

The two displayed e-maps are embeddings, but the intervening product quotient is not: its connected kernel is $\ker(\chi_{\gamma,T})^4 \cong T^{12}$*. Its intrinsic target slices have the pointed $\mathbb{R}^4$ limit in equation (3.20) below. Moreover*

$$\widetilde{H}_k(\mathcal{B}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^2, & k = 2, \\ \mathbb{Z}, & k = 3, \\ 0, & \textit{otherwise}, \end{cases} \tag{3.5}$$

and

$$\widetilde{H}_k(\mathcal{B}^{\wedge 4}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{16}, & k = 8, \\ \mathbb{Z}^{32}, & k = 9, \\ \mathbb{Z}^{24}, & k = 10, \\ \mathbb{Z}^{8}, & k = 11, \\ \mathbb{Z}, & k = 12, \\ 0, & \textit{otherwise}. \end{cases} \tag{3.6}$$

Proof. Monodromy invariance of ε_γ makes the interior formula single valued. The two endpoint-continuity and surjectivity arguments are the proof of theorem 2.1 with compact cross-section T^2 in place of T^3. Compact-to-Hausdorff again makes the map a quotient map. The base point lies in W, so the fourth smash power exists. The radial source and target product slices avoid the closed fat wedges. As in the proof of theorem 2.1, the smash projections restrict to embeddings on those compact slices, and functoriality gives equation (3.4). Finally, suspension shifts the free homology of T^2. In the fourth smash power, choosing j of the four factors in degree 3 and the remaining factors in degree 2 gives degree $8 + j$ and rank $\binom{4}{j} 2^{4-j}$, proving the formulas. □

3.2 Full-base Seifert gluing and cusp coning

Choose pairwise disjoint closed coordinate discs D_0, D_1, D_2 and put

$$P = B \setminus \bigcup_{j=0}^{2} \operatorname{int} D_j. \tag{3.7}$$

Glue the two solid-torus targets by their exact overlap maps:

$$\mathcal{M} = (P \times S^1) \underset{h_1^{S^1}}{\cup} C_1 \underset{h_2^{S^1}}{\cup} C_2. \tag{3.8}$$

The overlap maps restrict to homeomorphisms of the closed boundary tori. Consequently this is a compact orientable three-manifold with its one remaining boundary

$$\partial\mathcal{M} = \partial D_0 \times S^1 \cong T^2. \tag{3.9}$$

Attaching a cone along its closed boundary gives the compact Hausdorff finite CW space

$$\mathcal{C} = \mathcal{M} \cup_{\partial\mathcal{M}} \operatorname{Cone}(\partial\mathcal{M}), \tag{3.10}$$

and denote the cone vertex by c_0.

Lemma 3.2 (The cone attachment as a collapsed-boundary quotient)**.** *There is a based homeomorphism*

$$(\mathcal{C}, c_0) \cong (\mathcal{M}/\partial\mathcal{M}, [\partial\mathcal{M}]). \tag{3.11}$$

Moreover $\mathcal{C} \setminus \{c_0\} \cong \operatorname{int}\mathcal{M}$.

Proof. Choose a collar $h : \partial\mathcal{M} \times [0,1] \hookrightarrow \mathcal{M}$ with $h(y,0) = y$. Write the attached cone as

$$\frac{\partial\mathcal{M} \times [-1,0]}{\partial\mathcal{M} \times \{-1\} \text{ collapsed}},$$

with its 0-face glued to $h(\partial\mathcal{M} \times \{0\})$. Together with the internal collar, this is

$$\frac{\partial\mathcal{M} \times [-1,1]}{\partial\mathcal{M} \times \{-1\} \text{ collapsed}}.$$

The linear reparametrization $t \mapsto (t+1)/2$ identifies it with

$$\frac{\partial\mathcal{M} \times [0,1]}{\partial\mathcal{M} \times \{0\} \text{ collapsed}},$$

the image of the collar in $\mathcal{M}/\partial\mathcal{M}$, and it is the identity on the common $t = 1$ face. Extending by the identity on the complement of the open collar gives equation (3.11). Removing the collapsed point leaves $\mathcal{M}$ with an external open collar attached; the same collar reparametrization absorbs it into the internal collar and gives $\mathcal{C} \setminus \{c_0\} \cong \operatorname{int}\mathcal{M}$. □

Lemma 3.3 (Homology of the Seifert target and its cone)**.** *For boundary loops ρ_0, ρ_1, ρ_2 of P, oriented so that $\rho_0 + \rho_1 + \rho_2 = 0$, and the product-circle class η, the filling relations are*

$$3\rho_1 - \eta = 0, \qquad 4\rho_2 + \eta = 0. \tag{3.12}$$

Consequently

$$H_1(\mathcal{M};\mathbb{Z}) \cong \mathbb{Z}\langle g\rangle, \qquad \rho_0 = -g, \quad \rho_1 = 4g, \quad \rho_2 = -3g, \quad \eta = 12g, \tag{3.13}$$

$H_2(\mathcal{M};\mathbb{Z}) = 0$, *and*

$$\widetilde{H}_k(\mathcal{C};\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 2,3, \\ 0, & \textit{otherwise.} \end{cases} \tag{3.14}$$

Proof. The solid-torus charts equation (1.36)–equation (1.37) identify the two meridian slopes, with the displayed orientation, as $3\rho_1 - \eta$ and $4\rho_2 + \eta$. Hence

$$H_1(\mathcal{M};\mathbb{Z}) = \frac{\mathbb{Z}\langle \eta, \rho_1, \rho_2\rangle}{\langle 3\rho_1 - \eta, 4\rho_2 + \eta\rangle}.$$

The relation matrix has rank two and its 2×2 minors have greatest common divisor one, so the quotient is torsion free of rank one. The assignments in equation (3.13) satisfy the two relations and are primitive, which proves the exact identification. They also show that $H_1(\partial\mathcal{M}) \to H_1(\mathcal{M})$ is surjective.

An orientable compact three-manifold with one torus boundary has $H_3(\mathcal{M}) = 0$ and $\chi(\mathcal{M}) = \frac{1}{2}\chi(\partial\mathcal{M}) = 0$. Since $b_0 = b_1 = 1$, it follows that $b_2 = 0$. Integrally, Poincaré–Lefschetz duality and the universal coefficient theorem give

$$H_2(\mathcal{M};\mathbb{Z}) \cong H^1(\mathcal{M}, \partial\mathcal{M};\mathbb{Z}) \cong \mathrm{Hom}(H_1(\mathcal{M}, \partial\mathcal{M};\mathbb{Z}), \mathbb{Z}),$$

because $H_0(\mathcal{M}, \partial\mathcal{M};\mathbb{Z}) = 0$. Thus $H_2(\mathcal{M};\mathbb{Z})$ is free. Its rank is $b_2 = 0$, so it vanishes.

The cone attachment is the homotopy cofiber of the boundary inclusion, so

$$\widetilde{H}_k(\mathcal{C};\mathbb{Z}) \cong H_k(\mathcal{M}, \partial\mathcal{M};\mathbb{Z}).$$

The universal coefficient theorem gives $H^0(\mathcal{M}) = H^1(\mathcal{M}) = \mathbb{Z}$ and $H^2(\mathcal{M}) = H^3(\mathcal{M}) = 0$. Poincaré–Lefschetz duality now gives equation (3.14). □

Theorem 3.4 (Exact cone-compactified S^{24} route)**.** *There is a continuous surjective based quotient map*

$$q_1 : (X, x_0) \longrightarrow (\mathcal{C}, c_0) \tag{3.15}$$

whose restrictions are:

(i) $(f, \theta_\gamma) : f^{-1}(P) \to P \times S^1$, *equivalently* $\chi_\gamma|_{f^{-1}(P)}$;

(ii) *the finite-filling map* $\overline{\chi}_j : N_j \to C_j$ *over* D_j, $j = 1, 2$;

(iii) *on* $N_0 \setminus W$, *with* $t(f(x)) = re^{i\phi}$,

$$x \longmapsto [r, (e^{i\phi}, \theta_\gamma(x))] \in \mathrm{Cone}(\partial D_0 \times S^1), \tag{3.16}$$

while all of W *maps to* c_0.

Its fourth smash power is

$$q_1^{\wedge 4} : X^{\wedge 4} \longrightarrow \mathcal{C}^{\wedge 4}, \qquad X^{\wedge 4} \cong S^{24}. \tag{3.17}$$

On four synchronized radial fibres define the target-product embedding

$$e^{(4)}_{\mathcal{C},T}(c_1, \dots, c_4) := \bigwedge_{i=1}^{4} [r_T, (1, \vartheta_T^{-1}(c_i))] : C_T^4 \hookrightarrow \mathcal{C}^{\wedge 4}.$$

Then the radial restriction factors exactly as

$$q_1^{\wedge 4} \circ e^{(4)}_{X,T} = e^{(4)}_{\mathcal{C},T} \circ \chi_{\gamma,T}^{\times 4}. \tag{3.18}$$

The two e-maps are embeddings; the middle product quotient has the nontrivial connected kernel T^{12} *already identified above.*

Equip $C_T^4 = \mathbb{R}^4/\ell_T\mathbb{Z}^4$ with the Euclidean ℓ^2-product quotient distance

$$d_{C_T^4}([x],[y]) = \min_{n\in\mathbb{Z}^4} \|x - y + \ell_T n\|_2. \tag{3.19}$$

This intrinsic slice metric is assigned independently of any ambient metric on $\mathcal{C}^{\wedge 4}$. With this metric,

$$\mathrm{sys}(C_T^4) = \ell_T = T + H - B + O(T^{-1}), \qquad (C_T^4, 0) \longrightarrow (\mathbb{R}^4, 0) \tag{3.20}$$

in pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff topology. Once $\ell_T > 4R$, the universal quotient maps the Euclidean radius-R ball isometrically onto the corresponding metric ball in C_T^4.

Finally,

$$\widetilde{H}_k(\mathcal{C}^{\wedge 4};\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 8, 12, \\ \mathbb{Z}^4, & k = 9, 11, \\ \mathbb{Z}^6, & k = 10, \\ 0, & \textit{otherwise.} \end{cases} \tag{3.21}$$

Proof. The first two formulas agree on their punctured overlaps by equation (1.41). On ∂D_0, the base of the cone is glued to the remaining boundary of $P \times S^1$, so the first and third formulas agree there. If U is a neighborhood of c_0, its inverse image in the cone contains $[0, \delta) \times T^2$ for some $\delta > 0$, because the cone base is compact. Thus $q_1^{-1}(U)$ contains $f^{-1}(|t| < \delta)$, a neighborhood of W. Continuity elsewhere comes from the input maps; the closed pasting lemma proves continuity on X.

The circle quotients and both finite Seifert maps are surjective, and W maps onto the cone vertex. Hence q_1 is a continuous surjection from a compact space to the Hausdorff space $\mathcal{C}$ and is a quotient map. It is based, so its fourth smash power exists. Iterating $S^m \wedge S^n \cong S^{m+n}$ and using equation (1.2) identifies the source with S^{24}.

The base point belongs to W, whereas each radial regular fibre is disjoint from W. The source and target product slices therefore avoid the respective closed fat wedges. The smash projections restrict to embeddings on these compact slices, exactly as in theorem 2.1, and functoriality gives equation (3.18). The product target lattice is $\ell_T\mathbb{Z}^4$. Applying theorem 1.2 to equation (3.19) proves the exact onto-ball assertion and hence equation (3.20).

The homology in equation (3.14) is free. In the fourth smash power, choose j factors in degree 3 and $4 - j$ factors in degree 2. The result has degree $8 + j$ and rank $\binom{4}{j}$, which is exactly equation (3.21). □

4 Optimality and genuine sphere carriers

The constructions above land in conical smash targets. We now prove two stronger statements. First, within the class of connected fibrewise group quotients defined by fixed primitive sub-local systems and carrying the unscaled Euclidean quotient norm, the numbers of copies, and hence the sphere exponents 12 and 24, are forced by the ranks of the expanding native quotients. Second, canonical further quotients of the conical targets produce honest sphere codomains containing moving metric four-spheres whose pointed limits are exactly $\mathbb{R}^4$.

Theorem 4.1 (Optimality within native connected group quotients)**.** *For each $r \geq 1$, let $N \subset \Lambda^{\oplus r}$ be a fixed primitive sublattice which is invariant under the monodromy group on the base under consideration, so that it defines a connected fibrewise torus-group quotient of the native r-fold product. Equip that quotient with the Riemannian quotient norm induced by the unscaled coordinate Euclidean metric on the marked original fibres.*

(i) *For a connected torus quotient of r copies over the punctured cusp, divergence of the radial quotient systole forces the quotient lattice to have rank at most $2r$. A rank-four quotient therefore requires $r \geq 2$. Equality is attained by*

$$(\Lambda/K_\Lambda)^{\oplus 2}, \qquad L_T^{(12)} = \mathrm{diag}(Y_T, Y_T)\mathbb{Z}^4, \tag{4.1}$$

and the corresponding smash source is $X^{\wedge 2} \cong S^{12}$.

(ii) *For a connected torus quotient of r copies over the full regular base, or over any finite cover of that base, divergence of the radial quotient systole forces the quotient lattice to have rank at most r. A rank-four quotient therefore requires $r \geq 4$. Equality is attained by four copies of the full-monodromy-invariant character χ_γ, whose radial period lattice is*

$$L_T^{(24)} = \ell_T \mathbb{Z}^4, \qquad \ell_T = T + H - B + O(T^{-1}), \tag{4.2}$$

and the corresponding smash source is $X^{\wedge 4} \cong S^{24}$.

The asserted minimality is confined to this class. It does not exclude nonlinear or nongroup maps, disconnected kernels, parameter-dependent kernels or target rescalings, non-torus targets, or constructions which replace the native product family by a different source.

Proof. Fix the primitive kernel N from the statement. Every fixed element of

$$K_\Lambda^{\oplus r} = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle^{\oplus r}$$

has uniformly bounded physical norm on the radial cusp. If a vector v in that lattice had nonzero class in $\Lambda^{\oplus r}/N$, primitivity identifies that class with a nonzero period of the connected quotient lattice, whose quotient norm is at most the original norm and would therefore remain bounded. The quotient systole could not diverge. Hence

$$K_\Lambda^{\oplus r} \subset N$$

and

$$\mathrm{rank}(\Lambda^{\oplus r}/N) \leq 4r - 2r = 2r.$$

For $r = 2$, the primitive kernel $K_\Lambda^{\oplus 2}$ attains this bound. The cusp monodromy acts trivially modulo K_Λ, so the quotient descends over D_0^*. Its radial lattice is exactly equation (4.1), and

$$\mathrm{sys}(L_T^{(12)}) = \mathrm{sys}(Y_T \mathbb{Z}^2) \geq T - C \longrightarrow \infty.$$

For a quotient over the full regular base, monodromy invariance gives the stronger bound. The inclusion $K_\Lambda^{\oplus r} \subset N$, together with $A_1 \widehat{w} = \widehat{u} - \widehat{w}$, implies

$$L_\Lambda^{\oplus r} = \mathbb{Z}\langle \widehat{u}, \widehat{w}, \widehat{\delta} \rangle^{\oplus r} \subset N.$$

Consequently $\mathrm{rank}(\Lambda^{\oplus r}/N) \leq 4r - 3r = r$.

The conclusion survives every finite cover of B°. Put

$$G_* = A_2 A_1 = I + N_*, \qquad N_*^2 = 0.$$

Every finite-index subgroup of $\pi_1(B^\circ)$ contains a positive power of the loop with monodromy G_*. Direct multiplication gives

$$G_*^m \widehat{w} = \widehat{w} + m(\widehat{u} - \widehat{\delta}).$$

If N is invariant under the finite-index monodromy group, the already proved inclusion of $\widehat{w}, \widehat{\delta}$ gives $m\widehat{u} \in N$. Since N is primitive, $\widehat{u} \in N$, proving the same rank bound.

The quotient by L_Λ is precisely the rank-one character χ_γ. The constructions of theorems 3.1 and 3.4 show that four copies attain rank four, including the two finite Seifert fillings and the coned cusp. Finally, the based diffeomorphism equation (1.2) gives

$$X^{\wedge r} \cong (S^6)^{\wedge r} \cong S^{6r},$$

which proves the asserted sphere dimensions. □

Theorem 4.2 (Sphere-to-sphere carriers with moving four-spheres)**.** *There are continuous surjective based quotient maps*

$$\widehat{q}_3 : (X, x_0) \longrightarrow (S^3, *), \qquad \widehat{q}_\gamma : (X, x_0) \longrightarrow (S^2, *),$$

whose smash powers give continuous surjective quotient maps

$$\widehat{q}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6, \qquad \widehat{q}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8. \tag{4.3}$$

For every sufficiently large T, the first target contains a based embedded metric four-sphere

$$\mathbb{S}^4_{12,T} = S^2_T \wedge S^2_T \hookrightarrow S^6,$$

and the second contains a based embedded metric four-sphere

$$\mathbb{S}^4_{24,T} = C_T^{\wedge 4} \hookrightarrow S^8.$$

Here the embedded subspaces carry the T-dependent quotient length metrics specified below; no ambient metric on S^6 or S^8 is being silently imposed. On synchronized radial fibres, the exact factorizations are

$$F^2_T \xrightarrow{\pi_T \times \pi_T} Q^2_T \xrightarrow{\kappa_{12,T}} S^2_T \wedge S^2_T = \mathbb{S}^4_{12,T} \hookrightarrow S^6, \tag{4.4}$$

$$F^4_T \xrightarrow{\chi_{\gamma,T}^{\times 4}} C^4_T \xrightarrow{\kappa_{24,T}} C_T^{\wedge 4} = \mathbb{S}^4_{24,T} \hookrightarrow S^8, \tag{4.5}$$

Here

$$\kappa_{12,T}(y_1, y_2) = [c(\iota_T^{-1}(y_1)), c(\iota_T^{-1}(y_2))]$$

first collapses $K^{(1)}$ in each marked two-torus and then collapses the resulting product fat wedge, whereas $\kappa_{24,T} : C^4_T \to C_T^{\wedge 4}$ is the smash projection and collapses its fat wedge. The final arrows are topological embeddings. Neither radial map from the source product is an embedding: already its first arrow has connected kernel T^4 in the S^{12} row and T^{12} in the S^{24} row, and the following collapse has further nontrivial fibres. With the quotient length metrics defined in the proof and the specified midpoint basepoints,

$$(\mathbb{S}^4_{12,T}, \zeta_{12,T}) \longrightarrow (\mathbb{R}^4, 0), \qquad (\mathbb{S}^4_{24,T}, \zeta_{24,T}) \longrightarrow (\mathbb{R}^4, 0) \tag{4.6}$$

in pointed local-isometric and pointed Gromov–Hausdorff topology. For every $R > 0$, each radius-R ball is exactly Euclidean once $T - C > 4R$, respectively once $\ell_T > 4R$. The quotient metrics on these compact four-spheres are singular at their smash points, so no global pointed-smooth topology is invoked; the smooth assertion is exactly C^∞ convergence on the displayed, eventually Euclidean regular neighborhoods, which exhaust every fixed radius about the midpoint basepoints.

Proof. For a based compact space (Z, z_0), write

$$\Sigma_r(Z, z_0) = \frac{[0, \epsilon] \times Z}{(\{0, \epsilon\} \times Z) \cup ([0, \epsilon] \times \{z_0\})}$$

for its reduced suspension. Use the marked presentation $\overline{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$, put

$$K^{(1)} = ((\mathbb{R}/\mathbb{Z}) \times \{0\}) \cup (\{0\} \times (\mathbb{R}/\mathbb{Z})),$$

and let

$$c : \overline{T}^2 \longrightarrow \overline{T}^2/K^{(1)} \cong S^2$$

be the degree-one quotient which collapses the one-skeleton. The formula

$$\widehat{p}_2 : \mathcal{A} \longrightarrow \Sigma_r(\overline{T}^2/K^{(1)}, c(K^{(1)})) \cong S^3, \qquad [r, e^{i\phi}, u] \longmapsto [r, c(u)]$$

is well defined. Both suspension ends of $\mathcal{A}$, as well as the entire cylinder over $K^{(1)}$, map to the reduced-suspension basepoint. It is continuous and surjective. Define $\widehat{q}_3 = \widehat{p}_2 \circ q_2$.

Let $y_* = [(1/2, 1/2)] \in \overline{T}^2$. Pull the Euclidean quotient metric of Q_T back along ι_T; explicitly,

$$\bar{d}_T([a], [b]) = \min_{n \in \mathbb{Z}^2} \|Y_T(a - b + n)\|_2. \tag{4.7}$$

Give

$$S^2_T = \overline{T}^2/K^{(1)}$$

the quotient length metric of equation (1.46). From equation (1.25), for every $v \in \mathbb{R}^2$,

$$\|Y_T v\| \geq (T - C)\|v\|.$$

The standard flat distance from y_* to $K^{(1)}$ is $1/2$; therefore

$$d_T := \operatorname{dist}(y_*, K^{(1)}) \geq \frac{T - C}{2} \tag{4.8}$$

in the pulled-back metric. Translation of theorem 1.2 shows that the radius-R ball about y_* in $(\overline{T}^2, \bar{d}_T)$ is exactly Euclidean when $T - C > 4R$. Under the same inequality, $d_T > 2R$, so the collapsed-set part of that lemma maps this ball isometrically *onto* the radius- R ball about $z_T = c(y_*)$ in S^2_T. Thus that quotient ball is exactly a Euclidean two-ball.

At any fixed interior suspension coordinate, S^2_T is a based embedded subspace of S^3: the point $c(K^{(1)})$ is the global reduced-suspension basepoint. Consequently

$$\mathbb{S}^4_{12,T} = S^2_T \wedge S^2_T$$

embeds in $S^3 \wedge S^3 = S^6$. Equip $(S^2_T)^2$ with the Euclidean ℓ^2-product length metric

$$d_{2,T}((u_1, u_2), (v_1, v_2)) = \left(d_{S^2_T}(u_1, v_1)^2 + d_{S^2_T}(u_2, v_2)^2\right)^{1/2},$$

and give the smash its quotient length metric after collapsing

$$(S^2_T \times \{c(K^{(1)})\}) \cup (\{c(K^{(1)})\} \times S^2_T).$$

The point $\zeta_{12,T} = [z_T, z_T]$ has distance d_T from this fat wedge. Before the collapse, its product radius-R ball is the Euclidean four-ball. The collapsed-set part of theorem 1.2 maps that ball isometrically onto the radius-R ball in the smash when $T - C > 4R$. On a radial fibre,

$$\widehat{p}_2 \circ q_2 = [r, c(\iota_T^{-1}(\pi_T(\,\cdot\,)))],$$

which proves equation (4.4).

For the four-copy construction, choose $1 \in S^1$ as character basepoint and define

$$\widehat{p}_1 : \mathcal{B} \longrightarrow \Sigma_r(S^1, 1) \cong S^2, \qquad [r, e^{i\phi}, \theta] \longmapsto [r, \theta].$$

This formula forgets the base angle and collapses both suspension ends and the entire cylinder with $\theta = 1$ to the reduced-suspension basepoint. It is continuous and surjective. Put $\widehat{q}_\gamma = \widehat{p}_1 \circ q_{1,\mathrm{cusp}}$.

At an interior suspension coordinate the character circle is a based embedded $S^1 \subset S^2$, and therefore

$$\mathbb{S}^4_{24,T} = C_T^{\wedge 4} \subset (S^2)^{\wedge 4} = S^8.$$

Here the normalized circle $\mathbb{R}/\mathbb{Z}$ is identified with the arclength circle $C_T = \mathbb{R}/\ell_T\mathbb{Z}$ by $[\theta] \mapsto [\ell_T\theta]$. Equip C_T^4 with the Euclidean ℓ^2-product quotient metric equation (3.19), and give the smash the quotient length metric obtained by collapsing the fat wedge on which at least one coordinate is 0 modulo ℓ_T. The midpoint

$$\zeta_{24,T} = [(\ell_T/2, \ell_T/2, \ell_T/2, \ell_T/2)]$$

has distance $\ell_T/2$ from that fat wedge. If $\ell_T > 4R$, theorem 1.2 first identifies the product radius-R ball with the Euclidean four-ball and then maps it isometrically *onto* the radius-R ball in the smash quotient. The radial formula is equation (4.5).

Both one-copy maps are continuous surjections from compact X to Hausdorff spheres, hence quotient maps. Their smash powers are surjective; compact-to-Hausdorff again makes them quotient maps. Using equation (1.2),

$$X^{\wedge 2} \cong S^{12}, \quad (S^3)^{\wedge 2} \cong S^6, \qquad X^{\wedge 4} \cong S^{24}, \quad (S^2)^{\wedge 4} \cong S^8.$$

This proves equation (4.3). The exact onto-ball statements prove the pointed local-isometric and Gromov–Hausdorff limits in equation (4.6), as well as C^∞ convergence on the eventually Euclidean regular neighborhoods specified in the theorem. □

Corollary 4.3 (The full external smash pairing is zero)**.** *For every pair of based maps $x, y : S^6 \to S^3$,*

$$[x \wedge y] = 0 \qquad \text{in} \qquad \pi_{12}(S^6).$$

Consequently every repeated smash $g^{\wedge r} : S^{6r} \to S^{3r}$ is based null-homotopic for $r \geq 2$. In particular, the continuous surjective quotient $\widehat{q}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6$ in equation (4.3) is based null-homotopic.

Proof. Bifunctoriality of the based smash product supplies the exact factorization

$$S^6 \wedge S^6 \xrightarrow{\mathrm{id}_{S^6} \wedge y} S^6 \wedge S^3 \xrightarrow{x \wedge \mathrm{id}_{S^3}} S^3 \wedge S^3.$$

Under the displayed ordered sphere homeomorphisms, Toda's Barratt–Hilton formula identifies this composite exactly as

$$[x \wedge y] = (E^3[x]) \circ (E^6[y]) : S^{12} \longrightarrow S^9 \longrightarrow S^6.$$

Indeed, the sign in Toda's first formula is $(-1)^{(p+k)h} = (-1)^{18} = +1$ when $p = q = k = h = 3$ [Toda, 1962, Proposition 3.1]. Swapping the two smash inputs, rather than reversing an ill-typed composition, introduces the source–target twist sign $(-1)^{ph} = -1$. Thus the same ordered sphere conventions also give

$$[x \wedge y] = -(E^3[y]) \circ (E^6[x]).$$

No swap is used in the proof below.

The three exact groups needed are

$$\pi_6(S^3) \cong \mathbb{Z}/12, \qquad \pi_{12}(S^9) \cong \mathbb{Z}/24, \qquad \pi_{12}(S^6) \cong \mathbb{Z}/2,$$

[Hatcher, 2002, opening table, p. 339]; [Isaksen et al., 2023, Table 1, p. 113]. Put $\beta = E^6[y] \in \pi_{12}(S^9)$. Since $12[y] = 0$, functoriality gives $12\beta = 0$. In the cyclic group $\mathbb{Z}/24$, every element annihilated by 12 is even; hence $\beta = 2\gamma$ for some $\gamma \in \pi_{12}(S^9)$. Postcomposition with $E^3x : S^9 \to S^6$ is a group homomorphism, and its target has exponent two. Therefore

$$[x \wedge y] = (E^3[x])_*(2\gamma) = 2(E^3[x])_*(\gamma) = 0.$$

This generator-free argument deliberately avoids a common notation trap. In Toda's Chapter V, ν' denotes the order-four 2-primary component, not an integral generator of $\pi_6(S^3) \cong \mathbb{Z}/12$. If $\alpha_1(3)$ denotes the order-three component, an integral generator may instead be written

$$\omega = \nu' + \alpha_1(3),$$

as recorded explicitly by Maruyama–Oshima, p. 769. After choosing the stable generator ν_9 compatibly, the corresponding integral formula is $[E^6(a\omega)] = 2a\nu_9$, not the false assertion that Toda's order-four ν' alone generates the integral group. Taking $x = y = g$ proves $g^{\wedge 2} \simeq_* 0$. For $r \geq 2$, associativity writes $g^{\wedge r} = g^{\wedge 2} \wedge g^{\wedge(r-2)}$, and smashing the based null-homotopy with the remaining factors proves the final assertion. □

Remark 4.4 (The one-copy invariant and its computed value). The preceding vanishing theorem does not determine a one-copy class $[g] \in \pi_6(S^3) \cong \mathbb{Z}/12$. Stabilization identifies this group with the even subgroup of the stable three-stem. An integral Serre-spectral-sequence calculation gives the same injectivity and identifies twice the stable Hopf–ν generator with a generator of the suspended $\mathbb{Z}/12$-summand [Hatcher, n.d., Theorem 5.41(b), p. 573, and pp. 577–578]. Thus, after choosing a stable generator ν_9, there is a unique $a \in \mathbb{Z}/12$ such that

$$[E^6 g] = 2a\nu_9 \in \pi_{12}(S^9) \cong \mathbb{Z}/24. \tag{4.9}$$

Because $a \mapsto 2a$ embeds $\mathbb{Z}/12$ as the even subgroup of $\mathbb{Z}/24$, $E^6 g$ is null-homotopic if and only if g is null-homotopic. The one-copy class and its sixfold suspension therefore encode the same integer modulo 12, now computed as zero in Theorem 10.1.

For the specified quotient $g = \widehat{q}_3$, that integer has an exact geometric representative. Choose $0 < r_0 < \epsilon$ and $u_0 \in \overline{T}^2 \setminus K^{(1)}$, and let $y_0 = [r_0, c(u_0)]$ lie in the open three-cell of $\Sigma_r S^2$. The literal formula for q_2 gives

$$g^{-1}(y_0) = \{x \in f^{-1}(|t| = r_0) : \overline{\pi}_f(x) = u_0\}. \tag{4.10}$$

The fibre of $\overline{\pi}$ is the two-torus defined by $K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle$, and the cusp monodromy fixes both generators. Hence equation (4.10) is the mapping torus of an affine translation of the kernel torus, which is diffeomorphic to $S^1 \times T^2 \cong T^3$: if the translation is $k \mapsto k + v$, choose a real lift $\widetilde{v}$; the map $[\theta, k] \mapsto (e^{2\pi i\theta}, k + \theta\widetilde{v})$ respects $(1, k) \sim (0, k + v)$ and has the inverse obtained by subtracting $\theta\widetilde{v}$. After a smooth chart is chosen in the open target cell, the differential of g supplies its normal three-framing. Pontryagin–Thom identifies the stabilized class of g with the framed bordism class of this specifically framed T^3. Since stabilization on $\pi_6(S^3)$ is injective, computing that framing class computes a exactly. The unframed diffeomorphism type T^3, point-set surjectivity, and the zero ordinary homology map do not compute a.

Theorem 4.5 (Exact framed-fibre reduction for the one-copy map)**.** *Let $g = \widehat{q}_3 : S^6 \to S^3$, smooth it without changing it near the regular inverse image in equation (4.10), and orient source and target. For an oriented regular value y_0, put $M = g^{-1}(y_0) \cong T^3$. Then the following assertions hold.*

(i) *The derivative supplies an honest oriented normal framing φ_g of $M \subset S^6$, and its Pontryagin–Thom class is $[g] \in \pi_6(S^3)$. After adjoining the outward normal of $S^6 \subset \mathbb{R}^7$, the resulting stable normal framing represents*

$$\mathrm{PT}(M, \varphi_g^{\mathrm{st}}) = E^\infty[g] = 2a\nu \in \pi_3^{\mathrm{S}} \cong \mathbb{Z}/24, \tag{4.11}$$

where $a \in \mathbb{Z}/12$ is the same integer as in equation (4.9) and the stable generator ν is chosen compatibly with ν_9.

(ii) *For every closed connected oriented three-manifold N, the set $\mathrm{Fr}^{\mathrm{st}}_{\infty}(N)$ of direct-limit stable framing classes maps onto the spin structures. The latter form an affine $H^1(N;\mathbb{Z}/2)$-space, and, after a spin structure is fixed, its inverse image is an affine $H^3(N;\mathbb{Z})$-space. Equivalently, after one stable framing is chosen, differences lie in $[N, SO]$, whose three-dimensional obstruction filtration has precisely the $H^1(N;\mathbb{Z}/2)$ and $H^3(N;\mathbb{Z})$ stages. This splitting is not canonical.*

(iii) *Fix the spin component of φ^{st}_g. Changing a direct-limit stable framing by a local generator $\sigma \in \pi_3(SO) \cong \mathbb{Z}$ changes its Pontryagin–Thom class by $J(\sigma) = \nu$. Changing the honest rank-three normal framing in a small ball by a generator $\rho \in \pi_3(SO(3)) \cong \mathbb{Z}$ changes the stable difference by 2σ, and hence changes equation (4.11) by 2ν. Consequently the honest-framing $H^3(T^3;\mathbb{Z})$-coordinate, reduced modulo 12, can realize every possible change of the class a; neither the diffeomorphism $M \cong T^3$ nor its spin component determines a.*

Here direct-limit stable *means stabilization through $SO = \varinjlim SO(n)$. It must not be confused with a framing of the particular rank-four bundle $\varepsilon^1 \oplus TN$: the latter has difference torsor $[N, SO(4)]$, and, after its spin component is fixed, an affine $H^3(N;\pi_3(SO(4)))$-fibre. Since $\pi_3(SO(4)) \cong \mathbb{Z} \oplus \mathbb{Z}$, this fibre is $H^3(N;\mathbb{Z}\oplus\mathbb{Z}) \cong \mathbb{Z}\oplus\mathbb{Z}$ for connected closed oriented N, before passage to the direct limit.*

Proof. The literal quotient formula is smooth and a submersion on a neighborhood of the displayed regular fibre. Choose a smaller such neighborhood U. Its complement is compact and its image omits y_0, so a relative smooth approximation, fixed on U and sufficiently close on the complement, still omits y_0 there. It therefore has exactly the same regular inverse image and represents the same homotopy class.

Choose an oriented orthonormal basis (e_1, e_2, e_3) of $T_{y_0}S^3$. At $x \in M$, regularity makes

$$dg_x : \nu_x(M \subset S^6) \longrightarrow T_{y_0}S^3$$

is an isomorphism. Orient the normal bundle by pulling back the target orientation through dg, and then give M the unique orientation for which $TM \oplus \nu_{M/S^6} = TS^6|_M$ as oriented bundles. The three vector fields $(dg|_\nu)^{-1}e_i$ are therefore the asserted honest normal framing. The tubular-neighborhood collapse followed by this framed identification with $M \times \mathbb{R}^3$ is the usual Pontryagin–Thom representative of the regular-value map g; choosing the displayed orientation convention fixes its sign. Stabilizing $S^6 \subset \mathbb{R}^7$ sends this representative to $E^\infty[g]$. The injective stabilization $\pi_6(S^3) = \mathbb{Z}/12 \hookrightarrow \pi^{\mathrm{S}}_3 = \mathbb{Z}/24$ has image the even subgroup, as recalled in theorem 4.4; this proves (i).

After an ambient stable trivialization is fixed, a stable tangent framing determines a stable normal framing, and conversely, by taking the complementary summand; the two difference maps are inverse in SO. Thus the following obstruction calculation applies equally to the stable normal framing in (i), with only the already fixed sign convention changed. First note that every spin structure is realized by a stable framing. Indeed, if v_2 is the second Wu class, then for every $x \in H^1(N;\mathbb{Z}/2)$,

$$\langle v_2 x, [N]\rangle = \langle Sq^2 x, [N]\rangle = 0$$

by the instability axiom. Poincaré duality gives $v_2 = 0$, and $w_2(TN) = v_2 + w_1(TN)^2 = 0$ because N is oriented. For any chosen spin lift, its stable classifying map lands in $BSpin$, which is three-connected; obstruction theory on the three-complex N makes that lift null-homotopic, and a chosen null-homotopy is a compatible stable framing.

Now fix one direct-limit stable framing. Every other one is obtained by a map $N \to SO$, so the difference set is a torsor for $[N, SO]$. On a three-dimensional CW structure the only relevant homotopy groups are

$$\pi_1(SO) = \mathbb{Z}/2, \qquad \pi_2(SO) = 0, \qquad \pi_3(SO) = \mathbb{Z}.$$

The first obstruction changes the spin lift and lies in $H^1(N;\mathbb{Z}/2)$. Once that lift is fixed, there is no degree-two obstruction and the final difference lies in $H^3(N;\mathbb{Z})$. The fundamental group of a topological group acts trivially on its higher homotopy groups, so the last coefficient system is untwisted. This is the explicit obstruction tower stated by Matthey et al. [2010, Example 1.1, p. 245]; for honest three-framings, the fixed-spin affine $H^3(N;\pi_3 SO(3))$-description and the noncanonical union over $H^1(N;\mathbb{Z}/2)$ are given by Kirby and Melvin [1999, pp. 92–93].

For (iii), support the change of framing in an oriented embedded three-ball and collapse its boundary. The difference is then represented by a map $S^3 \to SO$. Pontryagin–Thom additivity identifies the changed framed manifold with the connected sum with the correspondingly framed S^3, so the change of class is exactly the stable J-image; this is Levine [1985, Lemma 5.1 and Corollary 5.2, pp. 89–90], and in direct-limit stable-parallelization language it is Matthey et al. [2010, Lemma 1.10, p. 247]. Adams's image-of-J theorem makes $J : \pi_3(SO) = \mathbb{Z} \twoheadrightarrow \pi_3^{\mathrm{S}} = \mathbb{Z}/24$ surjective [Adams, 1966, Theorem 1.5, p. 22]; choose signs so that $J(\sigma) = \nu$. Kirby and Melvin's explicit quaternionic generators

$$\rho(q)x = qxq^{-1}, \qquad \sigma(q)x = qx$$

satisfy $\pi_3(SO(3)) \to \pi_3(SO)$, $\rho \mapsto 2\sigma$ [Kirby and Melvin, 1999, pp. 91–92]. Thus an honest local twist changes the stable Pontryagin–Thom class by 2ν. Since 2ν generates the even subgroup of order 12, its coefficient modulo 12 is exactly the missing datum a.

For calibration, give each factor of $T^3 = (S^1)^3$ either its bounding stable framing or its Lie framing. If a factor is bounding, the product is framed-null-bordant, for instance by replacing that factor with D^2. If all three are Lie framed, multiplicativity in the framed-bordism ring gives η^3, and the integral stable-stem relation is

$$[T^3, \varphi_{\mathrm{Lie}}] = \eta^3 = 12\nu \quad \text{in } \pi_3^{\mathrm{S}} = \mathbb{Z}/24. \tag{4.12}$$

This relation is recorded in Hatcher [2002, Proposition 4.56 and the discussion on p. 385]. It supplies useful reference framings in the eight spin components, but does not identify the derivative framing φ_g with any one of them. The relative H^3-coordinate still has to be calculated from the literal differential of g. □

Corollary 4.6 (Null-homotopy of every repeated circle-quotient smash)**.** *Let $f = \widehat{q}_\gamma : S^6 \to S^2$ be the based sphere map in equation (4.3). Then, for every integer $r \geq 2$,*

$$[f^{\wedge r}] = 0 \in \pi_{6r}(S^{2r}).$$

In particular,

$$[\widehat{q}_\gamma^{\wedge 2}] = 0 \in \pi_{12}(S^4), \qquad [\widehat{q}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8).$$

Thus the displayed continuous surjection $\widehat{q}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8$ is based null-homotopic.

Proof. Bifunctoriality and associativity of the based smash product give the exact factorization

$$S^6 \wedge S^6 \xrightarrow{f \wedge \mathrm{id}_{S^6}} S^2 \wedge S^6 \xrightarrow{\mathrm{id}_{S^2} \wedge f} S^2 \wedge S^2.$$

Under the standard based homeomorphisms $S^a \wedge S^b \cong S^{a+b}$, the first arrow represents $\Sigma^6 f : S^{12} \to S^8$; with the opposite suspension convention it is conjugate to that map by the relevant twist homeomorphisms. Freudenthal's suspension theorem is in its isomorphism range because $12 \leq 2 \cdot 8 - 2$, and it remains in that range after every further suspension. Consequently

$$\pi_{12}(S^8) \cong \pi_4^{\mathrm{S}}.$$

Here the stable-range identification follows from Freudenthal [Hatcher, 2002, Corollary 4.24, p. 360], while the fourth stable stem is zero [Isaksen et al., 2023, Table 1, p. 113]. Hence $f \wedge \mathrm{id}_{S^6}$ is based null-homotopic, so its composite $f \wedge f$ is based null-homotopic.

For $r \geq 2$, associativity identifies

$$f^{\wedge r} = (f \wedge f) \wedge f^{\wedge(r-2)},$$

where $f^{\wedge 0} = \mathrm{id}_{S^0}$. Smashing a based null-homotopy with any based map gives a based null-homotopy, proving the general assertion. Null-homotopy does not contradict point-set surjectivity or the exact radial metric restrictions: those are properties of the specified representative, whereas its homotopy class forgets them. □

Corollary 4.7 (Smash, join, and Cartesian product are distinct tests)**.** *Let $g : S^6 \to S^3$ and $f : S^6 \to S^2$ be based maps. For every $r \geq 2$, the based reduced joins of the specified quotient maps satisfy*

$$\underbrace{g * \cdots * g}_{r} : S^{7r-1} \longrightarrow S^{4r-1}, \qquad g * \cdots * g \simeq_* 0,$$

$$\underbrace{f * \cdots * f}_{r} : S^{7r-1} \longrightarrow S^{3r-1}, \qquad f * \cdots * f \simeq_* 0,$$

provided $g = \widehat{q}_3$ in the first line and $f = \widehat{q}_\gamma$ in the second. In particular, the two-fold join of $\widehat{q}_3$ is a null map $S^{13} \to S^7$, and the four-fold join of $\widehat{q}_\gamma$ is a null map $S^{27} \to S^{11}$.

By contrast, for every $r \geq 1$ and every based map $h : A \to B$,

$$h^{\times r} : A^r \longrightarrow B^r \text{ is null-homotopic} \iff h \text{ is null-homotopic.} \tag{4.13}$$

Consequently the proven smash and join nullities do not determine the Cartesian product maps $\widehat{q}_3^{\times 2}$ or $\widehat{q}_\gamma^{\times 4}$: their nullity is exactly the one-copy question. Theorem 10.1 now proves all Cartesian powers of $\widehat{q}_3$ null; the one-copy question for $\widehat{q}_\gamma$ remains open.

Proof. The natural based equivalence between reduced join and suspended smash is

$$A * B \simeq_* \Sigma(A \wedge B).$$

Iterating it identifies the r-fold join map with $\Sigma^{r-1}(h^{\wedge r})$. The sphere dimensions are $r(6+1)-1 = 7r-1$, $r(3+1)-1 = 4r-1$, and $r(2+1)-1 = 3r-1$. The asserted nullities now follow from theorems 4.3 and 4.6, since suspension preserves a based null-homotopy.

If $h \simeq_* 0$, taking the Cartesian product of r based null-homotopies contracts $h^{\times r}$. Conversely, let $i_j : A \to A^r$ insert the basepoint in every factor except the j-th, and let $p_j : B^r \to B$ be projection. Then

$$p_j \circ h^{\times r} \circ i_j = h.$$

A null-homotopy of the product therefore contracts h, proving equation (4.13) without any cohomological or dimensional hypothesis. □

Corollary 4.8 (The literal open cells in the moving four-spheres)**.** *Put $K_T^{(1)} = \iota_T(K^{(1)}) \subset Q_T$, and denote by $0 \in C_T$ the character basepoint. Removing the smash point from each moving four-sphere gives*

$$\mathcal{U}_{12,T} := (Q_T \setminus K_T^{(1)})^2 \cong \mathbb{S}^4_{12,T} \setminus \{*\} \cong \mathbb{R}^4 \hookrightarrow S^6, \tag{4.14}$$

$$\mathcal{U}_{24,T} := (C_T \setminus \{0\})^4 \cong \mathbb{S}^4_{24,T} \setminus \{*\} \cong \mathbb{R}^4 \hookrightarrow S^8. \tag{4.15}$$

These are topologically embedded, nonclosed four-cells. They are open in the corresponding moving S^4, but not open in the ambient S^6 or S^8. Under any fixed compatible metric on the compact sphere target, the ambient restriction to either cell has finite diameter and therefore cannot be globally isometric to Euclidean $\mathbb{R}^4$. The exact metric statement is the exhaustion by the midpoint Euclidean balls of equation (4.6).

Proof. For any based Hausdorff spaces (Z_i, z_i), the smash projection restricts to a homeomorphism

$$\prod_i (Z_i \setminus \{z_i\}) \xrightarrow{\cong} \left(\bigwedge_i Z_i\right) \setminus \{*\},$$

because the complement of the fat wedge is saturated and the smash equivalence relation is equality there. Also

$$S^2_T \setminus \{c(K^{(1)})\} \cong Q_T \setminus K^{(1)}_T \cong (0,1)^2 \cong \mathbb{R}^2,$$

and

$$C_T \setminus \{0\} \cong (0, \ell_T) \cong \mathbb{R}.$$

Taking two and four products gives the two copies of $\mathbb{R}^4$. Restricting the established embeddings $\mathbb{S}^4_{12,T} \hookrightarrow S^6$ and $\mathbb{S}^4_{24,T} \hookrightarrow S^8$ gives the displayed embeddings. Each cell accumulates at the omitted smash point, so it is not closed. Its ambient nonopenness follows from invariance of dimension in local manifold charts, since $4 < 6$ and $4 < 8$. Every fixed compatible metric on the compact ambient sphere is bounded, and so is its restriction to the cell; Euclidean $\mathbb{R}^4$ is unbounded. □

Remark 4.9 (Null-homotopy of the moving four-spheres). For a fixed interior suspension coordinate, the inclusion $S^2_T \hookrightarrow S^3$ contracts through based maps by sliding that coordinate to a suspension endpoint. The collapsed one-skeleton remains at the global basepoint throughout. Hence $S^2_T \wedge S^2_T \hookrightarrow S^6$ is null-homotopic. The same sliding argument contracts $C_T \hookrightarrow S^2$, and therefore contracts $C_T^{\wedge 4} \hookrightarrow S^8$. Thus the moving metric four-spheres do not contradict the fixed-target orientation obstruction in theorem 6.2: their fundamental classes die in the ambient targets.

Remark 4.10 (Homotopical status of the two sphere maps). The maps in equation (4.3) induce zero on all positive-degree integral homology and cohomology groups and have zero ordinary Hurewicz image. They have no classical degree because source and target dimensions differ. The class of $\widehat{q}_\gamma^{\wedge 4} : S^{24} \to S^8$ is zero by theorem 4.6; the class of the distinct map $\widehat{q}_3^{\wedge 2} : S^{12} \to S^6$ is zero by theorem 4.3. These are theorems about the full specified sphere-valued quotient maps, not merely their Hurewicz images or their moving-S^4 inclusions. The one-copy class $[\widehat{q}_3] \in \pi_6(S^3)$ is zero by theorem 10.1, which computes the actual derivative framing. The one-copy class $[\widehat{q}_\gamma] \in \pi_6(S^2)$ remains open. The unstable based homotopy classes of the conical-target maps

$$q_2^{\wedge 2} : S^{12} \to \mathcal{A}^{\wedge 2}, \qquad q_{1,\mathrm{cusp}}^{\wedge 4} : S^{24} \to \mathcal{B}^{\wedge 4}, \qquad q_1^{\wedge 4} : S^{24} \to \mathcal{C}^{\wedge 4}$$

remain open. Nullity after the final one-skeleton or base-angle quotient does not imply nullity before that postcomposition.

4.1 Clutching classes and the exact nonextension statements

Theorem 4.11 (The $SU(3)$ clutching calculation and its scope)**.** *For each $k \in \mathbb{Z}$ there is a principal $SU(3)$-bundle $P_k \to S^4$ satisfying*

$$\langle c_2(P_k), [S^4] \rangle = k. \tag{4.16}$$

Here and below $c_2(P)$ means the second Chern class of the associated standard rank-three complex vector bundle. It may be placed on either moving four-sphere $\mathbb{S}^4_{12,T} \subset S^6$ or $\mathbb{S}^4_{24,T} \subset S^8$. With orientations chosen so that the corresponding quotient collapse

$$c_T : L_T \longrightarrow \mathbb{S}^4_{\nu,T}, \qquad L_T = Q^2_T \text{ or } C^4_T, \qquad \nu = 12 \text{ or } 24,$$

has degree $+1$, the pullback to the radial four-torus obeys

$$\langle c_2(c_T^* P_k), [L_T] \rangle = k. \tag{4.17}$$

For $k \neq 0$, these bundles have the nonextension properties (i) *and* (ii) *below. Independently, the full sphere maps satisfy* (iii)*.*

(i) *P_k on a moving S^4 is not the restriction of a principal $SU(3)$-bundle on the ambient S^6 or S^8.*

(ii) *$c_T^* P_k$ is not the restriction of a principal $SU(3)$-bundle on the fixed compact conical carrier $\mathcal{D}_\nu \cong \mathrm{Cone}(T^4)$.*

(iii) *If $F : S^N \to S^m$ is either full sphere map in equation (4.3) and $Q \to S^m$ is any principal $SU(3)$-bundle, then $F^*Q \to S^N$ is trivial. Thus neither P_k on its moving S^4 nor $c_T^* P_k$ on L_T is bundle data supplied by that pullback: those bundles live on different base spaces and are additional clutching choices.*

Moreover every principal $SU(3)$-bundle on an abstract S^8 is trivial, whereas the principal $SU(3)$-bundles on S^{12} form $\pi_{11}(SU(3)) \cong \mathbb{Z}/4$; those four source-side classes become trivial after stabilization to SU and are independent of the clutching integer in equation (4.16).

Proof. Write $S^4 = D^4_+ \cup_{S^3} D^4_-$, identify $S^3 \cong SU(2)$, and let $i : SU(2) \hookrightarrow SU(3)$ be $q \mapsto \mathrm{diag}(q, 1)$. Choose a smooth based degree-k map $d_k : S^3 \to S^3$ and use $g_k = i \circ d_k$ as the equatorial transition map. Since $SU(3)/SU(2) \cong S^5$, the long exact homotopy sequence and $\pi_4(S^5) = \pi_3(S^5) = 0$ show that $i_* : \pi_3(SU(2)) \to \pi_3(SU(3))$ is an isomorphism. Hence g_k represents $k \in \pi_3(SU(3)) \cong \mathbb{Z}$. The clutching construction gives P_k.

There is also a direct differential-form check of its integer. On the punctured hemisphere $D^4_- \setminus \{0\} \cong (0, 1] \times S^3$, let θ_k denote the radial pullback of $g_k^{-1} dg_k$. On D^4_+ take $A_+ = 0$. On the other hemisphere, choose a smooth radial cutoff χ, identically zero on a neighborhood of its center and one near its equator, and take $A_- = \chi\theta_k$, extended by zero across that neighborhood. On the equatorial collar the gauge-change formula is $A_- = g_k^{-1} A_+ g_k + g_k^{-1} dg_k$, so these are the local forms of a global connection. The Maurer–Cartan identity gives

$$F_- = d\chi \wedge \theta_k + (\chi^2 - \chi)\theta_k^2. \tag{4.18}$$

The square of the first term vanishes, the square of the second has four tangential legs on S^3 and vanishes, and the two cross terms agree. Since $2\int_0^1 (\chi^2 - \chi) d\chi = -1/3$, Chern–Weil and transgression give

$$\frac{1}{8\pi^2} \int_{S^4} \mathrm{tr}(F \wedge F) = -\frac{1}{24\pi^2} \int_{S^3} \mathrm{tr}\big((g_k^{-1} dg_k)^3\big) = k, \tag{4.19}$$

Here tr is the ordinary matrix trace in the defining three-dimensional representation; it is not rescaled. We choose the orientation and the sign of the generator so that the last integral is the degree k. Reversing both choices changes both displayed signs and not the classification. This proves equation (4.16).

For completeness, $BSU(3)$ is three-connected and

$$(\pi_4(BSU(3)), \pi_5(BSU(3))) = (\mathbb{Z}, 0).$$

Hence obstruction theory on any four-dimensional CW complex K identifies

$$[K, BSU(3)] \xrightarrow{c_2} H^4(K; \mathbb{Z}) \tag{4.20}$$

bijectionally. In the two-copy construction, $L_T = Q_T^2 \cong T^4$ and c_T collapses the product fat subcomplex; in the four-copy construction, $L_T = C_T^4 \cong T^4$ and it collapses the fat wedge. In either case the quotient is the displayed moving S^4, and its top cell is the product of the oriented top cells of the factors. Thus $(c_T)_*[L_T] = [S^4]$, and naturality of c_2 proves equation (4.17).

The moving-sphere inclusions are null-homotopic by theorem 4.9. If P_k extended over the ambient sphere, its classifying map restricted along that null inclusion would be null, making P_k trivial, contrary to

$k \neq 0$. Similarly, the radial slice inclusion $T^4 \hookrightarrow \mathrm{Cone}(T^4)$ contracts by sliding its radial coordinate to the cone vertex. An extending bundle would therefore have trivial restriction, contradicting equation (4.17). Finally, if $F : S^N \to S^m$ is either full sphere map and $b : S^m \to BSU(3)$ classifies a bundle, then the pullback is classified by $b \circ F$. The nullities in theorems 4.3 and 4.6 make this composite null. These three arguments prove (i)–(iii) and show explicitly that the clutching bundles are additional choices, not data descended through the quotient maps.

Clutching on S^n gives $[S^n, BSU(3)] = \pi_{n-1}(SU(3))$. The primary computations of Mimura and Toda give

$$\pi_7(SU(3)) = 0, \qquad \pi_{11}(SU(3)) = \mathbb{Z}/4 \tag{4.21}$$

[Mimura and Toda, 1964, opening table, p. 217]. This proves the S^8 and S^{12} assertions. Bott periodicity gives $\pi_{11}(SU) \cong \mathbb{Z}$; the stabilization homomorphism sends the finite group $\mathbb{Z}/4$ to this torsion-free group and is therefore zero. No principal bundle, connection, curvature form, or transport rule has been specified by the sphere quotient itself. □

Proposition 4.12 (Compact cone-on-T^4 carriers)**.** *Fix T_0 sufficiently large and put $r_0 = e^{-2\pi T_0} < \epsilon$. Define*

$$\mathcal{D}_{12} = \frac{[0, r_0] \times (\overline{T}^2)^2}{\{0\} \times (\overline{T}^2)^2 \sim d_{12}}, \qquad \mathcal{D}_{24} = \frac{[0, r_0] \times (S^1)^4}{\{0\} \times (S^1)^4 \sim d_{24}}.$$

Both spaces are homeomorphic to $\mathrm{Cone}(T^4)$*. There are topological embeddings*

$$\delta_{12} : \mathcal{D}_{12} \hookrightarrow \mathcal{A} \wedge \mathcal{A}, \qquad [r, u_1, u_2] \longmapsto [\,[r, 1, u_1], [r, 1, u_2]\,], \tag{4.22}$$

$$\delta_{24} : \mathcal{D}_{24} \hookrightarrow \mathcal{C}^{\wedge 4}, \qquad [r, \theta_1, \dots, \theta_4] \longmapsto \bigwedge_{j=1}^{4} [r, (1, \theta_j)]. \tag{4.23}$$

Here $1 \in S^1$ is the positive radial base angle. For $T \geq T_0$, the restrictions of the sphere quotient maps to the synchronized radial fibres are exactly

$$F_T^2 \xrightarrow{\pi_T \times \pi_T} Q_T^2 \cong \{e^{-2\pi T}\} \times T^4 \subset \mathcal{D}_{12}, \tag{4.24}$$

$$F_T^4 \xrightarrow{\chi_{\gamma,T}^{\times 4}} C_T^4 \cong \{e^{-2\pi T}\} \times T^4 \subset \mathcal{D}_{24}. \tag{4.25}$$

Every displayed four-torus slice is null-homotopic in its carrier. At the vertex d_ν, for $\nu \in \{12, 24\}$, the local integral homology is

$$H_k(\mathcal{D}_\nu, \mathcal{D}_\nu \setminus \{d_\nu\}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^4, & k = 2, \\ \mathbb{Z}^6, & k = 3, \\ \mathbb{Z}^4, & k = 4, \\ \mathbb{Z}, & k = 5, \\ 0, & \textit{otherwise.} \end{cases} \tag{4.26}$$

In particular, neither carrier is locally a four-manifold at its vertex.

Proof. The two displayed domains are cones on $(\overline{T}^2)^2 \cong T^4$ and $(S^1)^4 \cong T^4$. At $r = 0$, every factor on the right of equations (4.22) and (4.23) is the corresponding smash basepoint, so the formulas descend from the cylinders to the cones. For $0 < r \leq r_0$, none of the factors is a smash basepoint. Equality in the smash quotient is then equality in every factor, and the coordinate descriptions recover r and every torus coordinate. Each δ_ν is therefore injective. Its domain is compact and its target Hausdorff, so it is an embedding.

The restriction formulas are $q_2|_{F_T} = \iota_T^{-1} \circ \pi_T$ and $q_1|_{F_T} = \chi_{\gamma,T}$, applied in two and four factors. Sliding the cone coordinate from $e^{-2\pi T}$ to zero contracts each slice to the cone vertex. Finally, excision and the long exact sequence of the pair give, for $L = T^4$,

$$H_k(\mathrm{Cone}(L), \mathrm{Cone}(L) \setminus \{v\}; \mathbb{Z}) \cong \widetilde{H}_{k-1}(L; \mathbb{Z}).$$

The ranks $4, 6, 4, 1$ of H_1, H_2, H_3, H_4 of T^4 prove equation (4.26). □

Theorem 4.13 (Target homotopy types and point-set quotient maps)**.** *The smash maps are continuous surjective quotient maps*

$$q_2^{\wedge 2} : S^{12} \twoheadrightarrow \mathcal{A}^{\wedge 2}, \qquad q_1^{\wedge 4} : S^{24} \twoheadrightarrow \mathcal{C}^{\wedge 4}.$$

Their targets have homotopy types

$$\mathcal{A}^{\wedge 2} \simeq \bigvee^{9} S^4 \vee \bigvee^{18} S^5 \vee \bigvee^{15} S^6 \vee \bigvee^{6} S^7 \vee S^8, \tag{4.27}$$

$$\mathcal{C}^{\wedge 4} \simeq S^8 \vee \bigvee^{4} S^9 \vee \bigvee^{6} S^{10} \vee \bigvee^{4} S^{11} \vee S^{12}. \tag{4.28}$$

Both maps induce zero on every positive-degree integral homology and cohomology group and have zero ordinary Hurewicz image. Neither has a continuous homotopy section. These conclusions do not decide whether either unstable homotopy class is zero; both conical-target classes remain open. The same is true of the localized class $q_{1,\mathrm{cusp}}^{\wedge 4} : S^{24} \to \mathcal{B}^{\wedge 4}$.

Proof. If $g : (K, k_0) \to (L, \ell_0)$ is a surjective based map between compact Hausdorff spaces, every non-basepoint class of $L^{\wedge r}$ has a representative $(y_1, \dots, y_r)$ with all $y_i \neq \ell_0$. Choosing $x_i \in g^{-1}(y_i)$ proves that $g^{\wedge r}$ is surjective. Its source is compact, and its target is Hausdorff because the fat wedge is closed. Hence $g^{\wedge r}$ is a quotient map. Apply this to q_2 and q_1.

The standard suspension splitting

$$\Sigma(X_1 \times \cdots \times X_n) \simeq \bigvee_{\varnothing \neq I \subseteq \{1,\dots,n\}} \Sigma\left(\bigwedge_{i \in I} X_i\right)$$

gives

$$\mathcal{A} = \Sigma T^3 \simeq \bigvee^{3} S^2 \vee \bigvee^{3} S^3 \vee S^4.$$

Finite smash products distribute over finite wedges. Taking ordered degree sums gives multiplicities $9, 18, 15, 6, 1$ in dimensions $4, 5, 6, 7, 8$, proving equation (4.27).

For $\mathcal{C}$, write $x = \rho_1$, $y = \rho_2$, and $h = \eta$. The pair-of-pants product and the two Seifert fillings give

$$\pi_1(\mathcal{M}) = \langle x, y, h \mid [x, h] = [y, h] = 1,\ x^3 h^{-1} = 1,\ y^4 h = 1 \rangle.$$

The remaining boundary torus is generated by h and $\rho_0 = (xy)^{-1}$. Coning that torus kills their normal closure, so

$$\pi_1(\mathcal{C}) \cong \langle x, y \mid x^3 = 1,\ y^4 = 1,\ xy = 1 \rangle = 1,$$

because $y = x^{-1}$ and $\gcd(3, 4) = 1$. By theorem 3.3,

$$\widetilde{H}_2(\mathcal{C}; \mathbb{Z}) = \mathbb{Z}, \qquad \widetilde{H}_3(\mathcal{C}; \mathbb{Z}) = \mathbb{Z},$$

with no other reduced homology. Since $\mathcal{C}$ is a simply connected finite three-complex, Hurewicz supplies a sphere representing the H_2-generator, and the Whitehead exact sequence makes $\pi_3(\mathcal{C}) \to H_3(\mathcal{C})$ surjective. A wedge of representing maps $S^2 \vee S^3 \to \mathcal{C}$ is a homology equivalence between simply connected CW

complexes, hence a homotopy equivalence. Thus $\mathcal{C} \simeq S^2 \vee S^3$. Its fourth smash power gives equation (4.28) with binomial multiplicities $1, 4, 6, 4, 1$.

The first target has no reduced homology above degree eight and the second none above degree twelve. The source spheres have no positive homology below degrees twelve and twenty-four. All positive-degree homology and cohomology maps therefore vanish, as do the ordinary Hurewicz images. A homotopy section would split a nonzero lower homology group of the target into the zero lower homology of the source sphere, which is impossible. The displayed wedges can have nonzero higher homotopy groups, so this argument does not settle unstable null-homotopy. □

Theorem 4.14 (Exact decompactification spaces with boundary $\mathbb{R}^4$)**.** *For $\nu \in \{12, 24\}$, set*

$$A_T^{(12)} = \operatorname{diag}(Y_T, Y_T), \qquad A_T^{(24)} = \ell_T I_4, \qquad s_0 = T_0^{-1}.$$

On $[0, s_0] \times \mathbb{R}^4$, define a fibrewise equivalence relation $\sim_\nu$ by

$$(0, x) \sim_\nu (0, y) \iff x = y,$$

and, for $0 < s \le s_0$, by

$$(s, x) \sim_\nu (s, y) \iff x - y \in A_{1/s}^{(\nu)} \mathbb{Z}^4.$$

Let

$$\mathfrak{E}_\nu = ([0, s_0] \times \mathbb{R}^4)/\!\sim_\nu, \qquad p_\nu : \mathfrak{E}_\nu \longrightarrow [0, s_0].$$

Then $\mathfrak{E}_\nu$ is a Hausdorff, second-countable topological five-manifold with boundary, and

$$p_{12}^{-1}(s) \cong Q^2_{1/s} \quad (s > 0), \qquad p_{12}^{-1}(0) \cong \mathbb{R}^4, \tag{4.29}$$

$$p_{24}^{-1}(s) \cong C^4_{1/s} \quad (s > 0), \qquad p_{24}^{-1}(0) \cong \mathbb{R}^4. \tag{4.30}$$

The other boundary component is the compact four-torus over $s = s_0$.

Give each positive fibre the explicit Euclidean ℓ^2-quotient metric

$$d_{\nu,s}([x], [y]) = \min_{n \in \mathbb{Z}^4} \|x - y + A_{1/s}^{(\nu)} n\|_2 \qquad (s > 0), \tag{4.31}$$

and give the zero fibre its Euclidean metric. For every $R > 0$, there is $s_R > 0$ such that, for $0 \le s < s_R$, the universal quotient projection restricts to an exact isometry

$$\overline{B}_R(0) \xrightarrow{\cong} \overline{B}_R([0]) \subset p_\nu^{-1}(s)$$

onto the indicated metric ball. Thus the positive fibres converge to the zero fibre in pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff topology.

There are continuous surjective comparison maps

$$\mathbb{R}^4 = p_{12}^{-1}(0) \hookrightarrow \mathfrak{E}_{12} \xrightarrow{\beta_{12}} \mathcal{D}_{12} \xhookrightarrow{\delta_{12}} \mathcal{A}^{\wedge 2}$$

and

$$\mathbb{R}^4 = p_{24}^{-1}(0) \hookrightarrow \mathfrak{E}_{24} \xrightarrow{\beta_{24}} \mathcal{D}_{24} \xhookrightarrow{\delta_{24}} \mathcal{C}^{\wedge 4}.$$

For $s > 0$, β_ν identifies the fibre with the radial torus slice at $r = e^{-2\pi/s}$, and

$$\beta_\nu : \mathfrak{E}_\nu \setminus p_\nu^{-1}(0) \xrightarrow{\cong} \mathcal{D}_\nu \setminus \{d_\nu\} \tag{4.32}$$

is a homeomorphism. On $p_\nu^{-1}(0)$, it is constant with value d_ν, and $\beta_\nu^{-1}(d_\nu) = p_\nu^{-1}(0) \cong \mathbb{R}^4$. Nevertheless, β_ν is neither proper nor a quotient map. The induced continuous bijection

$$\overline{\beta}_\nu : \mathfrak{E}_\nu / p_\nu^{-1}(0) \longrightarrow \mathcal{D}_\nu \tag{4.33}$$

is not a homeomorphism. Thus $\mathbb{R}^4$ is an actual boundary fibre of $\mathfrak{E}_\nu$, but $\mathcal{D}_\nu$ is not the topological quotient obtained from $\mathfrak{E}_\nu$ through β_ν by collapsing that fibre; no topological blow-down or resolution is asserted.

Proof. Write

$$L_s = A^{(\nu)}_{1/s}\mathbb{Z}^4 \quad (s>0), \qquad L_0 = \{0\}.$$

The matrices vary continuously for $s>0$, and

$$\operatorname{sys}(L_s) \longrightarrow \infty \qquad (s \downarrow 0)$$

by equations (1.23) and (1.31). The equivalence relation is closed. Indeed, suppose $(s_j, x_j) \sim_\nu (s_j, y_j)$ and $(s_j, x_j, y_j) \to (s, x, y)$. If $s>0$, write $y_j - x_j = A^{(\nu)}_{1/s_j} n_j$. On a compact neighborhood of s, the inverse matrices are uniformly bounded, so boundedness of $y_j - x_j$ forces n_j into a finite set. After passage to a subsequence it is constant, and $x \sim_\nu y$. If $s=0$, every nonzero n_j would give

$$\|y_j - x_j\| \geq \operatorname{sys}(L_{s_j}) \longrightarrow \infty,$$

contrary to convergence. Hence $x=y$.

The quotient projection

$$q_\nu : [0, s_0] \times \mathbb{R}^4 \longrightarrow \mathfrak{E}_\nu$$

is open. If U is open, its saturation is U together with, for each $n \in \mathbb{Z}^4$, the image of $U \cap ((0, s_0] \times \mathbb{R}^4)$ under the homeomorphism

$$(s, x) \longmapsto (s, x + A^{(\nu)}_{1/s} n).$$

That saturation is open. A closed equivalence relation with an open quotient map has Hausdorff quotient: inequivalent points admit a product neighborhood disjoint from the relation, and the saturations of the two factor neighborhoods are disjoint. Since the source is second countable and the quotient is open, images of a countable base form a countable base of $\mathfrak{E}_\nu$.

Fix $x_0 \in \mathbb{R}^4$. Choose $a>0$ and $s_1>0$ such that $\operatorname{sys}(L_s) > 2a$ for $0<s<s_1$. The quotient projection is one-to-one on $[0, s_1) \times B_a(x_0)$, and its restriction is an open continuous bijection onto an open subset; hence it is a homeomorphism. These are half-space charts at $s=0$. Evenly covered small balls give product charts for $s>0$, and the corresponding one-sided charts at $s=s_0$ give the second boundary component. This proves the manifold and fibre claims.

For the metric assertion, choose s_R so that $\operatorname{sys}(L_s) > 4R$ for $s<s_R$. Formula equation (4.31) and theorem 1.2 show that the universal quotient maps $\overline{B}_R(0)$ isometrically onto the entire radius-R quotient ball. At $s=0$ the assertion is the identity. This proves all three pointed convergences.

For $s>0$, put $T = 1/s$ and $r = e^{-2\pi T}$. In the S^{12} case, write $x = (x_1, x_2) \in \mathbb{R}^2 \oplus \mathbb{R}^2$ and set

$$\beta_{12}([s, x]) = [r, \iota_T^{-1}([x_1]_{Y_T\mathbb{Z}^2}), \iota_T^{-1}([x_2]_{Y_T\mathbb{Z}^2})].$$

In the S^{24} case, write $x = (x_1, \dots, x_4)$ and set

$$\beta_{24}([s, x]) = [r, [x_1/\ell_T], \dots, [x_4/\ell_T]].$$

The formulas are well defined and continuous on positive fibres. Define $\beta_\nu([0, x]) = d_\nu$ for every $x \in \mathbb{R}^4$. Every neighborhood of d_ν contains all cone cross-sections with sufficiently small r, and $e^{-2\pi/s} \to 0$ as $s \downarrow 0$. Continuity at the zero fibre is therefore uniform in x. Each positive fibre maps onto its complete cross-section and the zero fibre onto the vertex, proving surjectivity.

The homeomorphism on the positive part is also explicit. The map

$$(0, s_0] \times \mathbb{R}^4 \longrightarrow (0, s_0] \times \mathbb{R}^4, \qquad (s, x) \longmapsto (s, (A^{(\nu)}_{1/s})^{-1} x)$$

is a homeomorphism and conjugates the varying lattice action to the constant translation action of $\mathbb{Z}^4$. It therefore induces

$$\Phi_\nu : \mathfrak{E}_\nu \setminus p_\nu^{-1}(0) \xrightarrow{\cong} (0, s_0] \times (\mathbb{R}^4/\mathbb{Z}^4).$$

The punctured cone has the corresponding product coordinates $\mathcal{D}_\nu \setminus \{d_\nu\} \cong (0, r_0] \times (\mathbb{R}^4/\mathbb{Z}^4)$. In these coordinates β_ν is

$$(s, [a]) \longmapsto (e^{-2\pi/s}, [a]).$$

Because $s \mapsto e^{-2\pi/s}$ is a homeomorphism from $(0, s_0]$ onto $(0, r_0]$, this proves equation (4.32). It also proves that the inverse image of d_ν is exactly the zero fibre. Since that fibre is $\mathbb{R}^4$, the compact singleton $\{d_\nu\}$ has noncompact preimage; hence β_ν is not proper.

It remains to prove that β_ν is not a quotient map. Fix

$$a = \tfrac{1}{2}e_1 = (\tfrac{1}{2}, 0, 0, 0) \in \mathbb{R}^4 \setminus \mathbb{Z}^4,$$

choose a strictly decreasing sequence $s_j \downarrow 0$, and set

$$z_j = [s_j, A^{(\nu)}_{1/s_j} a] \in \mathfrak{E}_\nu. \tag{4.34}$$

For every $n \in \mathbb{Z}^4$, the vector $e_1 + 2n$ is a nonzero lattice vector, so

$$\|A^{(\nu)}_{1/s_j}(a + n)\|_2 = \tfrac{1}{2}\|A^{(\nu)}_{1/s_j}(e_1 + 2n)\|_2 \geq \tfrac{1}{2}\,\mathrm{sys}(L_{s_j}) \longrightarrow \infty. \tag{4.35}$$

Consequently $Z = \{z_j : j \geq 1\}$ is closed in $\mathfrak{E}_\nu$. Indeed, its radial coordinates have no positive accumulation, and equation (4.35) says that it eventually misses every bounded half-space chart about every point of the zero fibre. On the other hand, in the product coordinates above,

$$\beta_\nu(z_j) = [e^{-2\pi/s_j}, [a]] \longrightarrow d_\nu,$$

while $d_\nu \notin \beta_\nu(Z)$. Thus $\beta_\nu(Z)$ is not closed. Because β_ν is injective away from the zero fibre and no z_j maps to d_ν,

$$\beta_\nu^{-1}(\beta_\nu(Z)) = Z.$$

If β_ν were a quotient map, closedness of this saturated inverse image would force $\beta_\nu(Z)$ to be closed, a contradiction.

The constant value on $p_\nu^{-1}(0)$ makes β_ν factor through the quotient projection to give the continuous bijection $\overline{\beta}_\nu$ in equation (4.33). Were that bijection a homeomorphism, its composite with the quotient projection would make β_ν a quotient map. Hence $\overline{\beta}_\nu$ is not a homeomorphism, completing every assertion. □

Corollary 4.15 (Exact global meaning of the Euclidean limit)**.** *The unmodified four-torus fibres themselves have native pointed limit $\mathbb{T}^2 \times \mathbb{R}^2$. Without finite-cover unspooling, the sphere-carrier constructions derived from that cusp produce $\mathbb{R}^4$ in three other precise senses:*

(i) *as the pointed intrinsic limit of the four-torus slices in the compact quotient carriers of S^{12} and S^{24}; and*

(ii) *as each nonclosed open cell $\mathcal{U}_{12,T} \subset S^6$ and $\mathcal{U}_{24,T} \subset S^8$ inside a moving four-sphere; and*

(iii) *as the decompactification boundary fibre of $\mathfrak{E}_{12}$ and $\mathfrak{E}_{24}$.*

It does not produce $\mathbb{R}^4$ as the whole compact sphere or compact conical target, as a continuous quotient target of either compact source sphere, as the quotient of $\mathfrak{E}_\nu$ through β_ν, or as the unrescaled ambient pointed limit of either compact conical target.

Proof. The positive assertions are theorems 4.8, 4.12 and 4.14. A continuous image of a compact sphere is compact, so it cannot be $\mathbb{R}^4$. No finite slice T^4 embeds in $\mathbb{R}^4$: invariance of domain would make its image open, while compactness would make it compact, and no nonempty open subset of $\mathbb{R}^4$ is compact. The local homology equation (4.26) shows that the carrier vertex is not a four-manifold point. The nonquotient conclusion for β_ν is equation (4.34)–equation (4.35). Finally, the unrescaled ambient collapse is proved in theorem 5.3 below. The comparison maps send the boundary $\mathbb{R}^4$ to the vertex, but do not give the vertex the topology of a collapsed boundary quotient. □

5 Intrinsic expansion and ambient collapse

The metric limit just proved is a statement about the varying intrinsic flat quotient metrics on the regular target slices. Here "flat" refers to the Riemannian metrics; no Sormani–Wenger intrinsic-flat convergence is being asserted. The limit cannot be interpreted as an unrescaled limit inside one fixed metric on a compact cone.

Two metrics on an embedded slice must therefore be distinguished. The intrinsic slice metric is the quotient or induced Riemannian path metric already specified on Q^2_T, C^4_T, or a moving S^4. If D is a fixed ambient metric and i_T is the topological slice embedding, the ambient restriction is instead

$$d^{\mathrm{amb}}_T(x,y) = D(i_T(x), i_T(y)).$$

It allows paths which leave the slice, including shortcuts toward a cone vertex or a collapsed fat wedge, and need not equal the intrinsic path metric. The next result concerns this ambient restriction; the preceding Euclidean-ball results concern the intrinsic metric.

Proposition 5.1 (Exact metric scope)**.** *Let fixed metrics compatible with the topologies be placed on $\mathcal{A}, \mathcal{B}, \mathcal{C}$ and on their relevant smash powers. If $j_T(Q^2_T)$ is the two-copy slice in $\mathcal{A}^{\wedge 2}$ and $k^{\mathcal{B}}_T(C^4_T)$, $k^{\mathcal{C}}_T(C^4_T)$ are the four-copy slices in the two circle targets, then*

$$\begin{aligned} \operatorname{diam}_{\mathrm{amb}} j_T(Q^2_T) &\longrightarrow 0, \\ \operatorname{diam}_{\mathrm{amb}} k^{\mathcal{B}}_T(C^4_T) &\longrightarrow 0, \\ \operatorname{diam}_{\mathrm{amb}} k^{\mathcal{C}}_T(C^4_T) &\longrightarrow 0. \end{aligned} \tag{5.1}$$

In the intrinsic quotient metrics, by contrast,

$$\operatorname{diam}(Q^2_T) \geq \tfrac{1}{2}(T - C) \longrightarrow \infty, \qquad \operatorname{diam}(C^4_T) \geq \tfrac{1}{2}\ell_T \longrightarrow \infty. \tag{5.2}$$

Proof. Every neighborhood of a cone vertex contains all sufficiently small radial cross-sections, uniformly in the compact angular and torus coordinates. In a compatible metric, given $\delta > 0$, an open $\delta/2$-ball about the vertex therefore contains each sufficiently small cross-section, whose diameter is then less than δ. Finite products and the continuous smash quotient preserve this uniform convergence, proving equation (5.1).

Let a flat torus have systole s, and choose a shortest nonzero period λ. It is primitive. For every period μ, the nonzero period $\lambda - 2\mu$ has norm at least s, so

$$\operatorname{dist}([\lambda/2], [0]) = \frac{1}{2} \inf_{\mu} \|\lambda - 2\mu\| \geq \frac{s}{2}.$$

Applying this to equations (1.23) and (1.31) and observing that a product torus contains each factor isometrically proves equation (5.2). □

Definition 5.2 (The radial target-slice embeddings)**.** Put $r_T = e^{-2\pi T}$, and define the arclength parametrization

$$\vartheta_T : \mathbb{R}/\mathbb{Z} \longrightarrow C_T = \mathbb{R}/\ell_T\mathbb{Z}, \qquad [\theta] \longmapsto [\ell_T\theta].$$

The three synchronized radial target slices are embedded by

$$\begin{aligned} j_T : Q_T^2 &\longrightarrow \mathcal{A}^{\wedge 2}, \\ (y_1, y_2) &\longmapsto [\,[r_T, 1, \iota_T^{-1}(y_1)], [r_T, 1, \iota_T^{-1}(y_2)]\,], \end{aligned}$$

$$\begin{aligned} k_T^{\mathcal{B}} : C_T^4 &\longrightarrow \mathcal{B}^{\wedge 4}, \\ (c_1, \ldots, c_4) &\longmapsto \bigwedge_{i=1}^{4} [r_T, 1, \vartheta_T^{-1}(c_i)], \end{aligned}$$

and

$$\begin{aligned} k_T^{\mathcal{C}} : C_T^4 &\longrightarrow \mathcal{C}^{\wedge 4}, \\ (c_1, \ldots, c_4) &\longmapsto \bigwedge_{i=1}^{4} [r_T, (1, \vartheta_T^{-1}(c_i))]. \end{aligned}$$

The notation in the final line uses the cusp cone $\mathrm{Cone}(\partial D_0 \times S^1) \subset \mathcal{C}$.

Theorem 5.3 (Topology-forced collapse for every fixed compatible metric)**.** *Let D_{12}, $D_{24}^{\mathcal{B}}$, and $D_{24}^{\mathcal{C}}$ be arbitrary fixed metrics compatible with the topologies of $\mathcal{A}^{\wedge 2}$, $\mathcal{B}^{\wedge 4}$, and $\mathcal{C}^{\wedge 4}$, respectively, and let $*$ be the corresponding smash basepoint. Then*

$$\begin{aligned} d_H^{D_{12}}\big(j_T(Q_T^2), \{*\}\big) &\longrightarrow 0, \\ d_H^{D_{24}^{\mathcal{B}}}\big(k_T^{\mathcal{B}}(C_T^4), \{*\}\big) &\longrightarrow 0, \\ d_H^{D_{24}^{\mathcal{C}}}\big(k_T^{\mathcal{C}}(C_T^4), \{*\}\big) &\longrightarrow 0. \end{aligned} \tag{5.3}$$

Thus, with metrics restricted from any one fixed ambient target, all three target slices converge in Gromov–Hausdorff topology to a point.

Proof. We prove the first assertion. Let

$$G : [0, r_0]^2 \times (T^3)^2 \longrightarrow \mathcal{A}^{\wedge 2}$$

be the product of the two inner-cylinder quotient maps followed by the smash quotient. It is continuous and $G(0, 0, z_1, z_2) = *$ for every (z_1, z_2). Compactness of the domain and uniform continuity with respect to D_{12} give

$$\sup_{(z_1,z_2)\in(T^3)^2} D_{12}\big(G(r, r, z_1, z_2), *\big) \longrightarrow 0 \qquad (r \downarrow 0).$$

The set $j_T(Q_T^2)$ is contained in the $r = r_T$ cross-section. This proves the first Hausdorff convergence. Replacing T^3 by T^2 and taking four factors proves the other two. Hausdorff convergence to a singleton implies Gromov–Hausdorff convergence to that singleton. □

Theorem 5.4 (One fixed cusp-adapted metric and its complete bubbles)**.** *There are fixed compatible length metrics*

$$D_{12} \text{ on } \mathcal{A}^{\wedge 2}, \qquad D_{24}^{\mathcal{B}} \text{ on } \mathcal{B}^{\wedge 4}, \qquad D_{24}^{\mathcal{C}} \text{ on } \mathcal{C}^{\wedge 4}$$

with the following properties. On the fixed marked torus $\overline{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$*, put*

$$h_T := \iota_T^* g_{Q_T} = dx^{\mathsf{T}} Y_T^{\mathsf{T}} Y_T dx, \qquad T(r) := -\frac{1}{2\pi}\log r.$$

On the inner cusp portion of $\mathcal{A}$*, in normalized base-angle coordinate* $u \in \mathbb{R}/\mathbb{Z}$*, the metric is induced by*

$$g_{\mathcal{A}} = dr^2 + r^2 du^2 + \frac{r^2}{T(r)^2} h_{T(r)}. \tag{5.4}$$

On the cusp cone of $\mathcal{B}$ *or* $\mathcal{C}$*, in normalized character coordinate* $\theta \in \mathbb{R}/\mathbb{Z}$*, it is induced by*

$$g_{\mathcal{D}} = dr^2 + r^2(du^2 + d\theta^2), \qquad \mathcal{D} \in \{\mathcal{B}, \mathcal{C}\}. \tag{5.5}$$

On every finite product use the ℓ^2*-product length metric*

$$d_2((x_i),(y_i)) = \left(\sum_i d_i(x_i,y_i)^2\right)^{1/2},$$

and on smash powers take its metric quotient after collapsing the closed fat wedge.

Let g_Z *denote the resulting Riemannian tensor on each open regular stratum, and put*

$$p_T^{12} = j_T(0,0), \qquad p_T^{24,\mathcal{D}} = k_T^{\mathcal{D}}(0,0,0,0).$$

Let $d_{12,T}^{\mathrm{sl}}$ *and* $d_{24,T}^{\mathcal{D},\mathrm{sl}}$ *denote the intrinsic Riemannian path metrics of* $j_T^* g_Z$ *and* $(k_T^{\mathcal{D}})^* g_Z$*, transported to the embedded slice images. These are not notation for the ambient restrictions of* D_{12} *and* $D_{24}^{\mathcal{D}}$*. The first fundamental forms satisfy the exact identities*

$$\left(\frac{T}{r_T}\right)^2 j_T^* g_Z = g_{Q_T} \oplus g_{Q_T}, \tag{5.6}$$

and

$$\left(\frac{\ell_T}{r_T}\right)^2 (k_T^{\mathcal{D}})^* g_Z = g_{C_T}^{\oplus 4}. \tag{5.7}$$

Consequently

$$\begin{aligned} \left(j_T(Q_T^2), \frac{T}{r_T} d_{12,T}^{\mathrm{sl}}, p_T^{12}\right) &\longrightarrow (\mathbb{R}^4, 0), \\ \left(k_T^{\mathcal{D}}(C_T^4), \frac{\ell_T}{r_T} d_{24,T}^{\mathcal{D},\mathrm{sl}}, p_T^{24,\mathcal{D}}\right) &\longrightarrow (\mathbb{R}^4, 0) \end{aligned} \tag{5.8}$$

in pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff topology. No equality with the ambient subspace distances is asserted.

The full ambient pointed blow-ups instead converge in pointed Gromov–Hausdorff topology as

$$\begin{aligned} \left(\mathcal{A}^{\wedge 2}, \frac{T}{r_T} D_{12}, p_T^{12}\right) &\longrightarrow (\mathbb{R}^8, 0), \\ \left(\mathcal{D}^{\wedge 4}, \frac{\ell_T}{r_T} D_{24}^{\mathcal{D}}, p_T^{24,\mathcal{D}}\right) &\longrightarrow (\mathbb{R}^{12}, 0), \qquad \mathcal{D} \in \{\mathcal{B}, \mathcal{C}\}. \end{aligned} \tag{5.9}$$

For every prescribed radius, the corresponding balls are eventually contained in unwrapped regular-stratum coordinate neighborhoods, and the pulled-back Riemannian tensors converge there in C^∞*. This is the only smooth-convergence assertion for the globally singular smash targets. The slice* $\mathbb{R}^4$ *is the coordinate subspace obtained by fixing the two radial and two base-angle coordinates in the first line, or the four radial and four base-angle coordinates in the second.*

Proof. By equation (1.25),

$$T^{-2}h_T = dx^{\mathsf{T}}(T^{-1}Y_T)^{\mathsf{T}}(T^{-1}Y_T)dx \longrightarrow dx^{\mathsf{T}}dx$$

in every C^k norm. Indeed, all nonleading cusp coefficients are holomorphic in $t = e^{-2\pi T}$, so their T-derivatives are exponentially small. Therefore the compact cross-section metrics in equation (5.4) remain uniformly bounded and converge to a positive definite flat metric. Their diameters are $O(r)$. The radial coordinate is 1-Lipschitz and a radial segment from level r to the vertex has length r; the length completion adds exactly one point at $r = 0$, with the prescribed cone topology. The same statements are immediate for equation (5.5). Interpolation on a closed annulus away from the vertex extends these tensors to fixed compatible length metrics on $\mathcal{A}, \mathcal{B}, \mathcal{C}$.

For a compact metric space (E, d) and closed subset $W \subset E$, the metric quotient collapsing W to one point is

$$d_{E/W}([x],[y]) = \min\{d(x,y), d(x,W) + d(y,W)\}.$$

Applying this formula to the fat wedges in the product length spaces gives fixed compatible metrics on the smash targets.

On the r_T-slice of $\mathcal{A}$, with $u = 0$, the first fundamental form is $r_T^2 T^{-2} h_T$. Since $h_T = \iota_T^* g_{Q_T}$, this proves equation (5.6). On the r_T-slice of $\mathcal{B}$ or $\mathcal{C}$, with $u = 0$, the form is $r_T^2 d\theta^2$. Under ϑ_T, the arclength metric of C_T pulls back to $\ell_T^2 d\theta^2$, proving equation (5.7). The systolic estimates

$$\operatorname{sys}(Q_T^2) \geq T - C, \qquad \operatorname{sys}(C_T^4) = \ell_T$$

then prove equation (5.8): every fixed-radius ball is eventually exactly Euclidean. These are the intrinsic path metrics of curves constrained to the slices; this argument does not identify them with the restrictions of the ambient distances.

For the ambient limits, use one $\mathcal{A}$-factor and put

$$\lambda_T = \frac{T}{r_T}, \quad s = \lambda_T(r - r_T), \quad \eta = Tu, \quad z = Y_T x$$

on the universal cover of the torus directions. On every fixed bounded set in (s, η, z),

$$\frac{r}{r_T} = 1 + \frac{s}{T}, \qquad T(r) = T + O(T^{-1}), \qquad Y_{T(r)} Y_T^{-1} = I + O(T^{-2}),$$

with the same estimates after any fixed number of derivatives. Hence $\lambda_T^2 g_{\mathcal{A}}$ converges to

$$ds^2 + d\eta^2 + dz_1^2 + dz_2^2.$$

The inward radial boundary is at distance T, the u-period is T, and the z-period lattice is $Y_T\mathbb{Z}^2$, with systole at least $T - C$. The outward interface $r = r_0$, beyond which the metric was interpolated, is at scaled distance

$$\frac{T}{r_T}(r_0 - r_T) \longrightarrow \infty.$$

Thus both inward and outward exits from the displayed regular coordinate neighborhood, as well as every periodic identification, recede to infinity. One factor converges to $\mathbb{R}^4$, and two factors to $\mathbb{R}^8$.

For one $\mathcal{B}$- or $\mathcal{C}$-factor, put

$$\lambda_T = \frac{\ell_T}{r_T}, \quad s = \lambda_T(r - r_T), \quad \eta = \ell_T u, \quad \zeta = \ell_T \theta.$$

Then

$$\lambda_T^2 g_{\mathcal{D}} = ds^2 + \left(\frac{r}{r_T}\right)^2 (d\eta^2 + d\zeta^2)$$

converges smoothly on bounded coordinate sets to the Euclidean metric on $\mathbb{R}^3$. Both circle periods and the inward radial boundary are at scaled distance $\ell_T \to \infty$. The outward interface is at distance

$$\frac{\ell_T}{r_T}(r_0 - r_T) \longrightarrow \infty,$$

so four factors converge to $\mathbb{R}^{12}$.

The product-metric distance from a synchronized point to the fat wedge is exactly r_T. After the displayed rescalings it is T, respectively ℓ_T. Thus every fixed-radius ball eventually avoids the fat wedge. Nor can a distance between two points of such a ball be shortened by an outward excursion: if $\mathcal{O}_T$ is the outward interface and the basepoint-to-$\mathcal{O}_T$ distance is a_T, any such excursion between points in the radius-R ball has length at least $2(a_T - R) > 2R$ for all sufficiently large T. The identical estimate with the fat wedge, whose scaled distance tends to infinity, excludes the inward collapsed-set shortcut. Hence all relevant metric balls remain inside the unwrapped regular neighborhoods, the smash quotient does not change their local calculations, and the C^∞ statement has exactly the restricted scope asserted in the theorem. This proves equation (5.9). □

Theorem 5.5 (Complete scalar-rescaling trichotomy)**.** *For the fixed metrics of theorem 5.4, let $\lambda_T > 0$ be arbitrary and put*

$$\beta_T = \lambda_T r_T.$$

On the Q_T^2 slice or either C_T^4 slice, pass to a subsequence so that β_T has a limit in $[0, \infty]$. The pointed limit is

$$\begin{cases} \{*\}, & \beta_T \longrightarrow 0, \\ (\mathbb{R}/L\mathbb{Z})^4, & \beta_T \longrightarrow L \in (0, \infty), \\ \mathbb{R}^4, & \beta_T \longrightarrow \infty. \end{cases} \tag{5.10}$$

In the middle line the lattice is orthogonally equivalent to $L\mathbb{Z}^4$. At the same time,

$$\lambda_T D(p_T, *) = \beta_T \tag{5.11}$$

for every synchronized basepoint used above. Therefore every scalar rescaling which unrolls the slice to $\mathbb{R}^4$ sends the cone/fat-wedge basepoint to infinite scaled distance. Such a limit is not a tangent cone based at that point.

Proof. For the Q_T^2 route, the rescaled period matrix in each two-dimensional factor is

$$\frac{\lambda_T r_T}{T} Y_T = \beta_T \left(\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \frac{1}{T} \begin{pmatrix} 6m_T & H_T \\ B_T - H_T & m_T \end{pmatrix} \right).$$

If $\beta_T \to L \in (0, \infty)$, this lattice converges to $LJ\mathbb{Z}^2$, where $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$; two factors give the square four-torus in equation (5.10). If $\beta_T \to 0$, the lengths of the displayed lattice basis tend to zero, and so does the diameter. If $\beta_T \to \infty$, the reverse triangle inequality gives systole

$$\frac{\beta_T}{T}(T - C) = \beta_T \left(1 - \frac{C}{T}\right) \longrightarrow \infty.$$

Every fixed-radius ball is then eventually Euclidean.

For a C_T^4 slice, its four rescaled periods all have length $\lambda_T r_T = \beta_T$, so the same three cases follow directly. Before the smash quotient, a synchronized point in the product lies at distance exactly r_T from the fat wedge: moving any one radial coordinate to zero realizes this distance, and the dr^2-term proves the lower bound. The metric-quotient formula in the proof of theorem 5.4 proves equation (5.11). □

Corollary 5.6 (Unit-volume normalization does not give $\mathbb{R}^4$)**.** *The four-volume of Q_T^2 is $(\det Y_T)^2 \sim T^4$. Its unit-volume length factor is $(\det Y_T)^{-1/2} \sim T^{-1}$, and its unit-volume limit is a square flat four-torus, not $\mathbb{R}^4$. Likewise, the four-volume of C_T^4 is ℓ_T^4, and its unit-volume normalization is exactly the square unit four-torus.*

Proposition 5.7 (Vertex tangents and compact asymptotic cones)**.** *The compact target topology alone determines no metric tangent cone. For an exact conic length metric built from fixed piecewise-Riemannian metrics on the links, however, the vertex links have topological and Hausdorff dimensions 7 for $\mathcal{A}^{\wedge 2}$ and 11 for both fourfold circle targets. The resulting vertex cones have dimensions 8 and 12, and are not $\mathbb{R}^4$.*

More precisely, one has the based topological models

$$\mathcal{A}^{\wedge 2} \cong \operatorname{Susp}(T^3 * T^3), \qquad \mathcal{B}^{\wedge 4} \cong \operatorname{Susp}((T^2)^{*4}).$$

At their suspension basepoints the conic links are $T^3 * T^3$ and $(T^2)^{*4}$. Also

$$\mathcal{C}^{\wedge 4} \cong \mathcal{M}^4/\partial(\mathcal{M}^4),$$

so an exact conic metric at the collapsed boundary has link $\partial(\mathcal{M}^4)$. If $*$ is its collapsed smash point, its complete local integral homology is

$$H_i(\mathcal{C}^{\wedge 4}, \mathcal{C}^{\wedge 4} \setminus \{*\}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^4, & i = 2, \\ \mathbb{Z}^6, & i = 3, \\ \mathbb{Z}^4, & i = 4, \\ \mathbb{Z}, & i = 5, \\ \mathbb{Z}, & i = 8, \\ \mathbb{Z}^4, & i = 9, \\ \mathbb{Z}^6, & i = 10, \\ \mathbb{Z}^4, & i = 11, \\ \mathbb{Z}, & i = 12, \\ 0, & \textit{otherwise.} \end{cases} \tag{5.12}$$

In particular the collapsed point is not a manifold point of any dimension. For the other two exact conic models, the top local homology occurs in degree 8, respectively 12, rather than degree 4.

Finally, scaling the distances of a fixed compact metric target by R_n^{-1}, where $R_n \to \infty$, gives a point:

$$\operatorname{diam}(Z, R_n^{-1} D) = R_n^{-1} \operatorname{diam}(Z, D) \longrightarrow 0.$$

Thus equations (5.8) and (5.9) describes moving-basepoint blow-ups, or bubbles, not vertex-based tangent cones and not asymptotic cones of the compact targets.

Proof. For compact spaces V_i with closed subspaces A_i, the quotient definitions give a canonical homeomorphism

$$(V_1/A_1) \wedge \cdots \wedge (V_m/A_m) \cong \frac{V_1 \times \cdots \times V_m}{\bigcup_i V_1 \times \cdots \times A_i \times \cdots \times V_m}.$$

Take $V_i = \operatorname{Cone}(L_i)$ and $A_i = L_i$, its outer boundary. A product of m cones is homeomorphic to the cone on the iterated join $L_1 * \cdots * L_m$, and the displayed union of faces is the outer link. Collapsing that link gives its suspension. Hence

$$\dim(T^3 * T^3) = 3 + 3 + 1 = 7, \qquad \dim((T^2)^{*4}) = 4 \cdot 2 + 3 = 11.$$

By theorem 3.2 and the same quotient identity, if $N = \mathcal{M}^4$ and

$$\partial_\times N = \bigcup_{j=1}^{4} \mathcal{M}^{j-1} \times \partial\mathcal{M} \times \mathcal{M}^{4-j},$$

then

$$\mathcal{C}^{\wedge 4} \cong N/\partial_\times N.$$

Rounding the product corners replaces $\partial_\times N$ by a homeomorphic eleven-dimensional closed manifold ∂N. A collar of ∂N descends to a cone neighborhood of the collapsed point, so its link is exactly ∂N.

The collar also makes $\operatorname{int}\mathcal{M} \hookrightarrow \mathcal{M}$ a homotopy equivalence. Hence, for

$$U = \mathcal{C}^{\wedge 4} \setminus \{*\} \cong (\operatorname{int}\mathcal{M})^4,$$

the Künneth theorem and theorem 3.3 give

$$H_j(U;\mathbb{Z}) \cong H_j(N;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{\binom{4}{j}}, & 0 \le j \le 4, \\ 0, & \text{otherwise.} \end{cases}$$

On the other hand, equation (3.21) gives the absolute homology of $\mathcal{C}^{\wedge 4}$: its reduced groups have ranks $1, 4, 6, 4, 1$ in degrees $8, 9, 10, 11, 12$ and vanish otherwise. The long exact sequence of the pair $(\mathcal{C}^{\wedge 4}, U)$ therefore yields

$$H_i(\mathcal{C}^{\wedge 4}, U;\mathbb{Z}) \cong H_{i-1}(U;\mathbb{Z}) \quad (2 \le i \le 5)$$

and

$$H_i(\mathcal{C}^{\wedge 4}, U;\mathbb{Z}) \cong H_i(\mathcal{C}^{\wedge 4};\mathbb{Z}) \quad (8 \le i \le 12).$$

The map on H_0 is an isomorphism and all groups in the remaining degrees vanish. This proves equation (5.12).

For comparison, the top relative homology groups of the two suspension models and of $N/\partial N$ are nonzero:

$$H_8(\operatorname{Cone}(T^3 * T^3), T^3 * T^3;\mathbb{Z}) \cong \mathbb{Z}$$

and, by orientation and Poincaré–Lefschetz duality,

$$H_{12}(\mathcal{M}^4, \partial(\mathcal{M}^4);\mathbb{Z}) \cong \mathbb{Z}.$$

The $\mathcal{B}$ calculation is identical in degree 12. Euclidean $\mathbb{R}^4$ has local homology $\mathbb{Z}$ only in degree 4, so none of these conic vertex models is $\mathbb{R}^4$. The diameter computation proves the final scaling assertion. □

Metric consequence. For each fixed cusp-adapted compact-codomain metric, the corresponding moving-base-point rescalings have a Euclidean bubble containing the native four-dimensional decompactification as its vertical $\mathbb{R}^4$ factor. The complete bubbles are $\mathbb{R}^8$ and $\mathbb{R}^{12}$. The factor $\beta_T = \lambda_T r_T$ proves that a different scalar rescaling of the same fixed cusp-adapted metric cannot remove this dimensional discrepancy: the slice unrolls only after the smash basepoint has gone to infinite scaled distance. No metric on a source sphere is supplied. Pulling a codomain metric back through $q_2^{\wedge 2}$ or $q_1^{\wedge 4}$ gives, where differentiable, a degenerate quadratic form along the positive-dimensional quotient kernels.

6 Sharp obstructions

Proposition 6.1 (Three exact almost-complex obstructions on S^{12})**.** *Let M be a smooth manifold homeomorphic to S^{12}. There is no bundle endomorphism $J : TM \to TM$ satisfying $J^2 = -\mathrm{Id}_{TM}$. Two independent contradictions—the Borel–Serre reduced-power calculation at the single prime 5, and Bott integrality—use only the homotopy type of M. For the standard smooth sphere there is also an independent stable-tangent Pontryagin-class contradiction.*

These statements concern an unstabilized complex structure on the actual rank-twelve tangent bundle. A stable complex structure on $TS^{12} \oplus \varepsilon^r_{\mathbb{R}}$, a complex structure on an open subset or on the fibres of a map, or an endomorphism which fails along a singular set does not satisfy this hypothesis.

Proof. Assume that J exists and put $E = (TM, J)$, a complex vector bundle of rank six. Orient M by the interleaved complex frame

$$(v_1, Jv_1, \dots, v_6, Jv_6),$$

write $[M]_J$ for the resulting fundamental class, and choose $u_J \in H^{12}(M;\mathbb{Z})$ by $\langle u_J, [M]_J\rangle = 1$. Since M has the integral cohomology of S^{12},

$$c_1(E) = c_2(E) = c_3(E) = c_4(E) = c_5(E) = 0. \tag{6.1}$$

The top-Chern/Euler identity uses exactly this orientation, and Poincaré–Hopf therefore gives

$$c_6(E) = e(E_{\mathbb{R}}) = e(TM) = 2u_J. \tag{6.2}$$

Reversing a previously chosen orientation changes the displayed generator, not the coefficient two.

For the first contradiction, specialize the original Borel–Serre calculation [Borel and Serre, 1953, §12, Proposition 12.4, Lemma 12.5, and Proposition 12.6, p. 430; §15, Proposition 15.1, p. 434]. They write C_{2j} for the universal class now denoted c_j. At $j = 6$, the admissible prime is exactly $p = 5 < 6$, with $5 \nmid 6$. In $H^*(BU(6);\mathbb{F}_5)$, pass to a splitting space, write $x_1, \dots, x_6$ for the degree-two Chern roots, and put $s_r = \sum_i x_i^r$. Since $\mathcal{P}^1_5(x_i) = x_i^5$, the Cartan formula and Newton identities give the complete universal identity

$$\begin{aligned}\mathcal{P}^1_5(\bar c_2) &= \sum_{i<j}(x_i^5 x_j + x_i x_j^5) = s_5 s_1 - s_6 \\ &= 6\bar c_6 + \bar c_1^4\bar c_2 - \bar c_1^3\bar c_3 - 4\bar c_1^2\bar c_2^2 + \bar c_1^2\bar c_4 + 7\bar c_1\bar c_2\bar c_3 - \bar c_1\bar c_5 \\ &\quad + 2\bar c_2^3 - 6\bar c_2\bar c_4 - 3\bar c_3^2.\end{aligned} \tag{6.3}$$

Modulo the lower-Chern ideal $I = (\bar c_1, \dots, \bar c_5)$, this says

$$\mathcal{P}^1_5(\bar c_2) \equiv 6\bar c_6 = \bar c_6 \pmod{I}.$$

Pulling back along the classifying map of E, using equation (6.1) and naturality, gives

$$\overline{c_6(E)} = \mathcal{P}^1_5(\overline{c_2(E)}) = 0.$$

But equation (6.2) gives $\overline{c_6(E)} = 2\overline{u}_J \neq 0$ in $H^{12}(M;\mathbb{F}_5)$, a contradiction.

For the second contradiction, choose an orientation-preserving homotopy equivalence $f : S^{12} \to M$. Bott integrality says that the sixth Chern class of every complex bundle on S^{12} is divisible by $(6-1)! = 120$; see Husemoller, *Fibre Bundles*, Chapter 20, §9, Theorem 9.6 and Corollary 9.8, p. 308. There is no normalization ambiguity here: Newton's sixth identity and equation (6.1) give

$$\mathrm{ch}_6(f^*E) = \frac{s_6(f^*E)}{6!} = -\frac{c_6(f^*E)}{5!}.$$

Hence $c_6(f^*E) = 120kf^*u_J$ for some $k \in \mathbb{Z}$, while equation (6.2) gives $c_6(f^*E) = 2f^*u_J$. The equation $120k = 2$ has no integral solution.

Finally take the standard smooth sphere $S^{12} \subset \mathbb{R}^{13}$. Its normal field gives the explicit stable trivialization

$$TS^{12} \oplus \varepsilon^1_{\mathbb{R}} \xrightarrow{\cong} \varepsilon^{13}_{\mathbb{R}}, \qquad (x; v, a) \longmapsto (x; v + ax), \tag{6.4}$$

so $p_3(TS^{12}) = 0$. On the other hand,

$$E_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \cong E \oplus \overline{E}, \qquad p_3(E_{\mathbb{R}}) = -c_6(E \oplus \overline{E}).$$

Equation (6.1) and $c_6(\overline{E}) = c_6(E)$ yield

$$p_3(TS^{12}) = -2c_6(E) = -4u_J, \tag{6.5}$$

contradicting stable triviality in the torsion-free group $H^{12}(S^{12}; \mathbb{Z})$. This last calculation is independently recorded in Konstantis–Parton, Lemma 4.1 and Theorem 4.2.

Each contradiction requires a globally defined J on the actual bundle TM. Stabilizing, deleting points, restricting to regular fibres, or allowing failure at a singular locus removes the hypothesis rather than producing an exception. □

Theorem 6.2 (Obstructions to stronger globalizations)**.** *The preceding constructions have the following exact limitations.*

(i) *Let $r \geq 1$ and let $N \subset \Lambda^{\oplus r}$ be a fixed primitive, monodromy-invariant kernel defining a connected fibrewise torus-group quotient of the native product over the full regular base, or over a finite cover of it. Equip it with the quotient norm of the unscaled marked Euclidean source metric. If its radial quotient systole tends to infinity, then*

$$\operatorname{rank}(\Lambda^{\oplus r}/N) \leq r. \tag{6.6}$$

Four copies are therefore minimal for a full-base, four-dimensional, all-expanding quotient within this class. This bound says nothing about nonlinear maps, varying kernels, rescaled metrics, or non-torus targets. The S^{12} construction avoids the bound by localizing to D_0^* and collapsing the exterior.

(ii) *No nontrivial finite base cover makes K_Λ a sub-local system over the whole regular base. This remains true for a compact cover allowed to branch over the original singular values. Moreover, a local cyclic change $t = s^M$ cannot extend as a connected nontrivial cover of $\mathbb{P}^1$ branched only at p_0.*

(iii) *There is no map from X^2, $X \wedge X$, X^4, or $X^{\wedge 4}$ to a fixed T^4 whose restriction retains the four target character classes of the corresponding displayed product quotient. There is no map to a fixed S^4 whose restriction, for even one regular fibre, is a degree-one collapse of Q^2_T or of C^4_T.*

(iv) *Cartesian products are not the sphere sources:*

$$X^2 \simeq S^6 \times S^6 \not\simeq S^{12}, \qquad X^4 \simeq (S^6)^4 \not\simeq S^{24}. \tag{6.7}$$

The synchronized two- and fourfold fibre products have real dimensions 10 and 18. Joins give $X * X \cong S^{13}$ and $X^{*4} \cong S^{27}$, not the required dimensions.

This product obstruction does not obstruct the preceding smash-source maps. The global quotient $q_2 : (X, x_0) \to (\mathcal{A}, v_0)$ is based and hence $q_2 \times q_2$ is constant on the global fat wedge $X \vee X \subset X \times X$. It therefore induces $q_2^{\wedge 2} : X \wedge X \to \mathcal{A} \wedge \mathcal{A}$, whose source is S^{12}. Since x_0 lies on the central fibre, the regular radial product F^2_T is disjoint from that fat wedge; the product-to-smash quotient is injective there and the restriction is exactly $\pi_T \times \pi_T$ under the markings. This differs from trying to factor the torus map $\pi_T \times \pi_T$ through $F_T \wedge F_T$ with the fibre identity as basepoint, which is impossible because the map is nonconstant on that fibrewise fat wedge.

(v) *Every connected unbranched finite cover of X, of a Cartesian power X^r, or of the smash spheres $X \wedge X$ and $X^{\wedge 4}$ is trivial. The local unspooling covers are not restrictions of nontrivial connected unbranched covers of these compact sources.*

(vi) *Neither S^{12} nor S^{24}, with any differentiable structure, admits an almost-complex structure. Hence neither smash source can be promoted to a compact complex sphere.*

Proof. Every fixed element of $K_\Lambda^{\oplus r}$ has uniformly bounded radial length because $P_T\widehat{w} = e_1$ and $P_T\widehat{\delta} = e_3$. If such an element were not in N, primitivity would make its class a nonzero period of the connected quotient lattice, with quotient norm no greater than its uniformly bounded source norm. Thus

$$K_\Lambda^{\oplus r} \subset N. \tag{6.8}$$

For full monodromy, $A_1\widehat{w} = \widehat{u} - \widehat{w}$ and invariance then put $\widehat{u}$ in N in every copy. Consequently $L_\Lambda^{\oplus r} \subset N$ and $\operatorname{rank}(\Lambda^{\oplus r}/N) \le 4r - 3r = r$. Taking $N = L_\Lambda^{\oplus 4}$ gives the four-circle quotient, so the bound is sharp.

For the finite-cover statements put

$$H_* = A_2A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ -5 & 0 & -1 & 1 \end{pmatrix} = I + N_*, \qquad N_*^2 = 0. \tag{6.9}$$

Hence

$$H_*^n\widehat{w} = \widehat{w} + n(\widehat{u} - \widehat{\delta}), \qquad H_*^n\widehat{\delta} = \widehat{\delta}. \tag{6.10}$$

Every finite-index subgroup of the base fundamental group contains a positive power of the loop with monodromy H_*. Equation equation (6.10) shows both that it never preserves K_Λ and, combined with equation (6.8), that $n\widehat{u} \in N$. Primitivity of N implies $\widehat{u} \in N$, so the same rank bound holds after every finite base cover. A branched compact cover restricts over B° to such a finite cover. Finally, $\mathbb{P}^1 \setminus \{p_0\} \cong \mathbb{C}$ is simply connected, so a connected cover branched only over p_0 has degree one.

For a fixed T^4 target, each target character class would pull back from $H^1(X^2;\mathbb{Z})$, $H^1(X^4;\mathbb{Z})$, or the first cohomology of the relevant smash sphere. Every one of these groups vanishes. On a radial fibre, however, a surjective torus homomorphism injects the target character lattice into the source character lattice; the four displayed target characters therefore pull back to four independent nonzero classes. This contradiction proves the fixed-torus assertion.

For the fixed S^4 assertion, let $\omega_T \in H^2(Q_T;\mathbb{Z})$ be the orientation class. The map $\pi_T^* : H^1(Q_T) \to H^1(F_T)$ is injective, and torus cohomology is the exterior algebra on H^1. Therefore

$$(\pi_T \times \pi_T)^*(\omega_T \times \omega_T) \neq 0 \quad \text{in } H^4(F_T^2;\mathbb{Z}). \tag{6.11}$$

A degree-one collapse $Q_T^2 \to S^4$ pulls the orientation generator back to $\pm(\omega_T \times \omega_T)$, while $H^4(X^2;\mathbb{Z}) = H^4(S^{12};\mathbb{Z}) = 0$. The four-circle case is identical: its orientation class pulls back to the exterior product of four independent circle characters, but $H^4(X^4;\mathbb{Z}) = H^4(S^{24};\mathbb{Z}) = 0$. The conical targets evade these contradictions because the fibre orientation and character classes are not restrictions of fixed global target classes.

The Künneth theorem gives $H_6(X^2) = \mathbb{Z}^2$, whereas $H_6(S^{12}) = 0$, and gives ranks $4, 6, 4$ in degrees $6, 12, 18$ for X^4, whereas S^{24} has none. An r-fold synchronized fibre product has dimension

$$6r - 2(r-1) = 4r + 2,$$

which gives 10 and 18. The standard join identity $S^m * S^n \cong S^{m+n+1}$ gives the stated join dimensions.

The spaces in the cover assertion are simply connected, so connected unbranched covers are trivial. Branched or normalized compactifications are different objects and have not been constructed here. Finally, theorem 6.1 proves the S^{12} assertion by three exact calculations. Borel and Serre's general theorem applies to S^{24}, with every differentiable structure, by taking $n = 12$ [Borel and Serre, 1953, §15, Proposition 15.1 and the following remark, p. 434]. □

7 The nine-dimensional Gram carrier and the surviving bridge problem

The higher-rung construction that produces a topological S^8 has a different source and a different quotient mechanism from every smash map above. We record its topology here so that the logical implications can be stated without identifying two constructions merely because their final boundaries are abstractly homeomorphic spheres. This is a statement about what has presently been constructed and proved, not a theorem that no deeper geometric or physical morphism exists.

Theorem 7.1 (The exact B^9/S^8 Gram carrier)**.** *Let*

$$\mathcal{V}_3 = \mathrm{Mat}_{3\times 3}(\mathbb{C}), \qquad \mathcal{P}_3 = \{H \in \mathrm{Herm}_3(\mathbb{C}) : H \succeq 0\},$$

and let $U(3)$ act on $\mathcal{V}_3$ by left multiplication. The Gram map

$$\Gamma : \mathcal{V}_3 \longrightarrow \mathcal{P}_3, \qquad \Gamma(A) = A^\dagger A,$$

induces homeomorphisms

$$\mathcal{V}_3/U(3) \cong \mathcal{P}_3, \qquad \mathcal{V}_3^+/U(3) \cong \mathcal{P}_3^+ \cong B^9, \tag{7.1}$$

where $+$ denotes one-point compactification. Under the last homeomorphism,

$$\partial B^9 = \{[A] : \mathrm{rank}\, A \le 2\} \cup \{[\infty]\} \cong S^8. \tag{7.2}$$

Thus the carrier is a nine-dimensional ball and its S^8 is the rank-loss-and-infinity boundary. The displayed orbit construction is not a principal $SU(3)$-bundle construction.

Proof. The Gram map is invariant because $(UA)^\dagger(UA) = A^\dagger A$, and it is surjective because $\Gamma(H^{1/2}) = H$ for $H \succeq 0$. Suppose $A^\dagger A = B^\dagger B$. The rule

$$T : \mathrm{im}\, A \longrightarrow \mathrm{im}\, B, \qquad T(Ax) = Bx,$$

is well-defined: $Ax = 0$ implies $\|Bx\|^2 = x^\dagger B^\dagger B x = x^\dagger A^\dagger A x = 0$. The same identity with two vectors shows that T is a complex-linear isometry. Extend T across the orthogonal complements to a unitary $U \in U(3)$; then $UA = B$. Thus Gram matrices separate orbits. Moreover

$$\|A\|_F^2 = \mathrm{tr}(A^\dagger A), \tag{7.3}$$

so Γ is proper. The induced map from the orbit space is a proper continuous bijection between locally compact Hausdorff spaces and hence a homeomorphism.

It remains to identify the compactified cone with a ball and its rank-loss locus with the boundary. Put

$$\mathcal{K}_3 = \{H \in \mathcal{P}_3 : \mathrm{tr}\, H = 1\}, \quad E = \{X \in \mathrm{Herm}_3(\mathbb{C}) : \mathrm{tr}\, X = 0\}, \quad c = I_3/3.$$

The real vector space E has dimension 8. For $u \in S(E)$, trace zero and $u \neq 0$ imply $\lambda_{\min}(u) < 0$; set

$$R(u) = -\frac{1}{3\lambda_{\min}(u)} > 0. \tag{7.4}$$

For $s \geq 0$, the eigenvalues of $c + su$ are $1/3 + s\lambda_j(u)$, and therefore

$$c + su \succeq 0 \quad \iff \quad 0 \leq s \leq R(u).$$

The continuous maps

$$\chi(c) = 0, \qquad \chi(c + su) = \frac{s}{R(u)} u, \tag{7.5}$$

$$\chi^{-1}(0) = c, \qquad \chi^{-1}(tu) = c + tR(u)u \quad (0 < t \leq 1) \tag{7.6}$$

are inverse homeomorphisms $\mathcal{K}_3 \cong B(E) \cong B^8$. Continuity at the origin follows because R is positive, continuous, and bounded above and below on the compact sphere $S(E)$. The equality $s = R(u)$ holds exactly on the singular locus, so χ takes that locus to $S(E)$.

Every nonzero $H \in \mathcal{P}_3$ has the unique form $H = r\rho$, where $r = \operatorname{tr} H > 0$ and $\rho \in \mathcal{K}_3$. Put

$$t = \frac{r-1}{r+1}, \qquad \mathcal{D}^9 = \{(y,t) \in E \oplus \mathbb{R} : \|y\|_F + |t| \leq 1\}.$$

Define on the one-point compactification

$$\Xi(r\rho) = ((1 - |t|)\chi(\rho), t), \qquad \Xi(0) = (0, -1), \qquad \Xi(\infty) = (0, 1). \tag{7.7}$$

For $|t| < 1$, its inverse is

$$\rho = \chi^{-1}\left(\frac{y}{1 - |t|}\right), \qquad r = \frac{1+t}{1-t}, \qquad H = r\rho. \tag{7.8}$$

The factor $1 - |t|$ tends uniformly to zero both as $r \downarrow 0$ and as $r \to \infty$. By equation (7.3), a set in $\mathcal{P}_3$ is unbounded exactly when its trace is unbounded. Thus equations (7.7) and (7.8) give a homeomorphism $\mathcal{P}_3^+ \cong \mathcal{D}^9$.

The latter is the unit ball of the norm $(y,t) \mapsto \|y\|_F + |t|$ on the nine-dimensional vector space $E \oplus \mathbb{R}$, hence is homeomorphic to the Euclidean B^9. Explicitly, for $p = (y,t) \neq 0$, the radial formula

$$p \longmapsto \frac{\|y\|_F + |t|}{\sqrt{\|y\|_F^2 + t^2}} p \tag{7.9}$$

and the origin is sent to the origin. For $q = (q_E, q_t) \neq 0$, the inverse is

$$q \longmapsto \frac{\|q\|_2}{\|q_E\|_F + |q_t|}\, q.$$

It too sends the origin to the origin. For finite $H = r\rho$, its image lies on the boundary precisely when $\|\chi(\rho)\|_F = 1$, equivalently when H and A are singular. The zero matrix and the point at infinity are the two boundary tips, and zero already belongs to the rank-loss locus. This proves the second homeomorphism and equation (7.2).

Because Γ is proper, it extends continuously by $\Gamma(\infty) = \infty$. The induced map $\mathcal{V}_3^+/U(3) \to \mathcal{P}_3^+$ is a continuous bijection from a compact space to a Hausdorff space, hence a homeomorphism. This completes equation (7.1).

Finally, this is a $U(3)$-orbit calculation. The Gram matrix alone does not separate strict $SU(3)$-orbits on the full-rank locus: if $B = UA$ with A invertible, then $\det B = (\det U) \det A$. A strict $SU(3)$-invariant must retain the additional determinant $z = \det A$, subject to $|z|^2 = \det(A^\dagger A)$. On the rank-loss locus the action also has nontrivial stabilizers. Hence neither the full quotient nor its boundary is being asserted to be a principal $SU(3)$-bundle. □

Theorem 7.2 (Exact obstruction scope and surviving bridges)**.** *Let $S^8_{\mathrm{Gram}} = \partial\mathcal{P}^+_3$ denote the boundary in theorem 7.1, and let $S^8_\wedge = (S^2)^{\wedge 4}$ denote the target of $\widehat{q}^{\wedge 4}_\gamma$. Then:*

(i) *A based homeomorphism $h : S^8_\wedge \to S^8_{\mathrm{Gram}}$ makes $h \circ \widehat{q}^{\wedge 4}_\gamma : S^{24} \to S^8_{\mathrm{Gram}}$ null-homotopic. This transfers only the homotopy class of that chosen map; it does not transfer the Gram orbit relation, rank stratification, a principal bundle, a connection, or a metric.*

(ii) *The nullity of the boundary inclusion $S^8_{\mathrm{Gram}} \hookrightarrow B^9$ follows immediately from the contractibility of B^9. It does not make the boundary a point and does not contradict equation (7.2). Likewise, the smash and join nullities and the Cartesian-product obstructions do not obstruct the existence of B^9 or its boundary, because the Gram carrier is not constructed as a Cartesian power, smash power, or join of X.*

(iii) *Every principal $SU(3)$-bundle on $S^8_{\mathrm{Gram}} \cong S^8$ is trivial by equation (4.21). This does not obstruct the orbit space: the quotient map $\mathcal{V}^+_3 \to \mathcal{V}^+_3/U(3)$ has stabilizers on the rank strata and is neither a principal $U(3)$-bundle globally nor a principal $SU(3)$-bundle.*

(iv) *If S^8_{Gram} is given any smooth structure, the resulting homotopy eight-sphere admits no almost-complex structure. This rules out a complex tangent bundle on the boundary, but not its topology as the boundary of B^9.*

(v) *Every fixed compatible metric on the compact S^8_{Gram} is bounded, so the unrescaled whole boundary is not Euclidean $\mathbb{R}^4$. For a fixed smooth Riemannian metric, every blow-up sequence with scale tending to infinity has, after passing to a basepoint subsequence, the full tangent-space limit $\mathbb{R}^8$ in pointed C^∞ Cheeger–Gromov topology, hence also in pointed Gromov–Hausdorff topology. The $\mathbb{R}^4$ vertical subspaces and the $\mathbb{R}^8/\mathbb{R}^{12}$ full bubbles in theorem 5.4 belong to the specifically defined conical smash-target metrics. Moreover theorem 5.5 shows that every rescaling which unrolls one of those slices to $\mathbb{R}^4$ sends its cone/fat-wedge vertex to infinite scaled distance, so it is not a vertex tangent cone. No metric identifying those bubbles or vertices with the Gram rank boundary has been specified.*

The strongest presently proved bridge is the based homeomorphism h in (i) *and its null composite. Each negative statement above rules out only the exact proposed identification or structure to which its proof applies. Constructing a stronger bridge remains an active problem: one would need a specified map from X, a smash/cone target, or its moving tori to B^9 or S^8_{Gram}, together with proofs that the map respects the relevant Gram orbit strata, descends the torus period data, intertwines any asserted principal-bundle or connection structures, and compares the fixed metrics in question.*

Proof. Composition with a based homeomorphism preserves a based null-homotopy, so (i) follows from theorem 4.6. A homeomorphism of underlying spheres contains none of the extra maps named there. The first claim of (ii) is the contraction of an inclusion into a contractible space. The remaining claims follow from the definitions: the source of the Gram construction is the real vector space $\mathrm{Mat}_{3\times 3}(\mathbb{C})$, and its equivalence relation is a left $U(3)$-orbit relation, not any of the product, fat-wedge, smash, or join relations in theorems 4.7 and 6.2. Part (iii) is exactly the first group in equation (4.21), together with the stabilizer observation at the end of the proof of theorem 7.1.

For (iv), let M be a smooth manifold homeomorphic to S^8_{Gram}, and suppose J made TM into a complex rank-four bundle E. With $u \in H^8(M;\mathbb{Z})$ oriented by J, sphere cohomology and the top-Chern/Euler identity give

$$c_1(E) = c_2(E) = c_3(E) = 0, \qquad c_4(E) = e(TM) = 2u.$$

The exact relation between Pontryagin and Chern classes gives

$$p_2(E_{\mathbb{R}}) = c_2(E)^2 - 2c_1(E)c_3(E) + 2c_4(E) = 4u \neq 0, \tag{7.10}$$

while $p_1(TM) = 0$ because $H^4(M;\mathbb{Z}) = 0$. Also $\operatorname{sign}(M) = 0$, since $H^4(M;\mathbb{R}) = 0$. Hirzebruch's eight-dimensional signature formula would then give

$$0 = \operatorname{sign}(M) = \left\langle \frac{7p_2(TM) - p_1(TM)^2}{45}, [M] \right\rangle = \frac{28}{45},$$

a contradiction. For the standard smooth sphere, the outward normal gives the independent identity $TS^8 \oplus \varepsilon^1_{\mathbb{R}} \cong \varepsilon^9_{\mathbb{R}}$, and hence $p_2(TS^8) = 0$, which also contradicts equation (7.10). These arguments use only a hypothetical complex structure on the actual tangent bundle and say nothing against the topological homeomorphism in equation (7.2).

For (v), compactness makes every compatible metric bounded. For a smooth Riemannian metric on a compact S^8, basepoints have a convergent subsequence, injectivity radius has a positive uniform lower bound, and all curvature derivatives are uniformly bounded. Normal coordinates then show that rescaling distances by factors tending to infinity converges on each fixed-radius ball to the Euclidean metric on the eight-dimensional tangent space. By contrast, the metrics of theorem 5.4 were defined from explicit cone and torus product coordinates and need not be smooth at their cone vertices. The equality $\lambda_T D(p_T, *) = \lambda_T r_T$ in equation (5.11) proves the asserted separation between an $\mathbb{R}^4$ slice-unrolling limit and a vertex tangent. No map in the Gram construction equates those metric tensors. This proves all five assertions and the final list of open bridges. □

8 An ancillary finite-cover route

The following construction is mathematically valid but answers a different question. It unrolls the two short periods of the original four-torus after simultaneous cyclic base change and finite fibre cover. It does not use the smash sources S^{12} or S^{24}, does not produce either sphere carrier, and is not a substitute for theorems 4.2 and 4.14.

Proposition 8.1 (Monodromy-compatible finite-cover unspooling)**.** *For $N \geq 1$, put*

$$S_N = \operatorname{diag}(1, 1, N, N), \qquad \Lambda_N = S_N\mathbb{Z}^4 = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, N\widehat{w}, N\widehat{\delta} \rangle.$$

The marked radial fibre

$$\widetilde{F}_{T,N} = \mathbb{R}^4 / P_T S_N \mathbb{Z}^4$$

is an N^2-sheeted holomorphic cover of F_T, with deck group $(\mathbb{Z}/N\mathbb{Z})^2$. After the cyclic cusp-base change $t = s^N$, these fibre covers assemble over the punctured cusp because

$$M_0^N \Lambda_N = \Lambda_N, \qquad S_N^{-1} M_0^N S_N = M_0. \tag{8.1}$$

Their systoles satisfy

$$\operatorname{sys}(\widetilde{F}_{T,N}) \geq \min\{T - C, N\}. \tag{8.2}$$

Consequently, for every sequence $T_j, N_j \to \infty$,

$$(\widetilde{F}_{T_j,N_j}, [0]) \longrightarrow (\mathbb{R}^4, 0)$$

in pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff topology.

Proof. The inclusion $P_T S_N \mathbb{Z}^4 \subset P_T \mathbb{Z}^4$ makes the identity on $\mathbb{R}^4$ descend to a finite holomorphic cover $\widetilde{F}_{T,N} \to F_T$. Its deck group is

$$\frac{P_T\mathbb{Z}^4}{P_T S_N \mathbb{Z}^4} \cong \frac{\mathbb{Z}^4}{S_N\mathbb{Z}^4} \cong (\mathbb{Z}/N\mathbb{Z})^2.$$

Write $M_0 = I + R$. The identity in equation (1.13) gives

$$R(a,b,c,d)^{\mathsf{T}} = (0,0,b,-a)^{\mathsf{T}}, \qquad R^2 = 0.$$

One turn around $s = 0$ after $t = s^N$ winds N times around the original cusp, so its monodromy is

$$M_0^N = (I+R)^N = I + NR.$$

In particular,

$$M_0^N \widehat{\gamma} = \widehat{\gamma} - N\widehat{\delta}, \qquad M_0^N \widehat{u} = \widehat{u} + N\widehat{w},$$

while $N\widehat{w}$ and $N\widehat{\delta}$ are fixed. This proves the first identity in equation (8.1); expressing the action in the displayed basis of Λ_N gives the second.

Reorder real coordinates as $(x_1, x_3; x_2, x_4)$ and put

$$A_T = \begin{pmatrix} 6\alpha_T & \eta_T \\ c_T & \alpha_T \end{pmatrix}.$$

Every vector of the covered period lattice has the exact form

$$\lambda_{T,N}(n,m) = \begin{pmatrix} A_T n + Nm \\ Y_T n \end{pmatrix}, \qquad n, m \in \mathbb{Z}^2.$$

If $n \neq 0$, then

$$\|\lambda_{T,N}(n,m)\| \geq \|Y_T n\| \geq (T - C)\|n\| \geq T - C.$$

If $n = 0$, then $m \neq 0$ and $\|\lambda_{T,N}(0,m)\| = N\|m\| \geq N$. This proves equation (8.2). Once the systole exceeds $4R$, theorem 1.2 says that the universal quotient sends $\overline{B}_R(0)$ isometrically onto the entire closed radius-R quotient ball. Letting R be arbitrary proves all three pointed convergences. □

Remark 8.2 (Why this route is ancillary). For $N = 2$, the fibre cover has four sheets after a twofold cusp-base cover. Sending $N \to \infty$ fills in the two momentum directions that remain periodic in the native $T^2 \times \mathbb{R}^2$ limit. This construction is useful for analysis on covers, but its source is an auxiliary covered cusp family. By contrast, the main theorems of this article derive their four-dimensional objects from smash powers of the original based candidate $X \cong S^6$, yielding the S^{12} and S^{24} source spheres without replacing the native family by a tower of covers.

9 Canonicity and exact conclusion

The lattice $K_\Lambda = \ker(M_0 - I) = \mathrm{im}(M_0 - I)$ and the quotient on the punctured cusp are intrinsic to the marked cusp monodromy. The compact map q_2 also depends on the chosen cusp coordinate, the closed cusp disc, and the decision to collapse its outer boundary and exterior. The map $q_{1,\mathrm{cusp}}$ has the analogous dependence. The map q_1 uses the invariant character over the full regular base, the two precise Seifert overlap maps, and the deliberate replacement of the entire cusp boundary torus by a cone. The based identifications of the smash powers with S^{12} and S^{24} depend on the based choice in equation (1.2).

Thus $q_2^{\wedge 2}$ is the exact two-copy construction from the native $\mathbb{T}^2 \times \mathbb{R}^2$ cusp limit. Its expanding quotient is cusp-local, but its source is the global real smash sphere S^{12}. The map $q_{1,\mathrm{cusp}}^{\wedge 4}$ is the corresponding localized one-circle construction. The stronger $q_1^{\wedge 4}$ uses the unique primitive invariant character over every regular fibre, the two exact finite Seifert fillings, and a cone at the nonnormal cusp. Two copies are minimal for the cusp-local rank-two route, while four copies are minimal for a full-base rank-four

expanding quotient, within the fixed-primitive- kernel, connected native torus-group quotients carrying the unscaled Euclidean quotient norm. No broader minimality is claimed.

Further quotienting gives the honest sphere maps

$$S^{12} \twoheadrightarrow S^6, \qquad S^{24} \twoheadrightarrow S^8.$$

Both full maps are based null-homotopic. For $\widehat{q}_\gamma^{\wedge 4}$, the smash square of $\widehat{q}_\gamma$ vanishes because $\pi_{12}(S^8) \cong \pi_4^{\mathrm{S}} = 0$. For $\widehat{q}_3^{\wedge 2}$, the stronger external-pairing calculation uses Toda's formula $[x \wedge y] = (E^3[x]) \circ (E^6[y])$. Since $\pi_6(S^3) = \mathbb{Z}/12$, the class $E^6[y] \in \pi_{12}(S^9) = \mathbb{Z}/24$ is even. Postcomposition by $E^3[x]$ is a homomorphism into $\pi_{12}(S^6) = \mathbb{Z}/2$, so it sends this even class to zero. Thus $x \wedge y \simeq_* 0$ for every based pair $x, y : S^6 \to S^3$, and all higher smash powers vanish as well. Their reduced joins vanish because join is suspended smash. Their Cartesian product maps vanish if and only if the corresponding one-copy map vanishes. The explicit framed filling now proves product nullity for $g = \widehat{q}_3$; it remains open for $f = \widehat{q}_\gamma$.

The one-copy class of $\widehat{q}_3$ is now zero by Theorem 10.1; that of $\widehat{q}_\gamma$ remains open. The general framing comparison retains the following exact structure. Its regular inverse image is T^3 with the honest normal framing supplied by the differential. Direct-limit stable framing components are indexed first by the spin torsor $H^1(T^3; \mathbb{Z}/2)$, and within a fixed spin component by the affine $H^3(T^3; \mathbb{Z})$-coordinate. An honest local rank-three twist maps to twice the generator of $\pi_3(SO)$ and changes the stable Pontryagin–Thom class by 2ν. Thus computing that relative integer modulo 12, not merely recognizing the unframed torus or its spin structure, computes $[\widehat{q}_3] \in \pi_6(S^3)$. The explicit cap in Theorem 10.1 carries out that computation and gives zero.

The based homotopy classes of $q_2^{\wedge 2}$, $q_{1,\mathrm{cusp}}^{\wedge 4}$, and $q_1^{\wedge 4}$ into their conically stratified targets remain open: postcomposition with a sphere quotient cannot be reversed. Surjectivity and the exact moving-slice restrictions survive the sphere-valued homotopical vanishing. Their targets contain moving embedded four-spheres whose intrinsic quotient metrics have exact Euclidean balls of every fixed radius sufficiently far into the cusp. Removing the smash point from each such sphere gives a nonclosed, topologically embedded $\mathbb{R}^4$ cell. Separately, the native radial four-tori admit the noncompact families $\mathfrak{E}_{12}$ and $\mathfrak{E}_{24}$, each a five-manifold with boundary $\mathbb{R}^4$. The continuous maps β_ν are homeomorphisms off that boundary and send it to the cone vertex, but they are nonproper and nonquotient; they are not topological blow-downs or resolutions. In a fixed cusp-adapted target metric, the same $\mathbb{R}^4$ appears as a vertical subspace of a moving-point bubble, but the full ambient bubbles have dimensions 8 and 12.

The $SU(3)$ clutching calculation is equally sharp. A class-k bundle on a moving S^4 has second Chern number k, and its pullback to the radial T^4 has the same number. When $k \neq 0$, it cannot extend over the ambient sphere or the contractible cone carrier. Conversely, every bundle pulled back through either null full sphere map is trivial. These statements neither construct nor obstruct the currently separately constructed Gram orbit space

$$\mathrm{Mat}_{3\times 3}(\mathbb{C})/U(3) \cong \mathrm{Herm}_3^+(\mathbb{C}), \qquad \mathrm{Herm}_3^+(\mathbb{C})^+ \cong B^9,$$

whose rank-deficient-and-infinity boundary is S^8. That S^8 is abstractly homeomorphic to the fourfold-smash target, and a chosen based homeomorphism gives a proved null composite from S^{24}. No structure-preserving refinement of that homeomorphism is presently constructed. The Gram carrier is not constructed as a product, smash, or join of X; its orbit map is not a principal $SU(3)$-bundle; and no period-lattice, connection, or fixed metric comparison map has been given. Its boundary inclusion into B^9 is null because the ball is contractible, but the boundary remains an S^8. No smoothing of this homotopy S^8 is almost complex, which excludes exactly that additional tangent-bundle structure and not the topological B^9/S^8 carrier or a different future bridge.

These statements do not produce a compact complex S^{12} or S^{24}, an unrescaled ambient $\mathbb{R}^4$ limit of a compact target, a Riemannian metric on a source sphere, or a physical spacetime. The finite-cover

tower of section 8 also has pointed limit $\mathbb{R}^4$, but it is logically separate from the sphere route. A global compactification of that auxiliary tower would require a specified global base cover, normalization and resolution of all pulled-back singular fibres, a lifted metric, and a new topology calculation; none is asserted here. Likewise, any proposed bridge to the Gram carrier still requires an explicit map respecting its rank strata and a separate proof of every bundle and metric compatibility claimed for that map. Its absence is the present proof boundary; it is not an ontological conclusion about the carrier or its possible physical interpretation.

10 The original cusp coefficient: an explicit framed nullbordism

Let X be the smooth compact source of the original construction, $f : X \to B$ its map, and t the specified cusp coordinate. Put $W = f^{-1}(0)$, choose the original $x_0 \in W$, and retain the ordered quotient lattice basis induced by $(\widehat{\gamma}, \widehat{u})$, with kernel generated by $(\widehat{w}, \widehat{\delta})$. Thus the quotient torus is $\overline{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$. The original periods are

$$Z(s) = sB_0 + C(t), \qquad e^{2\pi i s} = t, \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C(t) = \begin{pmatrix} 6\mu(t) & h(t) \\ b(t) - h(t) & \mu(t) \end{pmatrix}. \tag{10.1}$$

Every entry of C is holomorphic at $t = 0$. The dense torus of the original toric cusp has coordinates (x_1, x_2, t), with $x = \exp(2\pi i \zeta)$. The residual lattice action is exactly

$$\Psi_\lambda(x, t) = \left(t^{B_0 \lambda} \exp(2\pi i C(t)\lambda)x, t\right), \qquad \lambda \in \mathbb{Z}^2. \tag{10.2}$$

It extends freely and properly discontinuously across the original unimodular toric filling. In particular the quotient projection is a local biholomorphism at every toric fixed point. These are the reconstructed data and proved local properties of the original source [Alpöge, 2026, §4, source pp. 24–29].

Let

$$K^{(1)} = (\mathbb{R}/\mathbb{Z} \times \{0\}) \cup (\{0\} \times \mathbb{R}/\mathbb{Z}), \qquad c : \overline{T}^2 \longrightarrow \overline{T}^2/K^{(1)} \cong S^2$$

be the specified degree-one cell quotient. Its restriction to the open two-cell is taken with the smooth coordinates used in the original regular-fibre description. Both radial endpoints and the complete basepoint cylinder are collapsed in

$$\Sigma_r(S^2, *) = \frac{[0, \epsilon] \times S^2}{(\{0, \epsilon\} \times S^2) \cup ([0, \epsilon] \times \{*\})} \cong S^3.$$

The original based map is

$$g(x) = [|t|, c(\overline{\pi}_f(x))] \quad (0 < |t| < \epsilon), \qquad g(W) = g(X \setminus \operatorname{int} N_0) = *. \tag{10.3}$$

Theorem 10.1 (The original one-copy cusp map is null-homotopic)**.** *The based map $g : (X, x_0) \to (S^3, *)$ in (10.3) is based null-homotopic. More precisely, a regular inverse image with its actual ordered normal framing induced by (r, c_1, c_2) is framed nullbordant inside one original cusp chart times $[0, 1]$, disjoint from $\{x_0\} \times [0, 1]$.*

Proof. The proof constructs the complete inverse image, compares its actual framing with an explicitly extendible framing, and gives the extension.

1. The exact complete inverse image. Retain the lift

$$u^0 = \begin{pmatrix} -1/4 \\ 1/4 \end{pmatrix}, \qquad \overline{u}^0 = (3/4, 1/4) \pmod{\mathbb{Z}^2}, \qquad B_0 u^0 = \begin{pmatrix} 1/4 \\ 1/4 \end{pmatrix}.$$

Thus $\overline{u}^0 \notin K^{(1)}$, and $B_0 u^0$ lies in the interior of the original triangle $\operatorname{conv}(0, e_1, e_2)$. Use its original unimodular chart

$$z_0 = t/(x_1 x_2), \qquad z_1 = x_1, \qquad z_2 = x_2, \qquad t = z_0 z_1 z_2. \tag{10.4}$$

Set

$$\ell = \log|t| < 0, \qquad R(t) = -2\pi \operatorname{Im} C(t), \qquad A(t) = \ell B_0 + R(t).$$

For a representative $\zeta = Z(s)u + k$, where $u, k \in \mathbb{R}^2$, $\operatorname{Im} s = -\ell/(2\pi)$ gives the exact identity

$$\begin{pmatrix} \log|x_1| \\ \log|x_2| \end{pmatrix} = A(t)u. \tag{10.5}$$

The quotient value is $u \bmod \mathbb{Z}^2$. Since B_0 is orthogonal, for all $v \in \mathbb{R}^2$,

$$\|A(t)v\| \geq (|\ell| - \|R(t)\|)\|v\|. \tag{10.6}$$

The boundedness of R therefore makes $A(t)$ invertible on a sufficiently small punctured disc.

For any point of the fibre quotient over $\overline{u}^0$, subtracting the appropriate original period, equivalently applying $\Psi_{-\lambda}$, gives a representative with the exact lift $u = u^0$. This representative is unique: another residual translate has logarithmic moduli differing by $A(t)\lambda$, which vanishes only for $\lambda = 0$. Conversely every dense-torus point satisfying $\log|x| = A(t)u^0$ has this quotient value, since the remaining real parts of ζ are exactly the kernel coordinates k. For each fixed t, varying $k \bmod \mathbb{Z}^2$ gives all independent phases of x_1, x_2.

The following change of coordinates uses the full original period units:

$$h_j^0(t) = \exp(-2\pi i (C(t)u^0)_j), \qquad j = 1, 2, \tag{10.7}$$

$$\mathcal{F}(z_0, z_1, z_2) = (w_0, w_1, w_2) = \left(\frac{z_0}{h_1^0(t) h_2^0(t)},\, h_1^0(t) z_1,\, h_2^0(t) z_2 \right). \tag{10.8}$$

The functions h_j^0 are nowhere-zero holomorphic units, including at $t = 0$. The product $w_0 w_1 w_2$ is exactly t. The inverse of $\mathcal{F}$ is

$$(w_0, w_1, w_2) \longmapsto \left(w_0 h_1^0(t) h_2^0(t),\, \frac{w_1}{h_1^0(t)},\, \frac{w_2}{h_2^0(t)} \right), \qquad t = w_0 w_1 w_2.$$

Thus this is a biholomorphism of the original chart restricted to the cusp disc, not a replacement of $C(t)$.

For $r_0 \in (0, \epsilon)$, the entire inverse image of $y_0 = [r_0, c(\overline{u}^0)]$ becomes exactly

$$T_{r_0} = \{|w_0| = \rho_0,\ |w_1| = \rho_1,\ |w_2| = \rho_2\}, \qquad (\rho_0, \rho_1, \rho_2) = (r_0^{1/2}, r_0^{1/4}, r_0^{1/4}). \tag{10.9}$$

Indeed, $\log|h^0(t)| = 2\pi \operatorname{Im} C(t)u^0 = -R(t)u^0$, so (10.5) gives $\log|w_j| = \ell(B_0 u^0)_j$ for $j = 1, 2$; the product identity gives the exponent $1/2$ for w_0. Conversely these moduli recover $u = u^0$ by invertibility of $A(t)$. The angular map is explicitly

$$(\arg t, \arg w_1, \arg w_2) \longleftrightarrow (\arg w_0, \arg w_1, \arg w_2), \qquad \arg w_0 = \arg t - \arg w_1 - \arg w_2 \pmod{2\pi},$$

with inverse $\arg t = \arg w_0 + \arg w_1 + \arg w_2$. Its integral matrix has determinant one. This includes the complete angular mapping torus and all kernel phases, with their original signs.

Let o be the origin of (10.4). The original free properly discontinuous action supplies a neighborhood V of o on which the quotient projection is injective. To see the local injectivity directly, first choose a relatively compact neighborhood; proper discontinuity leaves only finitely many translates that can intersect it. None of those nonidentity translates fixes o. Shrinking finitely many times separates the neighborhood from all of them. The quotient map is a local biholomorphism, so its restriction to V is a chart of the original source. Choose a polydisc about 0 whose closure lies in $\mathcal{F}(V)$. For sufficiently small r_0, both T_{r_0} and

$$\{|w_0| \le \rho_0,\ |w_1| = \rho_1,\ |w_2| = \rho_2\} \tag{10.10}$$

lie in this polydisc. This proves that the whole inverse image embeds in one chart and supplies space for the bounding manifold below.

2. The actual normal framing, including its angle dependence. Put $L_j = \log|w_j|$, and use the unique real lift of the quotient coordinate near u^0. From the exact change of coordinates,

$$\begin{pmatrix} L_1 \\ L_2 \end{pmatrix} = \ell B_0 u^0 + A(t)(u - u^0), \qquad L_0 = \ell - L_1 - L_2. \tag{10.11}$$

Differentiating on $u = u^0$ annihilates the term $(dA(t))(u - u^0)$. Hence no derivative of $C(t)$ has been omitted: the exact differential there is

$$\begin{pmatrix} dL_0 \\ dL_1 \\ dL_2 \end{pmatrix} = J(t) \begin{pmatrix} d\ell \\ du_1 \\ du_2 \end{pmatrix}, \quad J(t) = \begin{pmatrix} 1/2 & -A_{11} - A_{21} & -A_{12} - A_{22} \\ 1/4 & A_{11} & A_{12} \\ 1/4 & A_{21} & A_{22} \end{pmatrix}. \tag{10.12}$$

Adding the second and third rows to the first gives $(1, 0, 0)$, and expansion in that row proves $\det J(t) = \det A(t)$.

Let $D = Dc_{\overline{u}^0}$ be the actual two-by-two derivative in the ordered open-cell target coordinates (c_1, c_2). It is a fixed invertible matrix; all its entries and signs are retained. Let

$$H = \begin{pmatrix} r_0 & 0 & 0 \\ 0 & D_{11} & D_{12} \\ 0 & D_{21} & D_{22} \end{pmatrix}.$$

Let n_j be the real radial vector w_j in the j-th complex coordinate, with other components zero. Then $dL_i(n_j) = \delta_{ij}$. Writing $N = (n_0, n_1, n_2)$ as the matrix with these vector columns, the actual dg-dual normal framing is, in the normal quotient bundle,

$$\mathfrak{n}(t) = NJ(t)H^{-1}. \tag{10.13}$$

Indeed, $dg(N) = HJ(t)^{-1}$, so multiplication by $J(t)H^{-1}$ gives the identity. In the Euclidean metric of this chart the radial vectors are perpendicular to the product torus; a different source metric changes their representatives by the canonical projection to its normal bundle and preserves this normal-quotient framing.

Put $\ell_0 = \log r_0$. Decrease r_0 further, if necessary, so that $|\ell_0| > 2\sup_{|t|\le r_0}\|R(t)\|$. For $0 \le \sigma \le 1$, define only a homotopy of frames by

$$A_\sigma(t) = \ell_0 B_0 + \sigma R(t),$$

and let $J_\sigma(t)$ be (10.12) with A_σ in place of A. For every $v \in \mathbb{R}^2$,

$$\|A_\sigma(t)v\| \ge (|\ell_0| - \|R(t)\|)\|v\| > 0 \quad (v \ne 0).$$

Thus all matrices are invertible. At $\sigma = 0$,

$$J_0 = \begin{pmatrix} 1/2 & \ell_0 & -\ell_0 \\ 1/4 & 0 & \ell_0 \\ 1/4 & -\ell_0 & 0 \end{pmatrix}, \qquad \det J_0 = \ell_0^2 > 0. \tag{10.14}$$

Continuity also proves $\det J_\sigma(t) > 0$ for every σ, t. The homotopy $NJ_\sigma(t)H^{-1}$, run from $\sigma = 1$ to $\sigma = 0$, compares the actual frame with the exact constant-matrix frame NP, where

$$P = J_0 H^{-1}. \tag{10.15}$$

The original embedded torus and original functions $C(t)$ have not changed. A framing homotopy is a framed cylinder: choose a smooth parameter function constant near its endpoints and use the displayed frames on $T_{r_0} \times [0, 1]$.

3. A collared framed nullbordism with the exact constant matrix. Use a new time coordinate $v \in [0, 1]$. Define

$$q(v) = \begin{cases} 0, & 0 \le v \le 1/4, \\ \rho_0^2 \exp(4 - 1/(v - 1/4)), & 1/4 < v \le 1. \end{cases} \tag{10.16}$$

This function is smooth: all right derivatives of $\exp(-1/(v - 1/4))$ at $1/4$ vanish. It is strictly increasing for $v > 1/4$, satisfies $q(1/2) = \rho_0^2$, and $q'(1/2) = 16\rho_0^2 > 0$. In the chart times $[0, 1]$, put

$$\mathcal{W} = \{|w_0|^2 + q(v) = \rho_0^2, \quad |w_1|^2 = \rho_1^2, \quad |w_2|^2 = \rho_2^2\}. \tag{10.17}$$

No point has $v > 1/2$. The three defining differentials are independent: the latter two have nonzero gradients in their separate complex factors; the first has nonzero w_0-gradient whenever $w_0 \neq 0$, and at $w_0 = 0$ it has the nonzero v-derivative $q'(1/2)$. Thus $\mathcal{W}$ is a smooth compact four-manifold. Its only boundary is $T_{r_0} \times \{0\}$, and on $0 \le v \le 1/4$ it is the exact product collar $T_{r_0} \times [0, 1/4]$. The north locus $w_0 = 0, v = 1/2$ is an interior cap: the implicit function theorem expresses v smoothly in terms of $(\operatorname{Re} w_0, \operatorname{Im} w_0)$ there. The first factor is a disk obtained by rotating its radial profile with this smooth cap and one boundary circle. Therefore $\mathcal{W} \cong D^2 \times T^2$. Its spatial coordinates lie in (10.10), so the complete manifold lies in the injective original chart.

In the Euclidean chart product metric define

$$V_0 = (w_0, 0, 0, q'(v)/2), \qquad V_1 = (0, w_1, 0, 0), \qquad V_2 = (0, 0, w_2, 0).$$

These are one half of the gradients of the three defining equations in (10.17). They are mutually orthogonal and never zero. For example, if $F_0 = |w_0|^2 + q(v) - \rho_0^2$, then

$$dF_0(V_0) = 2|w_0|^2 + \frac{1}{2} q'(v)^2 > 0$$

at every point of $\mathcal{W}$. On the entire collar $q' = 0$, so $(V_0, V_1, V_2) = (N, 0)$ exactly. Consequently

$$(V_0, V_1, V_2)P \tag{10.18}$$

is a framing of the normal three-plane bundle of $\mathcal{W}$, and its restriction to the collar is exactly the framing NP from (10.15), with every original target-coordinate sign retained. Concatenate the framing cylinder of the preceding paragraph with this nullbordism, rescaling the two interval lengths. Their framings agree exactly on the common product collar, so the concatenation is a smooth framed nullbordism of the actual inverse image.

4. The based map represented by this framed inverse image. Choose the injective chart to exclude x_0 if its center is not x_0. If its center is x_0, the constructed nullbordism still excludes x_0, since $|w_1| = \rho_1 > 0$ and $|w_2| = \rho_2 > 0$ everywhere on it. Compactness therefore gives tubular neighborhoods for the framed cylinder and cap disjoint from $\{x_0\} \times [0, 1]$.

The original continuous quotient map is smooth and a submersion near its entire inverse image of y_0. First make it constant in a neighborhood of x_0, by a based homotopy that changes nothing near that inverse image. Explicitly, take a target chart $\kappa : O \to \mathbb{R}^3$ with $\kappa(*) = 0$, whose image is a ball, and $y_0 \notin O$. By continuity choose a neighborhood U of x_0 with $g(U) \subset O$, and a smooth bump function β supported in U and equal to one near x_0. For $x \in U$, replace $g(x)$ during this homotopy by $\kappa^{-1}((1 - a\beta(x))\kappa(g(x)))$, $0 \le a \le 1$, and leave g fixed off U. Convexity of the target chart image makes the formula well-defined, the vanishing of β near the boundary of its support proves continuity of the pasted formula, and $\kappa(g(x_0)) = 0$ makes the homotopy based. It never introduces an inverse image of y_0. Its endpoint is smooth near x_0 and agrees with g near the entire old inverse image.

Denote the endpoint of the preceding homotopy by g_b. Choose one closed tubular neighborhood U_1 of the inverse image, contained in the smooth submersion region and with the entire inverse image contained in $\operatorname{int} U_1$. The compact set $X \setminus \operatorname{int} U_1$ has image under g_b missing y_0. In a fixed target metric put

$$d_0 = \min_{x \in X \setminus \operatorname{int} U_1} d(g_b(x), y_0) > 0.$$

Relative smooth approximation, fixed on the whole closed set U_1 and on a neighborhood of x_0, gives a smooth based representative g_{sm} whose error on $X \setminus \operatorname{int} U_1$ is less than $d_0/2$. It therefore has the same inverse image and differential on U_1, and $d(g_{\mathrm{sm}}(x), y_0) > d_0/2$ on its complement. Thus there is no uncontrolled intervening region and no additional inverse image. Make the approximation also closer than a fixed target injectivity radius; its short geodesics give the based homotopy.

The relative Pontryagin construction now applies to the explicit framed nullbordism. It produces a based homotopy to a map missing y_0; stereographic contraction of $S^3 \setminus \{y_0\} \cong \mathbb{R}^3$ to the image of $*$ is based and contracts that last map. For completeness, the dependence on the framed inverse image in this argument is precisely the local product-neighborhood construction and the interpolation lemma of [Milnor, 1965, §7, Theorem B and Lemma 4, pp. 43, 48–50]. Their relative versions fix the source basepoint because all tubular modifications are away from it and stereographic interpolation between equal values keeps that value fixed. A framed cylinder realizes a framing homotopy; the empty Pontryagin manifold realizes a map missing the chosen regular value. This proves the claimed based null-homotopy of the exact map g. □

Corollary 10.2. *Under the original fixed based identification $X \cong S^6$, the one-copy coefficient is*

$$[g] = 0 \in \pi_6(S^3) \cong \mathbb{Z}/12, \qquad a = 0 \pmod{12}, \qquad [E^6 g] = 0 \in \pi_{12}(S^9) \cong \mathbb{Z}/24.$$

The previously defined homomorphism $\Theta_g : \mathbb{Z}/12 \to \pi_6(S^3)$, $\Theta_g([n]) = n[g]$, and its composite with the marked peripheral character are zero.

Proof. The theorem states that the actual based map is null-homotopic. Precomposition with the fixed based identification preserves a based null-homotopy. Suspension preserves it as well. Thus the unique coefficient of $[g]$ in any generator of $\mathbb{Z}/12$ is zero. For every $[n]$, $\Theta_g([n]) = n \cdot 0 = 0$, and every composite with this homomorphism is zero. □

Exact scope. The proof computes this sphere-valued postcomposition. It does not compute the homotopy class of the earlier conical-target map $q_2 : X \to \operatorname{Susp}(S^1 \times \overline{T}^2)$, and it does not assert a null-homotopy for that different target. Point-set surjectivity of g and all of its stated metric restrictions

remain properties of the original map: a homotopy-class computation does not alter the defining formula (10.3).

11 The retained-angle cusp quotient and its nonzero top-cell class

The exact object. The notation refers to the specified threefold X, its fixed based smooth identification with S^6, and its cusp filling from [Alpöge, 2026, §§3.7,4.1–4.5]. The native quotient and its compact extension are the maps in Cusp sphere-route companion [2026]; their full formulas are recalled below. This calculation uses the original period matrix and native period lattices. It computes a new postcomposition of that particular quotient; it does not use the homotopy class of the angle-forgetting map $g : X \to S^3$.

11.1 Original coordinates and the specified sphere-valued map

The ordered quotient lattice is

$$\overline{\Lambda} = \mathbb{Z}\langle \widehat{\overline{\gamma}}, \widehat{\overline{u}} \rangle, \qquad K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle, \qquad \overline{T}^2 = \mathbb{R}^2/\mathbb{Z}^2.$$

In these ordered bases the original period data are

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Z(s) = sB_0 + C(t), \qquad t = e^{2\pi i s}, \qquad C(t) = \begin{pmatrix} 6\mu(t) & h_{\mathrm{per}}(t) \\ b(t) - h_{\mathrm{per}}(t) & \mu(t) \end{pmatrix}. \tag{11.1}$$

Here $b = \beta + \tau$, $s = \tau - h_{\mathrm{per}}(t)$, and every entry of C is holomorphic at $t = 0$, exactly as in the original source. The subscript in h_{per} distinguishes that period function from the sphere-valued map below. Write

$$R(t) = -2\pi \operatorname{Im} C(t), \qquad \ell = \log|t|, \qquad A(t) = \ell B_0 + R(t). \tag{11.2}$$

Choose a closed cusp disc $|t| \le \epsilon$ inside the original coordinate disc and small enough for the quotient construction. On a slightly larger closed disc put $M = \sup \|R(t)\| < \infty$. We will choose a regular value with $0 < r_0 < \epsilon$ and $|\log r_0| > M$. This chooses a point of the already specified map; it does not change that map or the cutoff radius ϵ.

The original covering model on its dense torus is

$$x_j = e^{2\pi i \zeta_j}, \qquad \Psi_n(x,t) = (e^{2\pi i C(t)n} t^{B_0 n} x, t), \qquad n \in \mathbb{Z}^2. \tag{11.3}$$

For a branch of s, writing $\zeta = Z(s)u + k$ with $u, k \in \mathbb{R}^2$ gives

$$\log|x| = A(t)u. \tag{11.4}$$

Indeed, $\operatorname{Im} s = -\ell/(2\pi)$, so $-2\pi \operatorname{Im}(Z(s)u + k) = \ell B_0 u + R(t)u$. Since B_0 is orthogonal,

$$\big|(\ell B_0 + vR(t))\xi\big| \ge (|\ell| - vM)|\xi| > 0 \quad (0 \le v \le 1,\ \xi \ne 0) \tag{11.5}$$

whenever $|\ell| > M$. Thus $A(t)$ is invertible there. Equation (11.3) changes $A(t)^{-1}\log|x|$ by n, so the native quotient is exactly

$$\overline{\pi}_f([x,t]) = [A(t)^{-1}\log|x|] \in \mathbb{R}^2/\mathbb{Z}^2. \tag{11.6}$$

This also follows from the flat marking in Cusp sphere-route companion [2026].

Let

$$T^3 = S^1_\phi \times \overline{T}^2, \qquad S^1_\phi = \mathbb{R}/(2\pi\mathbb{Z}),$$

with coordinate basepoints $\phi = 0$, $u_1 = 0$, $u_2 = 0$. Its coordinate two-skeleton is the following specified closed subset:

$$K^{(2)} = (\{0\} \times \overline{T}^2) \cup (S^1_\phi \times \{u_1 = 0\}) \cup (S^1_\phi \times \{u_2 = 0\}). \tag{11.7}$$

The product CW structure has one open three-cell, with coordinates $(\phi, u_1, u_2) \in (0, 2\pi) \times (0, 1)^2$. Consequently

$$c_3 : T^3 \longrightarrow T^3/K^{(2)} \cong S^3$$

is the degree-one quotient, where the orientation of S^3 is fixed by the displayed coordinate order. One explicit identification of its open cell with $\mathbb{R}^3$ is

$$(\phi, u_1, u_2) \longmapsto (-\cot(\phi/2), -\cot(\pi u_1), -\cot(\pi u_2)).$$

Each derivative in its own variable is strictly positive, and approach to any boundary face leaves every compact subset of $\mathbb{R}^3$. Extending by its point at infinity gives the stated based sphere identification and verifies its continuity on the collapsed skeleton.

The exact conical target and map are

$$\mathcal{A} = \frac{[0, \epsilon] \times T^3}{\{0\} \times T^3 \sim v_0, \quad \{\epsilon\} \times T^3 \sim v_1}, \tag{11.8}$$

$$q_2(x) = [r, e^{i\phi}, \overline{\pi}_f(x)] \qquad (t(f(x)) = re^{i\phi},\ 0 < r < \epsilon),$$

$$q_2(W) = v_0, \qquad q_2(X \setminus \operatorname{int} N_0) = v_1. \tag{11.9}$$

Fix the same source basepoint $x_{\text{base}} \in W$. Define

$$p_3 : \mathcal{A} \longrightarrow \Sigma_r(S^3, *), \qquad p_3([r, z]) = [r, c_3(z)], \qquad H = p_3 q_2 : (X, x_{\text{base}}) \longrightarrow (S^4, *). \tag{11.10}$$

The reduced suspension here collapses both endpoint copies of S^3 and the entire interval over $*$ to one point. Thus p_3 sends both v_0 and v_1 to $*$ and is well-defined and based. It is continuous by the defining quotient topology. An explicit identification of its open four-cell with $\mathbb{R}^4$ is

$$(r, \phi, u_1, u_2) \longmapsto (-\cot(\pi r/\epsilon), -\cot(\phi/2), -\cot(\pi u_1), -\cot(\pi u_2)). \tag{11.11}$$

This fixes the target orientation. Replacing r locally by $\ell = \log r$ preserves this orientation, since $dr/d\ell = r > 0$. The ordered coordinates in (11.11) specify the sphere identification used throughout this calculation.

Theorem 11.1. *For the exact map (11.10),*

$$[H] = \eta_4 \circ \eta_5 \neq 0 \quad \textit{in} \quad \pi_6(S^4) \cong \mathbb{Z}/2, \tag{11.12}$$

where $\eta_n : S^{n+1} \to S^n$ is the $(n-2)$-fold reduced suspension of the complex Hopf map. In particular q_2 is not based null-homotopic. Its stabilized top-cell postcomposition is the element $\eta^2 \in \pi_2^{\mathrm{S}}$.

11.2 The entire regular inverse image lies in one original chart

Choose the actual lift

$$u_0 = (-1/4, 1/4)^{\mathsf{T}}, \qquad p_0 = B_0 u_0 = (1/4, 1/4)^{\mathsf{T}}, \qquad \phi_0 = \pi. \tag{11.13}$$

Modulo $\mathbb{Z}^2$, u_0 is $(3/4, 1/4)$ in the open cell. Put $t_0 = r_0 e^{i\phi_0}$ and $y_0 = [r_0, c_3(e^{i\phi_0}, [u_0])]$. The numbers r_0 and t_0 will remain nonzero throughout the framing calculation.

The original triangle $\{0, e_1, e_2\}$ has toric chart

$$\mathcal{U}_\sigma \cong \mathbb{C}^3, \qquad (z_0, z_1, z_2) = (t/(x_1x_2), x_1, x_2), \qquad t = z_0z_1z_2. \tag{11.14}$$

These characters are dual to $(0,0,1), (1,0,1), (0,1,1)$, respectively, as direct pairing verifies. This is the chart of [Alpöge, 2026, Lemma 4.2(iii)], with its original coordinates.

For $j = 1, 2$, introduce the nowhere vanishing holomorphic functions

$$h_j(t) = \exp(-2\pi i(C(t)u_0)_j). \tag{11.15}$$

The following is an exact coordinate map, with the exact inverse displayed alongside it:

$$F(z) = (w_0, w_1, w_2) = \big(z_0/(h_1(t)h_2(t)),\ h_1(t)z_1,\ h_2(t)z_2\big), \tag{11.16}$$

$$F^{-1}(w) = \big(w_0h_1(t)h_2(t),\ w_1/h_1(t),\ w_2/h_2(t)\big), \quad t = w_0w_1w_2. \tag{11.17}$$

Both maps are holomorphic wherever $|t|$ is in the period disc, both preserve t, and substitution proves that both composites are the identity. In particular the maps extend across the origin and across every coordinate divisor. No period matrix is replaced by a different matrix in this coordinate change.

Lemma 11.2. *For sufficiently small $r_0 > 0$, the whole inverse image $M = H^{-1}(y_0)$ is represented injectively inside one contractible coordinate neighborhood in X. In its w-coordinates it is exactly the embedded two-torus*

$$w(a,b) = (\rho_0 e^{i(\phi_0 - a - b)}, \rho_1 e^{ia}, \rho_2 e^{ib}), \qquad \rho_0 = r_0^{1/2}, \quad \rho_1 = \rho_2 = r_0^{1/4}, \quad (a,b) \in \big(\mathbb{R}/(2\pi\mathbb{Z})\big)^2. \tag{11.18}$$

Proof. The point y_0 lies off the collapsed subset of the target, so (11.10) and (11.9) imply

$$M = \{x : t(f(x)) = t_0,\ \overline{\pi}_f(x) = [u_0]\}.$$

Any lift of such a point to $(\mathbb{C}^*)^2 \times \{t_0\}$ has $A(t_0)^{-1} \log|x| = u_0 + n$ for one $n \in \mathbb{Z}^2$. Applying the actual transformation Ψ_{-n} from (11.3) produces a lift with $\log|x| = A(t_0)u_0$. If two such lifts represent the same point, Ψ_n relates them and changes their lifted quotient coordinate by n; hence $n = 0$, and the lifts are identical. This proves both completeness and injectivity of this chosen representative.

Since $\log|h_j(t)| = 2\pi \operatorname{Im}(C(t)u_0)_j = -(R(t)u_0)_j$, its w-coordinates satisfy

$$(\log|w_1|, \log|w_2|)^{\mathsf{T}} = A(t_0)u_0 - R(t_0)u_0 = \ell_0 p_0, \qquad \ell_0 = \log r_0.$$

Thus $|w_1| = |w_2| = r_0^{1/4}$, and the unchanged product $w_0w_1w_2 = t_0$ forces $|w_0| = r_0^{1/2}$ and the displayed phase. Conversely every pair (a,b) in (11.18) gives one such lift. The original kernel coordinates k_j had period one in $x_j = e^{2\pi i\zeta_j}$; the phase coordinates a, b have period 2π, with no cover or change of lattice.

The covering $Y_\epsilon \to N_0$ is injective on a neighborhood of the origin in (11.14), because the original action is free and properly discontinuous. Choose a ball V centered at that origin with closure inside such a neighborhood and the period disc. Choose a ball V_w about the origin with closure contained in $F(V)$. Every coordinate in (11.18) tends uniformly to zero as $r_0 \downarrow 0$. Therefore for sufficiently small r_0 the entire torus lies in V_w. The inverse image of V_w under F, followed by the injective covering projection, is the required contractible chart in X. □

11.3 The full differential and its ordered normal framing

We calculate in a neighborhood of the compact torus (11.18). The quotient coordinates admit a unique local lift near u_0. Set

$$v = (\log|w_1|, \log|w_2|)^{\mathsf{T}}, \quad \ell = \log|w_0w_1w_2|, \quad \phi = \arg(w_0w_1w_2)$$

using the branch of ϕ near $\phi_0 = \pi$. Equations (11.4) and (11.16) give the exact identity

$$v = \ell B_0 u + R(t)(u - u_0), \qquad u = u_0 + A(t)^{-1}(v - \ell p_0). \tag{11.19}$$

Differentiate this identity before restricting to the inverse image:

$$du = -A(t)^{-1}(dA(t))A(t)^{-1}(v - \ell p_0) + A(t)^{-1}(dv - p_0\, d\ell). \tag{11.20}$$

On M the vector $v - \ell p_0$ is exactly zero. Therefore

$$du|_M = A_0^{-1}(dv - p_0\, d\ell), \qquad A_0 = A(t_0). \tag{11.21}$$

This explicitly accounts for all derivatives of the original bounded period terms; none are omitted or estimated away.

Identify $T_w\mathbb{C}^3$ with $\mathbb{C}^3$ in the given coordinate chart, and define real tangent vectors

$$L_j = (0, \dots, w_j, \dots, 0), \qquad Q_j = (0, \dots, iw_j, \dots, 0) \qquad (0 \le j \le 2). \tag{11.22}$$

Thus $d\log|w_i|(L_j) = \delta_{ij}$, $d\arg w_i(Q_j) = \delta_{ij}$, and the crossed evaluations vanish. The ordered tangent vectors of the parametrization (11.18) are

$$T_a = -Q_0 + Q_1, \qquad T_b = -Q_0 + Q_2. \tag{11.23}$$

They are independent because their w_1 and w_2 components, respectively, are nonzero. For the target order (ℓ, ϕ, u_1, u_2), define the following four lifts of its coordinate vectors:

$$N_\ell = \tfrac{1}{2}L_0 + \tfrac{1}{4}L_1 + \tfrac{1}{4}L_2, \qquad N_\phi = Q_0, \tag{11.24}$$

$$N_i = -\big((A_0)_{1i} + (A_0)_{2i}\big)L_0 + (A_0)_{1i}L_1 + (A_0)_{2i}L_2 \qquad (i = 1, 2). \tag{11.25}$$

Let $P = (\ell, \phi, u_1, u_2)$. Evaluation using (11.21) gives the entire differential table

	N_ℓ	N_ϕ	N_1	N_2	T_a	T_b
$d\ell$	1	0	0	0	0	0
$d\phi$	0	1	0	0	0	0
du_1	0	0	1	0	0	0
du_2	0	0	0	1	0	0

. (11.26)

In particular P, and therefore H in its open target cell, is a submersion along M. The classes of $(N_\ell, N_\phi, N_1, N_2)$ in $TX|_M/TM$ give the specified normal four-framing. Orthogonal representatives are not required for this quotient-bundle definition; projecting the displayed lifts along TM to any chosen orthogonal complement preserves the quotient frame.

If the normal frame is specified using the literal four real coordinates in (11.11), their differential at the chosen value is

$$dH_{\text{cell}} = K\, dP, \qquad K = \operatorname{diag}\left(\frac{\pi r_0}{\epsilon}\csc^2\frac{\pi r_0}{\epsilon}, \frac{1}{2}, 2\pi, 2\pi\right).$$

The first factor includes $dr = r_0\, d\ell$, the second uses $\phi_0 = \pi$, and the final two use the actual representatives $(3/4, 1/4)$ of u_0. Thus, writing $\mathcal{N} = (N_\ell, N_\phi, N_1, N_2)$, the literal dual frame is $\mathcal{N}K^{-1}$. All four entries of K are positive constants on M. The explicitly positive diagonal matrices $(1 - v)K^{-1} + vI_4$, $0 \le v \le 1$, give a homotopy from that literal frame to $\mathcal{N}$. This retains the original target coordinates and establishes that either coordinate description gives the same framed bordism class.

For completeness, the original period matrix admits the following explicit homotopy of this normal framing. Put

$$A_{0,v} = \ell_0 B_0 + vR(t_0), \qquad 0 \le v \le 1, \tag{11.27}$$

and replace A_0 by $A_{0,v}$ only in (11.25). Equation (11.5) proves invertibility for every v. Their images under the actual (du_1, du_2) are the columns of $A_0^{-1}A_{0,v}$, so these are normal frames throughout the homotopy. This is a homotopy of the computed frame on the same embedded torus, not a replacement of the original degeneration by a different one.

Lemma 11.3 (The exact orientation sign)**.** *Use the complex real-coordinate orientation, whose ordered coordinates are*

$$(\operatorname{Re} w_0, \operatorname{Im} w_0, \operatorname{Re} w_1, \operatorname{Im} w_1, \operatorname{Re} w_2, \operatorname{Im} w_2).$$

Then one has

$$\det(N_\ell, N_\phi, N_1, N_2, T_a, T_b) = -\det(A_0)|w_0 w_1 w_2|^2 < 0. \tag{11.28}$$

Hence the tangent orientation induced by the ordered normal frame is the orientation of $(-T_a, T_b)$.

Proof. The ordered frame $(L_0, Q_0, L_1, Q_1, L_2, Q_2)$ has determinant $|w_0|^2|w_1|^2|w_2|^2 > 0$ in real coordinates. Regroup its rows as $(L_0, L_1, L_2, Q_0, Q_1, Q_2)$; this permutation has three inversions and sign -1. Regroup the columns of the frame in (11.28) as $(N_\ell, N_1, N_2, N_\phi, T_a, T_b)$; this has two inversions and sign $+1$. The angular block is $\left(\begin{smallmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, of determinant one. The radial block has first column $(1/2, 1/4, 1/4)^{\mathsf{T}}$ and the other columns given by (11.25). Adding its last two rows to its first row gives first row $(1, 0, 0)$, with remaining lower-right block A_0. Its determinant is $\det A_0$. This proves the equality. Equation (11.27) joins A_0 to $\ell_0 B_0$ through invertible matrices, so $\operatorname{sign}\det A_0 = \operatorname{sign}\det(\ell_0 B_0) = +1$. The inequality and the tangent orientation follow. □

11.4 An explicit contraction of the full frame

Define a complex-linear matrix, also regarded as a real-linear matrix,

$$D(a, b) = \operatorname{diag}(e^{-i(a+b)}, e^{ia}, e^{ib}) \in \mathrm{SU}(3). \tag{11.29}$$

By (11.18), every L_j and Q_j at (a, b) is $D(a, b)$ times its value at $(0, 0)$. The coefficients in (11.24), (11.25), and (11.23) are constant on M, since $t = t_0$. It follows that the positively oriented full frame is exactly

$$(N_\ell, N_\phi, N_1, N_2, -T_a, T_b)_{(a,b)} = D(a, b)G_0, \tag{11.30}$$

where $G_0 \in \mathrm{GL}^+(6, \mathbb{R})$ is its single value at $(0, 0)$. Thus neither an unspecified phase nor a position-dependent real-linear matrix remains.

Here is a periodic contraction of D, with all parameters specified. For $v \in [0, 1]$ put

$$s_v = \sin(\pi v/2), \qquad c_v = \cos(\pi v/2), \qquad Q_v(\theta) = \begin{pmatrix} s_v e^{-i\theta} & -c_v \\ c_v & s_v e^{i\theta} \end{pmatrix}, \qquad K_v(\theta) = Q_v(\theta)Q_v(0)^*.$$

The two columns of Q_v have norm one and zero Hermitian inner product, and $\det Q_v = s_v^2 + c_v^2 = 1$. Thus $Q_v, K_v \in \mathrm{SU}(2)$. Direct substitution gives

$$K_0(\theta) = I, \quad K_1(\theta) = \operatorname{diag}(e^{-i\theta}, e^{i\theta}), \quad K_v(0) = I.$$

Let $\iota_{01}, \iota_{02} : \mathrm{SU}(2) \to \mathrm{SU}(3)$ act in the ordered coordinate planes $(0, 1)$ and $(0, 2)$, respectively, and as the identity in the unused coordinate. Then

$$D_v(a, b) = \iota_{01}(K_v(a))\, \iota_{02}(K_v(b)) \tag{11.31}$$

is a well-defined continuous map on $[0, 1] \times (\mathbb{R}/(2\pi\mathbb{Z}))^2$, with $D_0 = I$ and $D_1 = D$. This proves the needed null-homotopy directly, including both phase seams. In particular $D_v(a, b)G_0$ contracts the comparison matrix in (11.30) to one constant positive matrix.

Proposition 11.4. *The stable tangential framing induced by the actual normal four-frame of $H^{-1}(y_0)$ agrees, up to homotopy and the orientation sign (11.28), with the Lie framing (∂_a, ∂_b) on the native two-torus.*

Proof. In the coordinate chart, the stable tangent trivialization is the isomorphism

$$\mathbb{R}^4 \oplus TM \longrightarrow T\mathbb{C}^3|_M, \qquad (v, V) \longmapsto v_1 N_\ell + v_2 N_\phi + v_3 N_1 + v_4 N_2 + V.$$

Using the intrinsic frame $(-\partial_a, \partial_b)$ on TM, its matrix is precisely (11.30). The homotopy (11.31) therefore makes this comparison constant. A constant positive matrix changes no homotopy class of stable framing: Gram–Schmidt deforms it to an element of $SO(6)$, and an orthogonal matrix of determinant one can be reduced to the identity by finitely many oriented plane rotations, each joined to the identity by varying its angle. This proves the asserted comparison within the chart.

It remains to compare the chart with the ambient sphere framing. Pull back the usual stable tangent trivialization $TS^6 \oplus \mathbb{R} \cong \mathbb{R}^7$ under the fixed based diffeomorphism $X \cong S^6$. On the contractible w-ball containing M, it differs from the coordinate trivialization plus one real line by a continuous map into $\mathrm{GL}(7, \mathbb{R})$. Contracting the ball to its center contracts this comparison map to a constant. Thus its restriction to M contributes no varying stable framing class. If the chosen sphere orientation disagrees with the complex orientation, the remaining constant has negative determinant and reverses the resulting framed bordism class. Both that sign and the explicit sign (11.28) disappear in the order-two class computed below. This argument compares actual ambient trivializations on a contractible chart; it does not suppose that $\mathbb{C}^3$ globally trivializes TX. □

11.5 Nonzero framed bordism and the exact unstable class

We give the nonvanishing test for the resulting torus explicitly. A stable framing gives a spin structure by lifting its oriented frame to the trivial Spin-bundle. On the Lie-framed circle this is the disconnected double cover, whereas the spin structure induced from the boundary of a framed disc is the connected double cover: as the boundary tangent turns once, its 2π-rotation lifts from 1 to -1 in $\mathrm{Spin}(2)$. We call the former nonbounding and the latter bounding.

For a closed oriented spin surface S, the spin quadratic refinement

$$q_S : H_1(S; \mathbb{F}_2) \longrightarrow \mathbb{F}_2$$

takes a class represented by an embedded circle to zero or one according to whether the induced spin structure, using its normal line in the surface, is bounding or nonbounding. It satisfies

$$q_S(x + y) = q_S(x) + q_S(y) + (x \cdot y). \tag{11.32}$$

This is the usual spin quadratic form: resolving transverse intersections adds one tangent turn per intersection, and the 2π-spin-lift calculation above gives the last term. The well-defined correspondence between spin structures and such quadratic refinements is given in [Debray and Gunningham, 2018, Theorem 3.2 and Remark 3.3]; its original surface proof is Johnson [1980].

For a symplectic basis $(e_i, f_i)_{i=1}^g$, put

$$\mathrm{Arf}(S) = \sum_{i=1}^{g} q_S(e_i) q_S(f_i) \in \mathbb{F}_2. \tag{11.33}$$

The basis independence can be seen without selecting a preferred basis: factor the finite sum over the symplectic planes to obtain

$$\sum_{x \in H_1(S;\mathbb{F}_2)} (-1)^{q_S(x)} = \prod_{i=1}^{g} \sum_{a,b \in \mathbb{F}_2} (-1)^{a q_S(e_i) + b q_S(f_i) + ab} = 2^g (-1)^{\sum_i q_S(e_i) q_S(f_i)}.$$

The left side has no choice of basis, proving the assertion.

Here is also the bordism-invariance argument needed for the present obstruction. A spin bordism has a handle decomposition. An index zero or three handle creates or removes a spin sphere, whose first homology and Arf invariant are zero. An index one handle either connects two components, in which case the first-homology quadratic forms add orthogonally, or increases the genus of one component. In the latter case the new symplectic pair contains the meridian of the handle. That meridian bounds the cocore disc with its induced spin structure, so its quadratic value is zero; the pair contributes zero to (11.33), regardless of the value on its dual. Existing cycles can be represented outside the attaching discs, so their quadratic values are preserved. An index two handle is the reverse of an index one handle. Therefore each handle preserves the total Arf invariant of the boundary, and a spin boundary has Arf invariant zero. This also proves that a stably framed boundary has Arf invariant zero, because forgetting the framing gives a spin bordism. The full spin bordism statement is recorded in [Debray and Gunningham, 2018, Theorem 3.6].

For the intrinsic Lie framing on $M = (\mathbb{R}/(2\pi\mathbb{Z}))^2$, take

$$e = [a \mapsto (a, 0)], \qquad f = [b \mapsto (0, b)].$$

Both the tangent and normal vectors in the surface are constant in the Lie frame along either circle. Thus both circles have the nonbounding spin structure, and

$$e \cdot f = 1, \qquad q_M(e) = q_M(f) = 1, \qquad q_M(xe + yf) = x + xy + y, \qquad \operatorname{Arf}(M) = 1. \tag{11.34}$$

Reversing the first Lie tangent vector changes the orientation but does not change these mod-two values. In particular the specifically framed inverse image computed above is not a framed boundary.

To identify the element, let $\eta \in \pi_1^{\mathrm{S}}$ denote the stable class of the complex Hopf map. Its Pontryagin–Thom inverse image is the circle with its Lie framing. This can be verified directly on the Hopf map $S^3 \to \mathbb{CP}^1$: near the fibre $(z_1, z_2) = (e^{ia}, 0)$, the target coordinate is z_2/z_1, so the two horizontal normal vectors are $(0, e^{ia})$ and $(0, ie^{ia})$. Add the tangent vector $(ie^{ia}, 0)$ and the outward radial vector $(e^{ia}, 0)$. In $\mathbb{C}^2$ the resulting full frame makes one complete rotation in each of its two real coordinate planes. This loop is contractible explicitly: for the fixed real-linear involution $J(z_1, z_2) = (z_1, \overline{z}_2)$,

$$J \operatorname{diag}(e^{ia}, e^{ia}) J = \operatorname{diag}(e^{ia}, e^{-ia}) = K_1(-a),$$

and $JK_v(-a)J$ contracts the original loop to the identity using the periodic homotopy already constructed. Thus the induced stable tangent frame on the circle is the Lie frame. This is the basic framed-circle identification in Pontryagin [1955].

The product of two normally framed inverse images represents the product in stable homotopy: its tubular collapse is the smash of the two tubular collapse maps, as follows immediately from the product tubular neighborhood $\nu_1 \times \nu_2$ and its ordered normal coordinates. Consequently the product of the two Lie circles represents η^2. Equation (11.34) proves $\eta^2 \neq 0$ by an explicit bordism invariant. The stable group $\pi_2^{\mathrm{S}} = \mathbb{Z}/2$ is Pontryagin's computation Pontryagin [1950, 1955]; the stable table in [Hatcher, 2002, pp. 384–385] gives the same group. Thus the sign from (11.28) satisfies $-\eta^2 = \eta^2$.

Proof of the theorem. The globally continuous quotient H is smooth on a neighborhood of its compact inverse image M, by (11.19) and (11.11). First make it constant near the source basepoint, before choosing any approximation bound. Choose a target chart $\kappa : O \to \mathbb{R}^4$ about $*$ with $\kappa(*) = 0$, convex ball image, and $y_0 \notin O$. Choose a source neighborhood U of x_{base}, disjoint from M, with $H(U) \subset O$. Let β be a smooth function supported in U, equal to one near x_{base}, with $0 \leq \beta \leq 1$. On U use the based homotopy

$$H_s(x) = \kappa^{-1}\big((1 - s\beta(x))\kappa(H(x))\big), \qquad 0 \leq s \leq 1,$$

and leave H unchanged off U. Its endpoint H_b is smooth and constant near the basepoint; it agrees with H near M and has no new inverse image of y_0.

Choose one closed tubular neighborhood U_1 of M, with $M \subset \operatorname{int} U_1$, contained in the unchanged smooth region and disjoint from the basepoint neighborhood. Compactness gives the positive bound

$$d_0 = \min_{x \in X \setminus \operatorname{int} U_1} d(H_b(x), y_0) > 0.$$

Relative smooth approximation, fixing the same closed set U_1 and a neighborhood of the basepoint, produces a smooth based map H_{sm} whose error from H_b is less than $d_0/2$ on its complement. Choose the approximation closer also than a fixed target injectivity radius; the short target geodesics give a based homotopy. Thus the exact inverse image and its differential are unchanged on U_1, and the complement acquires no new inverse image. The relative Pontryagin construction applies as in [Milnor, 1965, §7, Theorem B and Lemma 4, pp. 43, 48–50].

Pontryagin–Thom therefore identifies the stabilization of $[H]$ with the bordism class of M carrying the four normal vectors (11.24)–(11.25). The comparison (11.30) and its explicit contraction identify that class with the Lie torus, with the harmless order-two orientation sign already computed. Hence

$$E^\infty[H] = \eta^2 \neq 0.$$

Freudenthal suspension is an isomorphism

$$E : \pi_{n+2}(S^n) \longrightarrow \pi_{n+3}(S^{n+1}) \qquad (n \geq 4),$$

because $n + 2 \leq 2n - 2$. Therefore stabilization from $\pi_6(S^4)$ is an isomorphism onto π_2^{S}. The composite $\eta_4 \circ \eta_5 : S^6 \to S^4$ stabilizes to η^2, so injectivity gives the exact equality (11.12). Finally, a based null-homotopy of q_2 would postcompose with p_3 to a based null-homotopy of H, contradicting (11.12). □

Exact scope of the comparison with the angle-forgetting map. The map here retains the angular coordinate before collapsing the coordinate two-skeleton. Its regular inverse image is the specific normally four-framed T^2 above. The angle-forgetting postcomposition has a normally three-framed T^3 inverse image. The calculation above is independent of its class and supplies an actual obstruction to null-homotopy of the original q_2, even when that other postcomposition is null-homotopic. The exact morphism carrying this obstruction is

$$(p_3)_* : [X, \mathcal{A}]_* \longrightarrow [X, S^4]_*, \qquad [q_2] \longmapsto [H] = \eta_4\eta_5.$$

No conclusion about an arbitrary earlier quotient follows merely from the diffeomorphism type of one inverse image; the specified four-framing is what proves this value.

Corollary 11.5 (The exact order-two quotient detected by the original cusp map)**.** *Let $d : (S^6, *) \to (X, x_{\mathrm{base}})$ be the fixed based diffeomorphism, and put*

$$\alpha = [q_2 \circ d] \in \pi_6(\mathcal{A}, v_0).$$

The original top-cell postcomposition induces a surjective homomorphism

$$\varepsilon = (p_3)_* : \pi_6(\mathcal{A}, v_0) \longrightarrow \pi_6(S^4, *) \cong \mathbb{Z}/2, \qquad \varepsilon(\alpha) = \eta_4 \circ \eta_5.$$

It gives the exact quotient isomorphism

$$\pi_6(\mathcal{A}, v_0)/\ker \varepsilon \xrightarrow{[\beta] \mapsto \varepsilon(\beta)} \mathbb{Z}/2.$$

Every odd multiple of α is nonzero. Its order is either infinite or an even positive integer; this calculation does not determine that order or assert a splitting of the displayed quotient.

Proof. Postcomposition distributes over the pinch sum on the source sphere, so $(p_3)_*$ is a group homomorphism. Theorem 11.1 gives $(p_3)_*(\alpha) = [p_3 q_2 d] = \eta_4\eta_5 \neq 0$. This is the nonzero element of the cyclic group of order two, proving surjectivity. The first isomorphism theorem gives the stated quotient with exactly the displayed map. For every integer n,

$$\varepsilon(n\alpha) = (n \bmod 2)\eta_4\eta_5.$$

An odd multiple therefore has nonzero image and cannot vanish. If $m\alpha = 0$, applying ε forces m to be even. These arguments prove all the assertions without inferring the exact order of α. □

12 A nonzero Hopf–cusp morphism to the stable mod–24 group

The retained-angle input used below is the complete Theorem 11.1 above.

12.1 The exact source maps and conventions

Retain the original marked source (X, x_0), its fixed based diffeomorphism $d : (S^6, *) \to (X, x_0)$, the cusp coordinate t, the fixed cutoff ϵ, and the native quotient $\overline{\pi}_f$ of the cusp fibres by $\mathbb{Z}\langle \widehat{w}, \widehat{\delta}\rangle$. The quotient coordinates are ordered by $(\widehat{\overline{\gamma}}, \widehat{\overline{u}})$. Thus the fixed continuous based map is

$$q_2 : X \longrightarrow \mathcal{A} = \mathrm{Susp}(S^1 \times \overline{T}^2), \qquad q_2(x) = [|t|, t/|t|, \overline{\pi}_f(x)] \quad (0 < |t| < \epsilon).$$

It sends the entire central fibre W to the first suspension vertex and $X \setminus \operatorname{int} N_0$ to the second. The unreduced suspension $\mathcal{A}$ is based at its first vertex. In the original product torus set

$$K^{(2)} = (\{1\} \times \overline{T}^2) \cup (S^1 \times \{u_1 = 0\}) \cup (S^1 \times \{u_2 = 0\}), \qquad c_3 : T^3 \longrightarrow T^3/K^{(2)} \cong S^3.$$

The retained-angle sphere map and its sphere-source representative are

$$p_3([r, z]) = [r, c_3(z)], \qquad H = p_3 q_2 : X \longrightarrow \Sigma_r(S^3, *) \cong S^4, \qquad h = H \circ d : S^6 \longrightarrow S^4. \tag{12.1}$$

Both suspension vertices and the whole cylinder over the basepoint are identified in $\Sigma_r(S^3, *)$. The preceding complete retained-angle framing theorem proves

$$[h] = \eta_4 \circ \eta_5 \neq 0 \quad \text{in} \quad \pi_6(S^4) \cong \mathbb{Z}/2. \tag{12.2}$$

Its proof identifies the actual derivative-framed inverse torus with the Lie-framed torus, with Arf invariant one. We use this already proved class, retaining the particular map (12.1); see Retained-angle cusp theorem [2026].

Fix the complex Hopf map

$$\eta_2 : (S^3, (1, 0)) \longrightarrow (\mathbb{CP}^1, [1 : 0]) \cong (S^2, *), \qquad (z_1, z_2) \longmapsto [z_1 : z_2].$$

Use ordered smash coordinates on spheres. In this note $E^k f = f \wedge \mathrm{id}_{S^k}$ denotes right reduced suspension; the usual ordered sphere homeomorphisms are understood. Put $\eta_n = E^{n-2}\eta_2$, and write η for the stable Hopf class. The order of each factor is retained below.

12.2 The Hopf–cusp homomorphism and its nonzero value

Theorem 12.1. *The assignment*

$$\mu_\eta : \pi_6(S^4) \longrightarrow \pi_9(S^6), \qquad \mu_\eta([x]) = [\eta_2 \wedge x] \tag{12.3}$$

is an injective group homomorphism. Its literal representative has the complete domain and codomain

$$\eta_2 \wedge x : S^3 \wedge S^6 \cong S^9 \longrightarrow S^2 \wedge S^4 \cong S^6.$$

Choose any generator ν_6 of $\pi_9(S^6) \cong \mathbb{Z}/24$. Then the particular original cusp map supplies

$$\kappa_H = [\eta_2 \wedge h] = \mu_\eta([h]) = 12\nu_6 \neq 0. \tag{12.4}$$

It has exact order two. Its value is independent of orientation reversals in the sphere identifications and of the choice of the generator ν_6.

Proof. A based homotopy of x smashes with the fixed based Hopf map, so (12.3) is well-defined. The exact ordered point-set factorization is

$$\eta_2 \wedge x = (\eta_2 \wedge \mathrm{id}_{S^4}) \circ (\mathrm{id}_{S^3} \wedge x) : S^3 \wedge S^6 \longrightarrow S^3 \wedge S^4 \longrightarrow S^2 \wedge S^4. \tag{12.5}$$

The first applied map has source dimension nine and target dimension seven; the second has source dimension seven and target dimension six. The second map is $E^4\eta_2$. To compare the first with $E^3 x = x \wedge \mathrm{id}_{S^3}$, interchange its sphere factors at source and target. The interchange degrees are respectively $(-1)^{3\cdot 6} = +1$ and $(-1)^{3\cdot 4} = +1$. A based map of a sphere of degree one is based homotopic to the identity, so

$$\mu_\eta([x]) = (E^4\eta_2)_*(E^3[x]). \tag{12.6}$$

Suspension is a group homomorphism. Postcomposition is also a group homomorphism on homotopy groups, since it distributes over the source pinch map defining the sum. Thus (12.6) proves additivity, including zero and inverses, before passage to stable homotopy.

Freudenthal makes

$$E : \pi_{n+3}(S^n) \longrightarrow \pi_{n+4}(S^{n+1})$$

an isomorphism for $n \geq 5$, because $n + 3 < 2n - 1$. In particular stabilization $\pi_9(S^6) \to \pi_3^{\mathrm{S}}$ is an isomorphism. Stabilization from $\pi_6(S^4)$ is likewise an isomorphism onto π_2^{S}, because $n + 2 < 2n - 1$ for $n \geq 4$. The exact factorization (12.5) shows that its stable class is the stable composition product of η with $[x]$. Using the proved original class (12.2),

$$E^\infty \kappa_H = \eta\, E^\infty[h] = \eta\, \eta^2 = \eta^3 = 12\nu \neq 0 \quad \text{in} \quad \pi_3^{\mathrm{S}} \cong \mathbb{Z}/24. \tag{12.7}$$

The integral relation here is the one explicitly stated in [Hatcher, 2002, p. 385], not the coefficient 4 in the two-primary quotient. More precisely, the two-primary relation is $\eta^3 = 4\nu$ modulo eight. Bilinearity and $2\eta = 0$ give $2\eta^3 = 0$ integrally. In $\mathbb{Z}/24$ the only elements killed by two are 0 and 12ν, and only the second has nonzero image 4ν modulo eight. This recovers the exact integral coefficient and its nonvanishing.

The inverse of the stabilization isomorphism therefore gives (12.4). Since the source of μ_η is cyclic of order two, generated by $[h]$, its nonzero value proves injectivity and identifies its image as $\{0, 12\nu_6\}$.

An orientation reversal at a sphere source or target multiplies the corresponding stable homotopy class by -1; here $-12\nu_6 = 12\nu_6$. A sign change in the chosen Hopf representative also leaves the stable product unchanged since $-\eta = \eta$. Finally, replacing the generator by $u\nu_6$, with u a unit modulo twenty-four, does not change the element: every such u is odd, and $12u \equiv 12 \pmod{24}$. All orientation and generator assertions consequently hold for the actual order-two element. □

Corollary 12.2 (The same nonzero class in the twelve-dimensional receiver)**.** *Put* $\nu_9 = E^3\nu_6$*. The particular based map*

$$E^3(\eta_2 \wedge h) : S^{12} \longrightarrow S^9$$

has class $12\nu_9 \neq 0$ *in* $\pi_{12}(S^9) \cong \mathbb{Z}/24$*.*

Proof. Each of these three suspensions is an isomorphism in the preceding stable range. Apply their composite to (12.4); it preserves the exact order two. □

12.3 The actual marked peripheral character

Retain the original three elliptic-kernel matrices, with their original signs and ordering:

$$C_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad C_1 = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}, \qquad C_2' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{12.8}$$

The existing peripheral theorem Higher torus wrap [2026] defines the surjective marked abelianization character

$$\chi : SL_2(\mathbb{Z}) \longrightarrow \mathbb{Z}/12, \qquad (\chi(C_0), \chi(C_1), \chi(C_2')) = (-1, 4, -3). \tag{12.9}$$

For an explicit check of this normalization, let

$$S_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T = C_0.$$

Then $C_2' = S_0^3$, $C_1 = T^{-1}S_0$, and direct matrix multiplication gives $C_1C_2' = C_0^{-1}$. In the modular presentation $S_0^4 = 1$, $S_0^2 = (S_0T)^3$, the abelianized generators satisfy $4s = 0$ and $s + 3t = 0$. Thus $s = -3t$ and $12t = 0$. The assignment $t \mapsto -1$, $s \mapsto 3$ satisfies both relations and maps onto $\mathbb{Z}/12$; consequently the order of the abelianized generator is exactly twelve. It gives, without a change of peripheral matrices, precisely $\chi(C_0) = -1$, $\chi(C_1) = 1 + 3 = 4$, and $\chi(C_2') = 3 \cdot 3 = -3 \pmod{12}$.

Theorem 12.3 (An exact nonzero peripheral homotopy morphism)**.** *Define*

$$\Theta_H : \mathbb{Z}/12 \longrightarrow \pi_6(S^4), \qquad \Theta_H([n]) = n[h], \qquad \Psi_H = \mu_\eta \circ \Theta_H \circ \chi : SL_2(\mathbb{Z}) \longrightarrow \pi_9(S^6). \tag{12.10}$$

These are group homomorphisms. The complete coefficient formula is

$$\Psi_H(M) = 12\,\widetilde{\chi(M)}\,\nu_6, \tag{12.11}$$

where $\widetilde{\chi(M)}$ *is any integer lift of the marked residue. In the original matrix order of* (12.8)*, its exact values are*

$$(\Psi_H(C_0), \Psi_H(C_1), \Psi_H(C_2')) = (12\nu_6, 0, 12\nu_6). \tag{12.12}$$

Its image and kernel are exactly

$$\operatorname{im}\Psi_H = \{0, 12\nu_6\}, \qquad \ker\Psi_H = \chi^{-1}(2\mathbb{Z}/12). \tag{12.13}$$

In particular its kernel has index two in $SL_2(\mathbb{Z})$*.*

Proof. The element $[h]$ has exact order two, so replacing n by $n + 12k$ leaves $n[h]$ unchanged. Addition in its abelian homotopy group proves that Θ_H is a group homomorphism. Moreover

$$\ker\Theta_H = 2\mathbb{Z}/12, \qquad \operatorname{im}\Theta_H = \pi_6(S^4).$$

The composite in (12.10) is consequently a group homomorphism, and (12.4) gives $\mu_\eta(n[h]) = 12n\nu_6$. Changing the integer lift by $12k$ changes its coefficient by $144k$, which is a multiple of twenty-four. This proves (12.11) with its independence of lifts. The three coefficients are respectively $-12 \equiv 12$, $48 \equiv 0$, and $-36 \equiv 12 \pmod{24}$, proving (12.12). Since $12n \equiv 0 \pmod{24}$ holds exactly when n is even, the asserted kernel is exact. Surjectivity of χ and nonvanishing at C_0 prove that the image is the full order-two subgroup, so the first isomorphism theorem gives the index assertion. □

Here is the complete parity factorization. Let $\mathrm{red}_2 : \mathbb{Z}/12 \to \mathbb{Z}/2$ be ordinary parity reduction, let $\iota_2([e]) = 12e \pmod{24}$, and let $c_{\nu_6} : \pi_9(S^6) \to \mathbb{Z}/24$ send ν_6 to one. Every arrow and its codomain in the diagram are fixed:

$$\begin{array}{ccc} SL_2(\mathbb{Z}) & \xrightarrow{\Psi_H} & \pi_9(S^6) \\ {\scriptstyle \mathrm{red}_2 \circ \chi}\downarrow & & \downarrow{\scriptstyle c_{\nu_6}} \\ \mathbb{Z}/2 & \xrightarrow{\iota_2} & \mathbb{Z}/24. \end{array} \tag{12.14}$$

Commutativity is exactly the coefficient calculation above; the bottom arrow is injective because $12e$ is zero modulo twenty-four exactly when e is zero modulo two.

12.4 Comparison with the earlier maps, with their exact types

Write $\psi_H = c_{\nu_6} \circ \Psi_H$. Ordinary reduction $r : \mathbb{Z}/24 \to \mathbb{Z}/12$ gives

$$r \circ \psi_H = 0, \qquad (r \circ \psi_H)(C_0) = 0 \neq -1 = \chi(C_0). \tag{12.15}$$

Thus this nonzero character is not a reduction-compatible lift of χ. Its natural commuting square is (12.14). The existing nonexistence theorem for a reduction-compatible lift remains valid: every homomorphism from $SL_2(\mathbb{Z})$ to an abelian group factors through its abelianization $\mathbb{Z}/12$, whereas a lift of $\chi(C_0) = -1$ would require C_0 to map to 11 or 23 modulo twenty-four, each of order twenty-four. The abelianized generator has order twelve and cannot have either image.

The universal even-subgroup injection is the separate map

$$j : \mathbb{Z}/12 \hookrightarrow \mathbb{Z}/24, \qquad j([a]) = [2a].$$

The new coefficient morphism factors through it by the exact identity

$$\psi_H = j \circ [6] \circ \chi, \qquad [6] : \mathbb{Z}/12 \longrightarrow \mathbb{Z}/12, \quad [a] \longmapsto [6a]. \tag{12.16}$$

Indeed $2(6a) = 12a \pmod{24}$. The map j is injective with image of order twelve; the composite (12.16) has image of order two because [6] retains precisely parity. The map j is realized by the already proved universal suspension $E^6 : \pi_6(S^3) \to \pi_{12}(S^9)$ after compatible generator choices. Equation (12.16) is a coefficient identity; the actual new geometric representative is $\eta_2 \wedge h$, followed, when desired, by three suspensions as above.

For the original angle-forgetting map $g = \widehat{p}_2 q_2 : X \to S^3$, the separate framed calculation has proved $[g \circ d] = 0$. Therefore the previous morphism

$$\Theta_g : \mathbb{Z}/12 \longrightarrow \pi_6(S^3), \qquad [n] \longmapsto n[g \circ d]$$

and $E^6 \Theta_g \chi$ are zero. In contrast, $E^3 \Psi_H(C_0) = 12\nu_9 \neq 0$ in this same stable receiver $\pi_{12}(S^9)$. These statements agree because the exact postcompositions from $\mathcal{A}$ are different: p_3 retains the angular coordinate in its top-cell map, while $\widehat{p}_2$ forgets it. The new homomorphism uses the already proved value of $p_3 q_2$ and the specified Hopf map, with all maps displayed in (12.1)–(12.10).

The constructed representative has source $S^3 \wedge X \cong S^9$; its threefold suspension has source S^{12}. The number twenty-four is the order of the receiving stable homotopy group. This calculation does not identify either source with a twenty-four-dimensional carrier, does not supply new periods or analytic overlap maps, and does not complete the global higher torus family. It supplies the exact nonzero homotopy and peripheral morphisms just proved.

Source checks. The three matrices and the marked values were checked directly against equations *elliptic-peripheral-matrices* and *peripheral-mod12-values* in the existing higher-rung draft Higher torus wrap [2026]. The calculation above reproduces their signs using the same S_0, T. The retained-angle input is the complete derivative-framing theorem in Retained-angle cusp theorem [2026]. For the stable facts, the author-hosted text Hatcher [2002] was read at Corollary 4.24, printed p. 360, and at printed pp. 384–385. Page 385 explicitly distinguishes the two-primary coefficient four from the integral coefficient twelve. Proposition 4.56 on that page, with its proof on p. 388, gives the stable composition product and its smash description. The fixed-factor proof (12.5)–(12.6) establishes the needed homomorphism here directly.

Supplementary references for the retained-angle and Hopf–cusp proofs

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